

72nd BPSC PRELIMS

प्रहार

Test series

Test- 04

Indian Geography 01

Solution

English



Khan Global Studies

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BPSC PRELIMS TEST - 4 (SOLUTION)

Q 1. Ans. A

Explanation:

- Damodar River:
 - ✓ River origin: Khamarpat Hill
 - ✓ Plateau: Chotanagpur Plateau
 - ✓ District: Palamu district
 - ✓ State: Jharkhand
 - ✓ River type: East-flowing peninsular river
 - ✓ River valley type: Rift valley
 - ✓ Geological structure: Gondwana rock region
 - ✓ States covered: Jharkhand, West Bengal
 - ✓ Final confluence: Hooghly River
 - ✓ Confluence location: Near Falta (south of Kolkata)
 - ✓ Flood characteristic: Frequent floods in the lower West Bengal plains
 - ✓ Nickname: Sorrow of Bengal
 - ✓ Damodar Valley Corporation (DVC)
 - Establishment year: 1948
 - Project type: Multipurpose river valley project
 - Model inspiration: Tennessee Valley Authority (USA)
 - ✓ Major dams on the Damodar system:
 - Panchet Dam — Damodar River
 - Konar Dam — Konar River
 - ✓ Tributaries of the Damodar River
 - Major tributary: Barakar River
 - ◇ Barakar origin: Padma area
 - ◇ District: Hazaribagh district
 - ◇ State: Jharkhand
 - ◇ Confluence with Damodar: Near Dishergarh
 - ◇ Location: Jharkhand–West Bengal border
 - ◇ Major dams on Barakar:
 - ◇ Tilaiya Dam
 - ◇ Maithon Dam
 - ✓ Other important tributaries:
 - Konar River
 - Bokaro River
 - ✓ Major coalfields:
 - Jharia Coalfield
 - Raniganj Coalfield
 - Bokaro Coalfield
- Subarnarekha River:
 - ✓ River type: East-flowing river
 - ✓ Origin: Nagri village in the Ranchi District of Jharkhand
 - ✓ Plateau: Ranchi Plateau
 - ✓ Final drainage: Bay of Bengal
 - ✓ Famous feature: Hundru Falls (Jharkhand)
 - ✓ Not a tributary of: Damodar River
- Ajay River:
 - ✓ River type: Tributary of the Bhagirathi River
 - ✓ Origin: south-west of Munger district
 - ✓ State: Bihar
 - ✓ Final confluence: Bhagirathi River
 - ✓ Confluence location: Near Katwa (West Bengal)
- Mayurakshi River:
 - ✓ River type: Tributary of the Hooghly system
 - ✓ Origin: Trikut Hill
 - ✓ Other name: Matihara
 - ✓ State: Jharkhand
 - ✓ Major dam: Massanjore dam (Canada dam)
- Location: Dumka district, Jharkhand

Q 2. Ans. A

Explanation:

- Trans-Himalaya:
 - ✓ Location: North of the Great Himalaya (Himadri)
 - ✓ Alternate name: Tibetan Himalaya
 - ✓ Major ranges:
 - Karakoram Range
 - Ladakh Range
 - Zaskar Range
 - Kailash Range
 - ✓ Major glacier: Siachen Glacier
 - Location: Karakoram Range
 - ✓ Major peak: K2 (Godwin Austen) — Karakoram Range
 - ✓ Structural boundary:
 - Separated from Himadri by Indo-Tsangpo Suture Zone (ITSZ)
 - ✓ Water divide significance:
 - Source region of the Indus River system
- Great Himalaya / Himadri:
 - ✓ Position: Northernmost continuous Himalayan range
 - ✓ Average elevation: About 6000 m

- ✓ Nature: Perennially snow-covered
 - ✓ Rock type: Granite core
 - ✓ Major glaciers:
 - Gangotri Glacier
 - Zemu Glacier
 - ✓ Major peaks:
 - Mount Everest
 - Kanchenjunga
 - Makalu
 - Dhaulagiri
 - ✓ Major rivers originating from glaciers:
 - Ganga system glaciers
 - Indus system glaciers
 - Brahmaputra system glaciers
 - ✓ Structural boundary:
 - Separated from the Lesser Himalaya by the Main Central Thrust (MCT)
 - Lesser Himalaya / Himachal:
 - ✓ Position: South of Himadri
 - ✓ Average width: About 50 km
 - ✓ Elevation range: 3700–4500 m
 - ✓ Major ranges:
 - Pir Panjal Range — the longest range
 - Dhauladhar Range
 - Mahabharat Range
 - ✓ Important valleys:
 - Kashmir Valley
 - Kangra Valley
 - Kullu Valley
 - ✓ Structural boundary:
 - Separated from the Shiwalik by the Main Boundary Thrust (MBT)
 - Shiwalik / Outer Himalaya:
 - ✓ Position: Southernmost Himalayan range
 - ✓ Elevation range: 900–1100 m
 - ✓ Width range: 10–50 km
 - ✓ Rock composition:
 - Unconsolidated sediments
 - Gravel
 - Alluvium
 - ✓ Hazard characteristics:
 - High landslide risk
 - Earthquake-prone zone
 - ✓ Structural boundary:
 - Separated from the Northern Plains by the Himalayan Frontal Fault (HFF)
 - Additional Facts:
 - ✓ Valley & Dun:
 - ✓ Dun's definition: Longitudinal valleys between the Lesser Himalaya and the Shiwaliks
 - ✓ Major Duns:
 - Dehradun
 - Kotli Dun
 - Patli Dun
 - Eastern equivalent term: Duars
 - ✓ Karakoram Range location: Ladakh region
 - ✓ Pir Panjal Range location: Jammu & Kashmir and Himachal Pradesh
 - ✓ Dhauladhar Range location: Himachal Pradesh
 - ✓ Mahabharat Range location: Nepal Himalayas
 - ✓ Shiwalik spread: Jammu to Arunachal Pradesh
- Q 3. Ans. C**
- Explanation:**
- Statement 1 is correct:
 - ✓ River origin: West of Rohtang Pass
 - ✓ Origin region: Kullu Hills
 - ✓ State: Himachal Pradesh
 - ✓ Tributary type: Left-bank tributary of the Indus
 - ✓ Major valley traversed: Chamba Valley
 - ✓ State coverage: Himachal Pradesh
 - ✓ River classification: Left-bank tributary
 - ✓ Main river system: Indus River system
 - Statement 2 is correct:
 - ✓ River basin location: Between Pir Panjal and Dhauladhar ranges
 - ✓ Northern boundary range: Pir Panjal Range
 - ✓ Southern boundary range: Dhauladhar Range
 - ✓ Drainage separation:
 - Ravi basin: Between Pir Panjal and Dhauladhar
 - Beas basin: South of Dhauladhar Range
 - Physiographic feature: Structural trough between the Lesser Himalayan ranges
 - Additional facts about the Ravi River:
 - ✓ Ancient name: Purushni / Iravati
 - ✓ Final confluence: Chenab River
 - ✓ Confluence location: Pakistan
 - ✓ Flows through states:
 - Himachal Pradesh
 - Punjab
 - ✓ Forms boundary:
 - ◇ India–Pakistan international boundary (approx. 80 km stretch)

- ◊ Entry into plains: Near Madhopur (Punjab)
- Tributary Facts:
 - ◊ Right-bank tributaries:
 - ▶ Budhil River
 - ▶ Nai (Dhodh) River
 - ▶ Seul River
 - ◊ Left-bank tributary:
 - ▶ Ujh River
 - ▶ Ujh origin: Kailash Mountains (Jammu region)
- Dam & Project:
 - ◊ Major multipurpose dam: Ranjit Sagar Dam (Thein Dam)
 - ▶ Location: Punjab–Jammu & Kashmir border
 - ◊ Major hydropower project: Chamera Dam
 - ▶ Location: Chamba district, Himachal Pradesh
 - ◊ Downstream project: Shahpurkandi Dam
 - ▶ Location: Punjab
- Treaty: Indus Water Treaty (1960)
 - ◊ Eastern Rivers allocated to India:
 - ▶ Ravi
 - ▶ Beas
 - ▶ Sutlej
 - Western Rivers allocated to Pakistan:
 - ◊ Indus
 - ◊ Jhelum
 - ◊ Chenab
 - ◊ Brokered by: World Bank
- Wind direction reverses from Northeast to Southwest
- ✓ Monsoon withdrawal:
 - Withdrawal begins from the Northern Plains
 - Typical start month: October
- ✓ Season characteristic:
 - Clear skies
 - High day temperature
 - Phenomenon name: October Heat
- Rainfall on Tamil Nadu Coast:
 - ✓ Primary rainy season for Tamil Nadu: Northeast Monsoon
 - ✓ Wind nature:
 - Northeast winds are mostly dry over land
 - ✓ Moisture source:
 - Winds cross the Bay of Bengal
 - Moisture absorbed over warm ocean waters
 - ✓ Rainfall zone:
 - Coromandel Coast
 - Tamil Nadu coast
 - South coastal Andhra Pradesh
 - ✓ Orographic effect:
 - Moist winds rise along the Eastern Ghats
 - Results in rainfall
 - ✓ Rainfall enhancement factor:
 - Tropical cyclones
 - ✓ Cyclone origin region:
 - Andaman Sea
 - Bay of Bengal
 - Peak cyclone season: October–November
- Punjab and Haryana:
 - ✓ Primary rainfall season: Southwest Monsoon (June–September)
 - ✓ Winter rainfall source: Western Disturbances
 - ✓ Western Disturbance origin: Mediterranean Sea
 - ✓ Winter rainfall importance:
 - Essential for Rabi crops
 - Major crop supported: Wheat
 - ✓ Reason for low NE monsoon rainfall:
 - Far inland location
 - Dry continental winds
- Western Rajasthan:
 - ✓ Climate type: Arid
 - ✓ Major desert: Thar Desert
 - ✓ Rainfall characteristic: Very low annual rainfall

Q 4. Ans. C

Explanation:

- Retreating Monsoon:
 - ✓ Season period: October–November
 - ✓ Cause: Apparent southward movement of the Sun
 - ✓ Sun position shift: Towards the Tropic of Capricorn
 - ✓ Pressure change:
 - Monsoon trough weakens over Northern Plains
 - High-pressure system develops over land
 - ✓ Wind change:
 - Southwest monsoon weakens

- ✓ Reason for low rainfall:
 - The Aravalli Range is aligned parallel to the SW monsoon winds
 - Poor interception of the Arabian Sea branch
- ✓ Northeast monsoon effect: Dry continental winds
- Gujarat Coast
 - ✓ Primary rainfall source: Arabian Sea branch of Southwest Monsoon
 - ✓ Rainfall months: June–September
 - ✓ Weather after monsoon: Dry conditions
 - ✓ Reason for low NE monsoon rainfall:
 - Northeast winds originate over land
 - Lack of a moisture source
- Additional Facts:
 - ✓ Monsoon Withdrawal Facts:
 - First region of withdrawal: Northwestern India
 - Last region of withdrawal: Tamil Nadu coast
 - Complete withdrawal time: By mid-December
 - ✓ Western Disturbance Facts:
 - Nature: Temperate cyclonic system
 - Origin region: Mediterranean Sea
 - Movement direction: West → East
 - Affected regions:
 - ◇ Punjab
 - ◇ Haryana
 - ◇ Western Uttar Pradesh
 - ◇ Northern Plains
 - Importance:
 - ◇ Winter rainfall
 - ◇ Snowfall in the Himalayas
 - ✓ Eastern Ghats Facts:
 - Location: Along the eastern coast of India
 - Nature: Discontinuous mountain range
 - Role in rainfall:
 - Causes the uplift of northeast monsoon winds
 - Enhances rainfall on the Tamil Nadu coast
 - ✓ Cyclone Facts (Highly Tested)
 - Major cyclone season: October–November
 - Major formation region: Bay of Bengal
 - Common landfall region:
 - ◇ Tamil Nadu coast

- ◇ Andhra Pradesh coast
- ◇ Odisha coast
- ◇ West Bengal coast

Q 5. Ans. A

Explanation:

- Koeppen Classification:
 - ✓ Classification basis: Temperature and precipitation
 - ✓ Indicator used: Natural vegetation
 - ✓ Key parameter: Monthly temperature and rainfall values
- Major Koeppen Climate Types in India:
 - ✓ Amw — Monsoon Climate with Short Dry Season:
 - Climate type: Tropical monsoon
 - Rainfall feature: Short dry season
 - Major regions:
 - ◇ West coast south of Goa
 - ◇ Konkan coast
 - ◇ Malabar coast
 - Rainfall source: Southwest Monsoon
 - ✓ As — Monsoon Climate with Dry Summer:
 - Climate type: Tropical monsoon
 - Dry season: Summer
 - Major region:
 - ◇ Coromandel Coast
 - State coverage:
 - ◇ Tamil Nadu coast
 - Primary rainfall season: Northeast Monsoon
 - ✓ Aw — Tropical Savanna Climate
 - Climate type: Tropical wet and dry
 - Rainfall pattern: Wet summer, dry winter
 - Major regions:
 - ◇ Peninsular Plateau south of Tropic of Cancer
 - States included:
 - ◇ Karnataka
 - ◇ Telangana
 - ◇ Parts of Maharashtra
 - Interior Tamil Nadu
 - ✓ BShw — Semi-Arid Steppe Climate
 - Climate type: Semi-arid
 - Rainfall feature: Low and irregular
 - Major regions:
 - ◇ North-western Gujarat
 - ◇ Western Rajasthan (marginal areas)

- ◇ Punjab (western/arid part)
 - Transition zone: Between desert and humid regions
 - ✓ BWhw — Hot Desert Climate
 - Climate type: Hot desert
 - Rainfall: Extremely low
 - Major region:
 - Extreme western Rajasthan
 - Major desert: Thar Desert
 - ✓ Cwg — Monsoon Climate with Dry Winter
 - Climate type: Subtropical monsoon
 - Winter condition: Dry winter
 - Temperature regime:
 - ◇ Hot summer before monsoon
 - Major regions:
 - ◇ Gangetic Plains
 - ◇ Eastern Rajasthan
 - ◇ Northern Madhya Pradesh
 - ◇ Most of North-East India
 - ◇ Most of Punjab
 - Important plains covered:
 - ◇ Upper Ganga Plain
 - ◇ Middle Ganga Plain
 - ◇ Lower Ganga Plain
 - Dfc — Cold Climate with Short Summer
 - Climate type: Cold humid
 - Winter condition: Severe cold
 - Summer duration: Short
 - Major region:
 - Arunachal Pradesh (high altitude areas)
 - Typical location: Higher Himalayan regions
 - ✓ E — Polar Type Climate
 - Climate type: Polar
 - Temperature: Extremely low
 - Major regions:
 - ◇ Jammu & Kashmir (Ladakh region)
 - ◇ Himachal Pradesh (high altitudes)
 - ◇ Uttarakhand (high Himalayan areas)
 - Typical terrain: Snow-covered high mountains
 - Additional Facts:
 - ✓ Map-Based Climate Distribution:
 - West coast climate: Amw
 - Coromandel coast climate: As
 - Peninsular plateau climate: Aw
 - Gangetic plains climate: Cwg
 - Western Rajasthan climate: BWhw
 - North-west transition zone climate: BShw
 - High Himalaya climate: E
 - ✓ Tropic of Cancer Climate Role
 - Latitude: 23°30' N
 - Divides India into:
 - ◇ Tropical region (south)
 - ◇ Subtropical region (north)
 - Major climate south of Tropic: Aw (Savanna)
 - Major climate north of Tropic: Cwg (Subtropical monsoon)
- Q 6. Ans. A**
- Explanation:**
- Diurnal Temperature Range
 - ✓ Definition: Difference between maximum daytime temperature and minimum nighttime temperature
 - ✓ Highest temperature time: Mid-afternoon
 - ✓ Lowest temperature time: Just before sunrise
 - ✓ Region with Lowest Diurnal Temperature Range
 - ✓ Andaman & Nicobar Islands
 - ✓ Location type: Oceanic island group
 - ✓ Surrounding water bodies:
 - Bay of Bengal
 - Andaman Sea
 - ✓ Latitude range: 6° N to 14° N
 - ✓ Temperature variation: Very low
 - ✓ Typical diurnal range: About 3°C–5°C
 - ✓ Primary controlling factors:
 - Strong maritime influence
 - High humidity
 - Frequent cloud cover
 - ✓ Climate type: Tropical maritime climate
 - ✓ Key reason:
 - Water has a high specific heat capacity
 - Slow heating and slow cooling of the surrounding seas
 - Thar Desert
 - ✓ Location: Western Rajasthan
 - ✓ Climate type: Hot desert climate
 - ✓ Diurnal temperature range: Very high
 - ✓ Typical variation: Up to 30°C–35°C
 - ✓ Key causes:
 - Low humidity
 - Clear skies
 - Sandy surface — low heat retention
 - Strong continentality
 - Western Coastal Plains:

- ✓ Location: Along the Arabian Sea coast
- ✓ Climate type: Tropical maritime
- ✓ Diurnal temperature range: Low but not the lowest
- ✓ Reason:
 - Moderate maritime influence
 - Connected to the mainland landmass
- ✓ Key winds:
 - Land breeze
 - Sea breeze
- Ladakh Region:
 - ✓ Location: Union Territory of Ladakh
 - ✓ Climate type: Cold desert
 - ✓ Average altitude: Above 3000 m
 - ✓ Diurnal temperature range: Very high
 - ✓ Key causes:
 - Thin atmosphere
 - Low humidity
 - Clear skies
 - Strong radiational cooling at night
- Additional Facts:
 - ✓ Factors reducing diurnal range:
 - High humidity
 - Cloud cover
 - Proximity to oceans
 - ✓ Factors increasing diurnal range:
 - Dry air
 - Clear skies
 - Desert surface
 - High altitude
 - ✓ Maritime vs Continental Climate:
 - Maritime climate characteristics:
 - ◇ Low temperature variation
 - ◇ High humidity
 - ◇ Moderate temperatures
 - ◇ Examples in India:
 - ▶ Andaman & Nicobar Islands
 - ▶ Lakshadweep Islands
 - ▶ Western coastal plains
 - Continental climate characteristics:
 - ◇ High temperature variation
 - ◇ Low humidity
 - ◇ Extreme seasonal temperatures
 - ◇ Examples in India:
 - ▶ Thar Desert
 - ▶ Interior Peninsular Plateau
 - ▶ Ladakh region
 - ✓ Lowest diurnal range region:
 - Andaman & Nicobar Islands
 - ✓ Highest diurnal range regions:

- Thar Desert
- Ladakh Plateau
- ✓ Moderate diurnal range regions:
 - Western Coastal Plains
 - Eastern Coastal Plains

Q 7. Ans. A

Explanation:

- Assertion (A) is correct:
 - ✓ Eastern coastal plain has fewer ports and harbours
 - ✓ Coast type: Emergent coast
 - ✓ Shelf characteristic: Broad continental shelf
 - ✓ Water depth near coast: Shallow waters
 - ✓ Port implication: Fewer natural harbours
 - ✓ Major east coast ports type: Mostly artificial
 - ✓ Examples:
 - Chennai Port — Artificial harbour
 - Visakhapatnam Port — Artificial harbour
 - Paradip Port — Artificial harbour
 - ✓ Maintenance needs: Frequent dredging due to heavy siltation
- Reason (R) is correct:
 - ✓ Eastern Coastal Plain is broader and is an emergent coast
 - ✓ Average width: About 80–100 km
 - ✓ Coast type: Emergent coast
 - ✓ Slope characteristic: Gentle slope into the Bay of Bengal
 - ✓ Formation process: Land uplift or sea-level fall
 - ✓ Shelf characteristic: Wide continental shelf
- Reason (R) correctly explains Assertion (A):
 - ✓ Emergent coast feature: Shallow coastal waters
 - ✓ Result: Deep-draft ships cannot approach shore naturally
 - ✓ Additional factor: Heavy river sedimentation
 - ✓ Major rivers depositing sediments:
 - Mahanadi
 - Godavari
 - Krishna
 - Kaveri
 - ✓ Effect of sedimentation:
 - Formation of large deltas
 - Filling of natural coastal indentations
 - Reduction of natural harbour formation
- Additional Facts:

- ✓ Eastern Coastal Plain Facts
- ✓ Coast type: Emergent coast
- ✓ Width: Wider than the western coastal plain
- ✓ Major deltas present:
 - Mahanadi Delta
 - Godavari Delta
 - Krishna Delta
 - Kaveri Delta
- ✓ River mouth type: Delta formation dominant
- Western Coastal Plain Facts:
 - ✓ Coast type: Submergent coast
 - ✓ Shelf characteristic: Narrow continental shelf
 - ✓ Slope: Steep coastal slope
 - ✓ Water depth: Deep coastal waters
 - ✓ River mouth type: Estuaries
 - ✓ Major estuarine rivers:
 - Narmada
 - Tapi
 - ✓ Natural harbour examples:
 - Mumbai Port
 - Mormugao Port
 - Kochi Port
- Confluence location: Harike (Punjab)
- ✓ Final drainage sequence:
 - Satluj joins Chenab
 - Forms the Panjnad River
 - Panjnad joins the Indus River (Pakistan)
- ✓ Major multipurpose project: Bhakra–Nangal Project
- ✓ Major reservoir: Gobind Sagar Reservoir
- ✓ Major barrage: Harike Barrage
- ✓ Canal origin point: Indira Gandhi Canal
- ✓ Canal significance: Irrigation to western Rajasthan
- Singi Khamban — Tibetan name of the Indus River
 - ✓ Meaning: Lion's Mouth
 - ✓ Indus origin: Near Lake Mansarovar
 - ✓ Region: Tibetan Plateau
 - ✓ Tsangpo / Yarlung Zangbo — Tibetan name of Brahmaputra
 - ✓ Brahmaputra origin: Chemayungdung Glacier
 - ✓ Mountain range: Kailash Range
 - ✓ Major bend: Namcha Barwa

Q 8. Ans. B

Explanation:

- Origin and antecedent nature of the Satluj:
 - ✓ River origin: Raksas Tal (Rakshastal) Lake
 - ✓ Nearby landmark: Lake Mansarovar
 - ✓ Location: Tibetan Plateau
 - ✓ Origin elevation: About 4555 m
 - ✓ Local Tibetan name: Langchen Khambab
 - ✓ Meaning: Elephant's Mouth
 - ✓ River type: Antecedent river
 - ✓ Antecedent feature:
 - River existed before the Himalayan uplift
 - Maintained course by vertical erosion
 - Forms deep gorges in the Himalayas
 - ◇ Example gorge location: Near Rupnagar (Rupar)
 - ✓ Initial flow: North-west in Tibet
 - ✓ Parallel flow: Along the Indus River (about 400 km)
 - ✓ Entry into India: Shipki La Pass
 - ✓ Entry district: Kinnaur district
 - ✓ State: Himachal Pradesh
 - ✓ Major right-bank tributary: Spiti River
 - ✓ Major confluence in plains:
 - Beas River joins the Satluj

Q 9. Ans. D

Explanation:

- Himalayan Mountains:
 - ✓ Role: Major climatic divide of India
 - ✓ Winter effect:
 - Blocks cold winds from Central Asia
 - Prevents severe freezing in the Northern Plains
 - ✓ Summer effect:
 - Intercepts Southwest Monsoon winds
 - Forces the uplift of moisture-laden winds
 - ✓ Monsoon deflection:
 - Bay of Bengal branch deflected westward along the Ganga Plains
 - ✓ Additional role:
 - Prevents monsoon's escape into Central Asia
- Distribution of Land and Water:
 - ✓ Role: Differential heating of land and sea
 - ✓ Heating property:
 - Land heats faster than water
 - Land cools faster than water
 - ✓ Summer condition:
 - Low-pressure system forms over the Northern Plains

- Pressure system: Monsoon Trough
 - Moisture source: Indian Ocean
 - Wind movement: High pressure (ocean) → Low pressure (land)
 - Result: Southwest Monsoon winds enter India
 - Distance from Sea:
 - ✓ Climate type inland: Continental climate
 - ✓ Temperature characteristic: High temperature range
 - ✓ Coastal areas: Equable climate
 - ✓ Temperature variation: Low
 - Relief features (Topography & Altitude):
 - ✓ Rainfall control factor: Orographic uplift
 - ✓ Windward side: Heavy rainfall
 - ✓ Leeward side: Low rainfall (Rain Shadow region)
 - ✓ Altitude effect: Temperature decreases with height
 - ✓ Normal Lapse Rate:
 - 1°C decrease per 165 m height
 - ✓ Hill station temperature: Cooler than the plains
 - Additional Facts:
 - ✓ Monsoon Branch Facts:
 - Two main branches of the Southwest Monsoon:
 - ◇ Arabian Sea Branch
 - ◇ Bay of Bengal Branch
 - Bay of Bengal branch path:
 - ◇ Moves along the Gangetic Plains
 - ◇ Supplies rainfall to Bihar
 - Arabian Sea branch path:
 - Strikes the Western Ghats first
 - ✓ Bihar Climate Facts:
 - Climate type: Tropical Monsoon Climate
 - Primary rainfall season: June–September
 - Main rainfall source: Bay of Bengal Monsoon Branch
 - Rainfall distribution:
 - ◇ North Bihar — Higher rainfall
 - ◇ South Bihar — Lower rainfall
 - Flood-prone rivers in Bihar:
 - ◇ Kosi
 - ◇ Gandak
 - ◇ Bagmati
 - ✓ Orographic Rainfall Facts:
 - Major rain-shadow region: Deccan Plateau (leeward of Western Ghats)
 - High rainfall region example:
 - ◇ Western Ghats windward slopes
 - ✓ Low rainfall region example:
 - Interior Maharashtra
 - Karnataka Plateau
 - ✓ Temperature Gradient Facts:
 - Temperature decreases with:
 - ◇ Increase in altitude
 - Temperature increases with:
 - ◇ Distance from the sea
- Q 10. Ans. B**
- Explanation:**
- Palghat Gap (Palakkad Gap):
 - ✓ Location: Western Ghats
 - ✓ Width: About 30 km
 - ✓ Position:
 - Between Nilgiri Hills (north)
 - Anaimalai Hills (south)
 - ✓ States connected:
 - Kerala — Tamil Nadu
 - ✓ Major cities connected:
 - Palakkad (Kerala)
 - Coimbatore (Tamil Nadu)
 - ✓ Physiographic significance:
 - Major break in Western Ghats
 - ✓ Climatic role:
 - Acts as a wind corridor
 - Allows Southwest Monsoon winds to cross the Western Ghats
 - Other Western Ghats Passes:
 - ✓ Thal Ghat (Kasara Ghat):
 - Location: Maharashtra
 - Connectivity:
 - ◇ Mumbai — Nashik
 - Transport route: NH-160
 - Physiographic role:
 - ◇ Connects the Konkan coast to the Deccan Plateau
 - ✓ Bhor Ghat:
 - Location: Maharashtra
 - Connectivity:
 - ◇ Mumbai — Pune
 - Transport route: NH-48
 - Importance:
 - ◇ Major historical trade and transport route
 - Physiographic role:
 - ◇ Links the Konkan coast with the Deccan Plateau

- ✓ Amboli Ghat:
 - Location: Maharashtra (near the Goa border)
 - Connectivity:
 - ◊ Sawantwadi — Belagavi (Karnataka)
 - Climate feature:
 - ◊ Receives very heavy monsoon rainfall
 - Hill station: Amboli
 - Rainfall feature: Among the wettest locations in Maharashtra
- Other Important Gaps & Passes:
 - ✓ Shencottah Gap (Sengottai Pass):
 - Location: Western Ghats (Cardamom Hills)
 - Position: South of Palghat Gap
 - Connectivity:
 - Madurai (Tamil Nadu) — Kollam (Kerala)
 - Rank: Second major gap in Western Ghats
 - ✓ Asirgarh Pass — Satpura Range:
 - Location: Madhya Pradesh
 - Mountain range: Satpura Range
 - Connectivity:
 - Narmada Valley — Tapi Valley
 - Strategic importance:
 - Historic route between North India and the Deccan Plateau
 - Nickname: Key to the Deccan
 - ✓ Haldighati Pass:
 - Location: Aravalli Range
 - State: Rajasthan
 - Connectivity:
 - Rajsamand — Pali districts
 - Historical event: Battle of Haldighati (1576)
 - ✓ Goram Ghat:
 - Location: Aravalli Range
 - State: Rajasthan
 - Connectivity:
 - Mewar — Marwar regions
 - ✓ Kumbharli Ghat:
 - Location: Western Ghats
 - State: Maharashtra
 - Connectivity:
 - Ratnagiri (Konkan coast) — Satara (Deccan Plateau)
 - ✓ Chorla Ghat:
 - Location: Western Ghats (Sahyadri Range)
 - Border location:
 - Goa — Karnataka — Maharashtra junction
 - Connectivity:
 - Links inland Karnataka with the Goa region
- Additional Facts:
 - ✓ Western Ghats — Structural Facts:
 - Alternate name: Sahyadri Range
 - Extent: Gujarat to Tamil Nadu
 - ✓ Northern limit: Tapi Valley
 - ✓ Southern limit: Kanyakumari
 - ✓ Average elevation: About 900–1600 m
 - ✓ Highest peak: Anamudi (2695 m) — Kerala
 - ✓ Rainfall & Gap Significance
 - Western Ghats role: Orographic barrier to the Southwest Monsoon
 - ✓ Windward side rainfall: Heavy rainfall (Kerala coast)
 - ✓ Leeward side: Rain-shadow region (Tamil Nadu plateau)
 - ✓ Gap role:
 - Allows moisture penetration inland
 - Major climatic gap: Palghat Gap

Q 11. Ans. C

Explanation:

- Lakshadweep Archipelago:
 - ✓ Origin type: Coral origin
 - ✓ Island type: Coral atolls and reefs
 - ✓ Location: Arabian Sea
 - ✓ Administrative status: Union Territory
 - ✓ Administrative Headquarters: Kavaratti
 - ✓ Area: About 32 sq km
 - ✓ Smallest Union Territory: Lakshadweep
 - ✓ Judicial jurisdiction:
 - Falls under the Kerala High Court
 - High Court location: Ernakulam
 - ✓ Lakshadweep Water Channels:
 - 8° Channel:
 - ◊ Separates Minicoy Island and the Maldives
 - 9° Channel:
 - ◊ Separates Minicoy Island from the main Lakshadweep group
 - 11° Channel:
 - ◊ Separates the Amindivi Islands (north)
 - ◊ From the Laccadive Islands (south)

- ✓ Important Lakshadweep Islands
 - Minicoy Island
 - ◇ Position: Southernmost island of Lakshadweep
 - ◇ Nearest country: Maldives
 - ◇ Language spoken: Mahl (Maldivian dialect)
 - Agatti Island
 - ◇ Key feature: Only airport in Lakshadweep
 - ◇ Transport importance: Main air entry point to Lakshadweep
 - Pitti Island
 - ◇ Habitation status: Uninhabited island
 - ◇ Protected area: Bird Sanctuary
 - ◇ Importance: Important nesting site for seabirds
- Andaman & Nicobar Islands:
 - ✓ Origin type: Tectonic origin
 - ✓ Geological nature: Submarine mountain extension
 - ✓ Mountain link: Extension of Arakan Yoma Range (Myanmar)
 - ✓ Location: Bay of Bengal
 - ✓ Important Channels
 - ✓ 10° Channel:
 - Separates Andaman Islands (north)
 - From the Nicobar Islands (south)
 - ✓ Major Physical Features
 - ✓ Highest peak: Saddle Peak
 - ✓ Location: North Andaman
 - ✓ Height: About 732 m
 - ✓ Active volcano: Barren Island
 - ✓ Status: Only active volcano in India
 - ✓ Dormant volcano: Narcondam Island
 - ✓ Extreme Points:
 - Southernmost point of India: Indira Point
 - Location: Great Nicobar Island
 - Former name: Pygmalion Point
 - Affected event: Submerged during 2004 Tsunami
 - ✓ Tribal Distribution:
 - Andaman tribes:
 - ◇ Jarawa
 - ◇ Onge
 - ◇ Sentinelese
 - ◇ Great Andamanese
 - Nicobar tribe:
 - ◇ Shompen
- Other Important Indian Islands:
 - ✓ Sriharikota Island
 - Location: Andhra Pradesh coast
 - Island type: Barrier island
 - Separates: Pulicat Lake and Bay of Bengal
 - Major facility: Satish Dhawan Space Centre (ISRO launch site)
 - ✓ Abdul Kalam Island (Wheeler Island)
 - Location: Odisha coast
 - Function: Missile testing site
 - Facility: Integrated Test Range
 - Missiles tested:
 - ◇ Agni
 - ◇ Prithvi
 - ✓ Pamban Island
 - Location: Gulf of Mannar
 - State: Tamil Nadu
 - Nearby country: Sri Lanka
 - Major town: Rameswaram
 - Famous temple: Ramanathaswamy Temple
 - ✓ Majuli Island
 - Location: Assam
 - River: Brahmaputra River
 - Island type: Riverine island
 - Global status: Largest inhabited riverine island in the world
 - Major issue: Severe riverbank erosion
 - ✓ New Moore Island
 - Location: Bay of Bengal
 - Region: India–Bangladesh maritime boundary
 - Island type: Sandbar island
 - Formation event: Bhola Cyclone (1970)
 - Current status: Submerged due to sea-level rise

Q 12. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ Geological age of the Peninsular plateau: Precambrian / Archaean era
 - ✓ Geological nature: Oldest and most stable landmass of India
 - ✓ Tectonic activity: Rarely affected by folding since formation
 - ✓ Rock stability: Forms a rigid crystalline shield

- Statement 2 is correct:
 - ✓ Dominant rock type: Crystalline rocks
 - ✓ Major rock groups:
 - Archaean gneisses
 - Schists
 - ✓ Rock nature: Highly metamorphosed rocks
 - ✓ Rock category: Igneous and metamorphic rocks are dominant
 - ✓ Geological feature: Very old crystalline basement rocks
- Statement 3 is correct:
 - ✓ Ancient landmass: Part of Gondwanaland
 - ✓ Plate movement: Drifted northward
 - ✓ Collision event: The Indian Plate collided with the Eurasian Plate
 - ✓ Result of collision: Formation of the Himalayas
 - ✓ Plate type: Continental plate movement
- Additional Facts:
 - ✓ Peninsular Plateau:
 - Shape: Irregular triangular plateau
 - Geological stability: Most stable landmass of India
 - General slope direction: West → East
 - Major boundaries:
 - Northwest — Delhi Ridge; East — Rajmahal Hills; West — Gir Range; South — Cardamom Hills
 - Northeastern Extension Facts
 - Separated plateaus: Meghalaya Plateau, Karbi-Anglong Plateau
 - Separation structure: Malda Fault
 - Alternate name: Garo–Rajmahal Gap
 - ✓ Major Rock Systems of the Peninsular Plateau
 - Dharwar System
 - ◇ Rock type: Metamorphosed sedimentary rocks
 - ◇ Major minerals: Iron ore, Gold, Manganese
 - ◇ Economic importance: Major metallic mineral belt
 - Vindhyan System
 - ◇ Rock type: Sedimentary rocks
 - ◇ Major rocks: Limestone, Sandstone
 - ◇ Economic use: Cement industry
 - ◇ Bihar districts: Rohtas, Kaimur
 - ◇ Key resource: Limestone
 - Gondwana System
 - ◇ Geological period: Carboniferous
 - ◇ Major resource: Coal
 - ◇ Coal reserve share: About 98% of India's coal
 - ◇ Major coal valleys: Damodar Valley, Mahanadi Valley, Godavari Valley
- Deccan Trap
 - ◇ Formation type: Fissure volcanic eruption
 - ▶ Geological period: Cretaceous
 - ▶ Rock type: Basalt
 - ▶ Soil formed: Black Soil (Regur)
 - ▶ Major crop suitability: Cotton
- ✓ Major Subdivisions of Peninsular Plateau
 - Central Highlands:
 - ◇ Location: North of the Narmada River
 - ◇ Slope direction: Southwest → Northeast
 - ◇ Major rivers: Chambal, Sind, Betwa, Ken
 - ◇ Drainage direction: Towards the Yamuna River
 - Deccan Plateau:
 - ◇ Location: South of the Narmada River
 - ◇ Boundaries: Eastern Ghats, Western Ghats
 - ◇ Slope direction: West → East
 - ◇ Major river flow: East-flowing rivers are dominant
 - Chotanagpur Plateau:
 - ◇ Location: Primarily Jharkhand
 - ◇ Bordering Bihar districts: Gaya, Nawada, Jamui, Banka
 - ◇ Nickname: Ruhr of India
 - ◇ Major river: Damodar River
 - ◇ Valley type: Rift valley
 - ◇ Major resources: Coal, Iron ore, Mica.

Q 13. Ans. D

Explanation:

- Statement A is correct:
 - ✓ Western Ghats as a biodiversity hotspot
 - ✓ Global status: One of 36 globally recognised biodiversity hotspots of the world
 - ✓ Endemism level: Very high plant endemism
 - ✓ Share of Indian flora: About one-third (27–33%) of India's flowering plant species

- ✓ Forest type dominance: Tropical evergreen forests on western slopes
- Statement B is correct:
 - ✓ Extent and state coverage of the Western Ghats
 - ✓ Total length: About 1600 km
 - ✓ Northern limit: Near the Tapi River valley (Gujarat)
 - ✓ Southern limit: Kanyakumari (Tamil Nadu)
 - ✓ States covered: Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu
 - ✓ Alternate name: Sahyadri Range
- Statement C is correct:
 - ✓ Monsoon and rainfall effects of the Western Ghats
 - ✓ Role: Orographic barrier to the Southwest Monsoon
 - ✓ Windward side: Western slopes — heavy rainfall
 - ✓ Leeward side: Eastern slopes — low rainfall
 - ✓ Rain-shadow region: Deccan Plateau
 - ✓ Vegetation pattern:
 - ✓ Windward — Evergreen forests
 - ✓ Leeward — Dry deciduous, scrub vegetation
- Statement D is incorrect:
 - ✓ Agasthyamalai BR — Western Ghats (Kerala, Tamil Nadu)
 - ✓ Simlipal BR — Odisha (Eastern India)
 - ✓ Simlipal location: Mayurbhanj district, Odisha
 - ✓ Conclusion: Simlipal is not located in the Western Ghats
- Additional Facts:
 - ✓ Western Ghats — Structural Features
 - Mountain type: Faulted edge of the Deccan Plateau
 - Not a fold mountain system
 - Local names:
 - ◇ Sahyadri — Maharashtra, Karnataka
 - ◇ Nilgiri Hills — Tamil Nadu
 - ◇ Anaimalai Hills — Kerala
 - ◇ Cardamom Hills — Kerala
 - Meeting point of Ghats: Nilgiri Hills (Western Ghats + Eastern Ghats junction)
 - Major Peaks of Western Ghats
 - ◇ Highest peak overall: Anamudi — 2695 m — Anaimalai Hills (Kerala)
 - ◇ Highest peak (Maharashtra): Kalsubai
 - ◇ Highest peak (Nilgiris): Doda Betta
 - ◇ Highest peak of Peninsular India: Anamudi
 - ✓ River Origin Facts:
 - Water divide role: Western Ghats act as a major peninsular watershed
 - Major east-flowing rivers originating:
 - ◇ Godavari — Trimbakeshwar
 - ◇ Krishna — Mahabaleshwar
 - ◇ Kaveri — Talakaveri
 - ✓ Biosphere Reserve Facts:
 - Total Biosphere Reserves in India: 18
 - UNESCO MAB recognised: 12
 - First Biosphere Reserve: Nilgiri BR (1986)
 - States covered (Nilgiri BR): Tamil Nadu, Kerala, Karnataka
 - Largest Biosphere Reserve: Gulf of Kachchh — Gujarat
 - Smallest Biosphere Reserve: Dibru-Saikhowa — Assam
 - Biosphere reserves in Western Ghats:
 - ◇ Nilgiri Biosphere Reserve
 - ◇ Agasthyamala Biosphere Reserve
 - ✓ Simlipal Biosphere Reserve:
 - Location — Mayurbhanj district, Odisha
 - Forest type — Dense Sal forests
 - Major fauna — Royal Bengal Tiger, Elephants
 - ✓ Biodiversity Hotspots in India:
 - Western Ghats,
 - Eastern Himalayas,
 - Indo-Burma, and
 - Sundaland (Andaman & Nicobar).

Q 14. Ans. D

Explanation:

- Statement 1 is incorrect:
 - ✓ Khadar:
 - Khadar is younger alluvium found in low-lying floodplains along river channels. It consists mainly of fine silt and clay and is renewed every year by flood deposits. Kankar nodules are absent, and the soil is highly fertile.
- Statement 2 is incorrect:
 - ✓ Bhangar:
 - Bhangar is located on elevated terraces above the floodplains. It contains old alluvium with kankar nodules and is less fertile than Khadar.

● Additional Facts:

- ✓ Correct sequence from the Himalayan foothills towards the plains:
 - Bhabar → Terai → Bhangar → Khadar
- ✓ Bihar-specific types of Bhangar:
 - Balsundari Soil:
 - ◇ Found in North Bihar — Saran, Champaran, Muzaffarpur, and Saharsa. It is a loamy old alluvial soil containing lime nodules.
 - Karail–Kewal Soil:
 - ◇ Found in South Bihar — Rohtas, Gaya, Patna, and Munger. It is a heavy, dark, clayey soil with high water-retention capacity.
- ✓ Bihar-specific features of Khadar:
 - Khadar soils dominate the plains of North Bihar due to active river systems such as the Kosi, Gandak, and Bagmati.
 - Local term — Diara Land:
 - ◇ Diara lands are fertile floodplain tracts and riverine islands that are flooded annually.
 - ◇ These areas are particularly suitable for crops such as maize and watermelon.
- ✓ Bhabar Zone — Piedmont Belt
 - The Bhabhar is a narrow belt located just south of the Shiwalik foothills.
 - It is about 8–10 km wide and consists mainly of coarse gravel and pebbles.
 - Due to its high porosity, rivers disappear underground in this zone, making it unsuitable for agriculture.
- ✓ Terai Zone — Marshy Belt
 - The Terai lies immediately south of the Bhabar zone.
 - In this region, underground streams re-emerge, creating marshy and waterlogged conditions.
 - The area supports dense forests and fertile soils.
- ✓ Bihar Terai belt:
 - Occurs as a narrow strip (3–8 km wide) along the northern districts of West Champaran, Sitamarhi, Araria, and Kishanganj.
 - These soils are suitable for crops such as sugarcane and jute.

Q 15. Ans. A

Explanation:

● Mt. Everest	–	Nepal	–	8848 m
● Kanchenjunga	–	India	–	8586 m
● Makalu	–	Nepal	–	8481 m
● Dhaulagiri	–	Nepal	–	8172 m
● Nanga Parbat	–	India	–	8126 m
● Annapurna	–	Nepal	–	8078 m
● Nanda Devi	–	India	–	7817 m
● Kamet	–	India	–	7756 m
● Namcha Barwa	–	India	–	7782 m
● Gurla Mandhata	–	Nepal	–	7728 m

Q 16. Ans. A

Explanation:

- UNCLOS Maritime Zones:
 - ✓ Territorial Sea: 12 nautical miles from baseline
 - Baseline: Low-water mark
 - Sovereignty: Full control over sea, seabed, airspace
 - Limitation: Right of Innocent Passage for foreign vessels
- Additional Information:
 - ✓ Contiguous Zone: Up to 24 nautical miles (nm)
 - Extent: 12 nm beyond Territorial Sea
 - Rights: Enforcement of
 - ◇ Customs laws
 - ◇ Fiscal laws
 - ◇ Immigration laws
 - ◇ Sanitation laws
 - Recent: Increased patrolling in the Palk Strait, Arabian Sea
 - ✓ Exclusive Economic Zone (EEZ): Up to 200 nautical miles
 - Rights: Exploration and exploitation of resources
 - Resources: Living and non-living
 - Coverage: Water column and seabed
 - India EEZ Area: 2.37 million sq km
 - Recent: Blue Economy 2.0 focus
 - Deep Sea Mining: Under Deep Ocean Mission
 - ✓ Continental Shelf: Up to 350 nautical miles
 - Provision: Article 76 of UNCLOS
 - Condition: Proof of natural extension of landmass

- Rights: Seabed and subsoil resources
- Recent: India claims to CLCS for Bay of Bengal extension
- Legal Framework India
 - ✓ Act: Territorial Waters, Continental Shelf, EEZ and Other Maritime Zones Act, 1976
 - ✓ Alignment: UNCLOS III (1982)
- Geopolitical Context
 - ✓ Policy: SAGAR (Security and Growth for All in the Region)
 - ✓ Focus: Maritime security, regional cooperation
 - ✓ Concern: Foreign research vessels in the Indian Ocean Region

- Zoji La (Ladakh): Situated at an altitude of 3,528m. It connects Srinagar to Leh (NH-1). It is often called the “Mountain of Blizzards.”

- ✓ Himachal Pradesh
 - Bara-Lacha La: Located in the Zaskar range. it connects Lahaul (HP) to Leh (Ladakh). It is a key point on the Manali-Leh highway.
 - Debsa Pass: Connects the Kullu and Spiti valleys.
- ✓ Uttarakhand
 - Lipu Lekh: Located at the tri-junction of India, Nepal, and China. This is the primary route for pilgrims going for the Kailash Mansarovar Yatra.
 - Mana Pass & Niti Pass: These connect Uttarakhand with Tibet. Mana Pass is often cited as one of the highest vehicle-accessible passes in the world.
- ✓ Sikkim & Arunachal Pradesh
 - Jelep La (Sikkim): Much like Nathu La, it connects Sikkim with Lhasa. It passes through the Chumbi Valley.
 - Diphu Pass (Arunachal): Located at the tri-junction of India, China, and Myanmar. It is a strategic pass on the McMahon Line.
 - Pangsau Pass (Arunachal): Connects Arunachal Pradesh with Myanmar (on the Patkai Bum hills).
 - Nathu La (Sikkim): Situated on the Indo-China border at 4,310m. It was a part of the ancient Silk Road. It was closed after the 1962 war and reopened in 2006 for border trade.

Q 17. Ans. D

Explanation:

- **Muling La:** Connects Uttarakhand and Tibet north of Gangotri.
- **Shipki La (Himachal Pradesh):** Located in the Kinnaur district, Himachal Pradesh, on the main Himalayan range boundary with Tibet.
- **Mig La Pass,** located in eastern Ladakh near the Line of Actual Control, lies at an altitude of about 19,400 feet above sea level.
 - ✓ The Border Roads Organisation, through Project Himank, achieved a new world record by constructing the world’s highest motorable road at Mig La Pass, surpassing its earlier record at Umling La set in 2021.
- **Bomdi La (Arunachal Pradesh):** Located in the West Kameng district. It connects India with Lhasa (Tibet). It lies to the east of the Bhutan border.
- **Additional Facts:**
 - ✓ Ladakh & Jammu-Kashmir (UTs)
 - Karakoram Pass: Located in the Karakoram Range. It is one of the highest passes in India (over 5,500m) and was a major part of the ancient Yarkand trade route.
 - Banihal Pass: Located in the Pir Panjal range. It connects Jammu to Srinagar. The famous Jawahar Tunnel is constructed under this pass.
 - Burzil Pass: Connects Srinagar with the Deosai Plains of Astore and Gilgit.
 - Aghil Pass: Situated north of K2 in the Karakoram, connecting Ladakh with the Xinjiang province of China.

Q 18. Ans. D

Explanation:

- India’s total coastline length (revised GIS measurement)
 - ✓ Total coastline of India: 11,098.81 km
 - ✓ Mainland coastline: 7,870.51 km
 - ✓ Island coastline (Andaman–Nicobar + Lakshadweep): 3,228.30 km
 - ✓ Measurement method: GIS-based digital mapping
 - ✓ Reason for increase: Inclusion of detailed indentations, estuaries, deltas, and creeks.

- State-wise coastline ranking (Longest → Shortest)
 - ✓ 1st — Gujarat: 2,340.62 km
 - Reason: Highly indented coastline — Gulf of Kachchh, Gulf of Khambhat
 - ✓ 2nd — Tamil Nadu: 1,068.69 km
 - Tamil Nadu overtook Andhra Pradesh
 - ✓ 3rd — Andhra Pradesh: 1,053.07 km
 - ✓ 4th — Maharashtra: 883.33 km
 - ✓ 5th — West Bengal: 721.28 km
 - Reason: Detailed mapping of the Sundarbans delta
 - ✓ 6th — Kerala: 559.08 km
 - ✓ 7th — Odisha: 540.35 km
 - ✓ 8th — Karnataka: 316.03 km
 - ✓ 9th (Shortest) — Goa: 160.50 km
- Union Territory coastline rankings
 - ✓ Longest UT (and longest overall entity):
 - Andaman & Nicobar Islands — 1,962 km
 - ✓ Shortest UT:
 - Puducherry — 30.60 km
 - Puducherry coastline reduced in revised GIS mapping
- Geopolitical and strategic significance:
 - ✓ Ocean which is uniquely named after a country: Indian Ocean
 - ✓ Strategic position: India is located at the northern head of the Indian Ocean
 - ✓ Major international sea routes near India:
 - Suez Canal route (westward connection to the Mediterranean Sea)
 - Strait of Malacca route (eastward connection to the Pacific region)
 - Strategic importance: Control and monitoring of major global trade routes.
- Additional Facts:
 - ✓ Pacific Ocean: Largest and deepest ocean; location — between Asia/Australia and the Americas
 - ✓ Atlantic Ocean: Second largest ocean; location — between the Americas and Europe/Africa
 - ✓ Arctic Ocean: Smallest ocean; location — around the North Pole
 - ✓ Indian Ocean: Third largest ocean; northern boundary — the Indian subcontinent
 - ✓ India's Island Neighbours in the Indian Ocean:
 - Sri Lanka: Southeast of India; separated by the Palk Strait and the Gulf of Mannar
 - Maldives: Southwest of India; separated from Lakshadweep by 8° Channel
 - Mauritius: Located southwest of India in the Indian Ocean
 - Seychelles: Located northwest of Mauritius in the Indian Ocean

Q 19. Ans. D

Explanation:

- Geographical definition and surrounding water bodies
 - ✓ Peninsula definition: A landmass surrounded by water on three sides and connected to land on the fourth side.
 - ✓ Peninsular India surrounding water bodies:
 - West — Arabian Sea
 - East — Bay of Bengal
 - South — Indian Ocean
 - ✓ Ocean system: All three water bodies form part of the Indian Ocean system.
- Southernmost mainland point and meeting of seas:
 - ✓ Southernmost tip of the Indian mainland: Kanyakumari (Cape Comorin)
 - ✓ State: Tamil Nadu
 - ✓ Geographical significance: Meeting point of three water bodies —
 - Arabian Sea
 - Bay of Bengal
 - Indian Ocean
 - ✓ Note: This is the southernmost point of mainland India, not of the entire India.

Q 20. Ans. B

Explanation:

- Sikkim: International Boundaries
 - ✓ West: Nepal
 - ✓ East: Bhutan
 - ✓ North: China (Tibet region)
 - ✓ Unique Fact: Only the Indian state shares borders with Nepal, Bhutan and China
 - ✓ Geographical Features
 - Type: Landlocked state
 - Location: Eastern Himalayas
 - Key Passes:
 - ◇ Nathu La: India–Tibet trade route
 - ◇ Jelep La: Strategic trade and defence route

- ✓ Strategic Importance
 - Region: Chumbi Valley (tri-junction area)
 - Sensitivity: One of the world's most strategic zones
- Additional Information
 - ✓ Arunachal Pradesh:
 - Borders: Bhutan, China, Myanmar
 - ✓ Recent:
 - Vibrant Villages Programme
 - Sela Tunnel: World's longest twin-lane tunnel above 13,000 ft
 - Connectivity: Tawang year-round
 - ✓ West Bengal:
 - Borders: Nepal, Bhutan, Bangladesh
 - Key Feature: Siliguri Corridor
 - Importance: Connects Northeast India
 - Recent: Increased BSF-Army coordination
 - ✓ Assam:
 - Borders: Bhutan, Bangladesh
 - **Recent:**
 - ◇ Border Settlement Agreement with Arunachal Pradesh
 - ◇ Significance: Model for interstate dispute resolution
- Historical Background
 - ✓ Status before 1975: Protectorate of India
 - ✓ Statehood: 22nd state
 - ✓ Amendment: 36th Constitutional Amendment Act, 1975
 - ✓ **Geopolitical Significance**
 - Doklam Plateau: Tri-junction of India, Bhutan, China
 - Importance: Overlooks Siliguri Corridor
 - ✓ Economic Features
 - Status: India's first 100% Organic State
 - Focus: Organic agriculture exports
 - Connectivity: National Highway 10 (NH10)
 - Aim: Integration with Global Value Chains (GVCs)
- ✓ They are major blue carbon ecosystems, storing large amounts of carbon in both biomass and deep waterlogged soils.
- ✓ Carbon storage zones: Above-ground biomass, below-ground roots, and anaerobic muddy soils.
- ✓ Environmental significance: Important for climate change mitigation due to long-term carbon sequestration.
- Statement 2 is correct:
 - ✓ Mangroves are salt-tolerant plants (halophytes) found in coastal saline and tidal environments.
 - ✓ Typical habitat: Estuaries, deltas, tidal creeks, and coastal mudflats.
 - ✓ They grow in brackish water zones, where freshwater mixes with seawater.
- Additional Facts:
 - ✓ Mangrove Adaptations:
 - Halophytes: Salt-tolerant plants adapted to saline coastal environments.
 - Pneumatophores: Vertical breathing roots found in waterlogged, oxygen-deficient soils.
 - Vivipary: Seeds germinate on the parent tree before falling into water or mud.
 - ✓ Mangrove Cover in India — ISFR 2023:
 - Total mangrove cover: 4,992 sq km
 - Major decrease:
 - ◇ Gujarat — Decrease of 36.39 sq km
 - Major increases:
 - ◇ Andhra Pradesh — Increase of 13.01 sq km
 - ◇ Maharashtra — Increase of 12.39 sq km
 - ✓ India Forest & Tree Cover — ISFR 2023:
 - Total Forest + Tree Cover: 8,27,356.95 sq km (25.17%)
 - Forest Cover Only: 7,15,342.61 sq km (21.76%)
 - Tree Cover Only: 1,12,014.34 sq km (3.41%)
 - Net Increase since 2021: +1,445.81 sq km
 - ✓ State Rankings — ISFR 2023:
 - Maximum increase in total green cover:
 - ◇ 1st — Chhattisgarh (+684 sq km)
 - ◇ 2nd — Uttar Pradesh/Odisha (+559 sq km)
 - Maximum increase in forest cover only:

Q 21. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ Mangrove forests hold the highest average carbon stocks per unit area among coastal/marine ecosystems.

- ◇ Mizoram (+242 sq km)
- Largest decrease in forest cover:
 - ◇ Madhya Pradesh (-612.41 sq km)
 - ◇ Second largest decrease: Karnataka
- Largest forest area (total):
 - ◇ 1st — Madhya Pradesh
 - ◇ 2nd — Arunachal Pradesh
 - ◇ 3rd — Chhattisgarh
- Highest forest cover percentage:
 - ◇ 1st — Lakshadweep (91.33%)
 - ◇ 2nd — Mizoram (85.34%)
 - ◇ 3rd — Andaman & Nicobar Islands (81.62%)

Q 22. Ans. B

Explanation:

- Tropic of Cancer in India
 - ✓ Latitude: 23°30'N
 - ✓ Does not pass through: Andhra Pradesh
 - ✓ Reason: Andhra Pradesh latitude range 12° N to 19° N
 - ✓ States through the Tropic of Cancer: 8
 - Gujarat
 - Rajasthan
 - Madhya Pradesh
 - Chhattisgarh
 - Jharkhand
 - West Bengal
 - Tripura
 - Mizoram
- Additional Information
 - ✓ Gujarat:
 - Position: Westernmost state on the Tropic
 - ✓ Recent:
 - Semicon India 2024 hosted
 - Khavda: World's largest renewable energy park
 - ✓ Jharkhand:
 - On Tropic
 - ◇ City: Ranchi, closest to Tropic in India
 - Recent:
 - ◇ PM-JANMAN mission focus
 - ◇ Target: PVTGs upliftment
 - ✓ Tripura:
 - On Tropic
 - ◇ NE states on line: Tripura, Mizoram
 - Recent:

- ◇ Maitri Setu over the Feni River
- ◇ Link: Access to Chittagong Port
- ◇ Role: Gateway to the Northeast

● Key Facts

- ✓ Mahi River:
 - Unique: Crosses the Tropic of Cancer twice
 - First: Madhya Pradesh (northward flow)
 - Second: Gujarat (southward flow)

Q 23. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ The Central Highlands are wider in the western part and become narrower towards the east.
 - ✓ Western part: Includes the Malwa Plateau, which is the broadest section.
 - ✓ Eastward trend: The plateau gradually tapers and merges into the Chotanagpur Plateau region.
 - ✓ This west-to-east narrowing is a standard physiographic characteristic of the Central Highlands.
- Statement 2 is correct:
 - ✓ The eastward extensions of the Central Highlands are locally known as Bundelkhand and Baghelkhand.
 - ✓ Bundelkhand Plateau:
 - Located between the Yamuna River and the Vindhyan Hills and lies across southern Uttar Pradesh and northern Madhya Pradesh.
 - ✓ Rock type: Ancient granitic rocks.
 - ✓ Topography: Highly eroded surface with ravines and drought-prone terrain.
 - ✓ Baghelkhand Plateau:
 - Located east of Bundelkhand, mainly in eastern Madhya Pradesh.
 - Represents the further eastward continuation of the Central Highlands.
- Additional Facts:
 - ✓ Central Highlands — Core Features:
 - Location: North of the Narmada River
 - Boundaries:
 - ◇ Northwest — Aravalli Range
 - ◇ South — Vindhyan Range
 - Slope direction: South-West → North-East

- Major rivers flow towards the Yamuna River
 - ◇ Major rivers:
 - ◇ Chambal
 - ◇ Sind
 - ◇ Betwa
 - ◇ Ken
- Subdivisions of Central Highlands (West → East Sequence)
 - ◇ Malwa Plateau — Westernmost and broadest part;
 - ▶ rock type: Deccan Trap basalt;
 - ▶ soil: Black Soil (suitable for cotton cultivation).
 - ◇ Bundelkhand Plateau — Located along the UP–MP border;
 - ▶ rock type: Granitic rocks; terrain: Ravine-cut and drought-prone.
 - ◇ Baghelkhand Plateau — Located east of Bundelkhand; forms the eastern continuation of the Central Highlands.
 - ◇ Dandakaranya Plateau — Eastward extension; Dense forests; tribal region
- Central Highlands slope direction: South-West to North-East
- Western boundary feature: Aravalli Range
- Southern boundary feature: Vindhyan Range
- Eastern termination of the Peninsular block: Rajmahal Hills
- Geological separation feature: Malda Fault
 - Length: Approx. 720 km

● Additional Information

- ✓ Kali River:
 - Boundary: Kumaon and Nepal Himalaya
 - West of Kali: Kumaon
 - East of Kali: Nepal
- ✓ Teesta River:
 - Boundary: Nepal and Assam Himalaya
 - West of Teesta: Nepal
 - East of Teesta: Assam
- ✓ Dihang River:
 - Meaning: Name of the Brahmaputra in Arunachal Pradesh
 - Use: Technical term in exams instead of Brahmaputra

● Key Facts

- ✓ Peak Distribution:
 - Nepal Himalaya:
 - Mt. Everest
 - Lhotse
 - Makalu
- ✓ Kumaon Himalaya:
 - Nanda Devi
 - Badrinath

● Geopolitical Issue:

- ✓ Kali River Dispute:
 - Countries: India, Nepal
 - Issue: Origin at Limpiyadhura
 - Impact: Boundary at Lipulekh Pass
 - Significance: India–Nepal–China tri-junction

● Economic Importance

- ✓ Teesta River:
- ✓ Issue: India–Bangladesh water-sharing dispute
- ✓ Use: Hydropower potential
- ✓ Method: Run-of-the-River projects
- ✓ Technology: ISRO satellite monitoring of glacial melt

● Environmental Issue

- ✓ Region: Kumaon Himalaya
- ✓ Place: Joshimath
- ✓ Event: Land subsidence (2023–24)
- ✓ Impact: Infrastructure risk

Q 25. Ans. B

Explanation:

- Purvachal Himalayas (Eastern Hills and Mountains)

Q 24. Ans. B

Explanation:

- Himalayan Regional Divisions (West to East)
 - ✓ Punjab Himalaya:
 - Extent: Indus to Satluj rivers
 - Length: Approx. 560 km
 - ✓ Kumaon Himalaya:
 - Extent: Satluj to Kali (Sarda) rivers
 - Length: Approx. 320 km
 - ✓ Nepal Himalaya:
 - Extent: Kali to Teesta rivers
 - Length: Approx. 800 km
 - Longest Himalayan section
 - ✓ Assam Himalaya:
 - Extent: Teesta to Dihang (Brahmaputra) rivers

- ✓ Location: Southward extension of the Himalayas beyond the Dihang Gorge
- ✓ Alignment: Along the eastern edge of India
- ✓ Boundary: India–Myanmar border
- ✓ Geology: Strong sandstones (sedimentary rocks)
- ✓ Vegetation: Dense forests
- Additional Information
 - ✓ Assam Himalayas:
 - Extent: Teesta to Dihang rivers
 - Covers: Northern Arunachal Pradesh, Bhutan
 - Not: Southward-trending hill system
 - ✓ Trans-Himalayas:
 - Location: North of the Great Himalayas
 - Ranges: Karakoram, Ladakh, Zaskar
 - Formation: Older than the main Himalayas
 - ✓ Karbi Anglong Plateau:
 - Origin: Extension of Peninsular Plateau
 - Link: Meghalaya (Shillong) Plateau
 - Separation: Malda Fault
- Key Facts
 - ✓ Highest Peak:
 - Mount Saramati
 - Location: Naga Hills, India–Myanmar border
 - ✓ Infrastructure
 - Programme: Vibrant Villages Programme
 - Region: For villages on India's northern border with China/Tibet, covering:
 - ◇ Himachal Pradesh
 - ◇ Uttarakhand
 - ◇ Sikkim
 - ◇ Arunachal Pradesh
 - ◇ Ladakh
 - Aim: Connectivity, to prevent border migration

Q 26. Ans. C

Explanation:

- Tarai: South of Bhabar
 - ✓ Marshy, swampy, waterlogged
 - ✓ Streams re-emerge (disappear in Bhabar)
 - ✓ High water table
- Ecological Significance
 - ✓ Tarai: Originally dense forests
 - ✓ Post-1947: Forests cleared

- ✓ Purpose: Settlement (Partition migrants), agriculture
- ✓ Crops: Sugarcane, Rice
- ✓ Wildlife Areas:
 - Dudhwa National Park (UP)
 - Jim Corbett National Park (Uttarakhand)

Q 27. Ans. A

Explanation:

- Eastern Coastal Plain: Between Eastern Ghats and Bay of Bengal
 - ✓ Division:
 - Northern Circars: Mahanadi → Krishna
 - Coromandel Coast: Krishna → Kaveri
 - Northern Circars: Coastal Odisha, North Andhra Pradesh
- Additional Information
 - ✓ Krishna–Kaveri: Coromandel Coast
 - Region: South Andhra Pradesh, Tamil Nadu
 - Recent: Sagarmala Project
 - Focus: Chennai, Ennore ports
 - Purpose: Boost electronic exports (SE Asia)
 - ✓ Tapi–Narmada: Western Coastal Plain
 - Region: Gujarat Plains
 - Recent: Gulf of Khambhat
 - Project: Kalpasar Project
 - Purpose: Freshwater reservoir, solve Saurashtra water scarcity
 - Indus–Narmada: Kachchh & Kathiawar
 - ✓ Region: Gujarat
 - ✓ Recent: Rann of Kachchh
 - ✓ Project: Hybrid Renewable Energy Park
 - ✓ Capacity: 30 GW
 - ✓ Significance: Green energy transition
 - Chilika Lake: Odisha
 - ✓ Type: Brackish lagoon
 - ✓ Status: Largest in Asia
 - ✓ Fact: India's first Ramsar Site
 - Pulicat Lake: Near TN–AP coast
 - ✓ Feature: Sriharikota (ISRO launch site)
 - Northern Circars: Key for Act East Policy
 - ✓ Major Ports:
 - Paradip (Odisha)
 - Visakhapatnam (Andhra Pradesh)
 - Exports: Iron ore, Coal
 - Destination: East Asia
- Science & Environment

- ✓ Zone: Mahanadi–Krishna coast
- ✓ Risk: High cyclone vulnerability
- ✓ Recent (2025): IMD AWS deployment
- ✓ Purpose: Improve last-mile cyclone warnings
- ✓ Target: Fishing communities
- Recent Event
 - ✓ Paradip Port: Top-performing major port
 - ✓ Criteria: Cargo volume
 - ✓ Declared by: Ministry of Ports (2026)

Q 28. Ans. A

Explanation:

- North–South Orientation
 - ✓ Purvachal Hills: North–South alignment
 - ✓ Regions: Nagaland, Manipur, Mizoram
 - ✓ Formation
 - Cause: Syntaxial bend beyond Dihang Gorge
 - Process: Indian Plate–Eurasian Plate collision
 - Result: Southward deflection of eastern Himalayas
 - Feature: Parallel ridges and valleys
 - Location: India–Myanmar border
- Additional Information
 - ✓ East–West: Main Himalayan ranges
 - Includes: Great Himalaya, Lesser Himalaya, Shiwaliks
 - Region: India, Nepal, Bhutan
 - Northwest–Southeast: Western Himalayas
 - Examples: Zaskar, Ladakh ranges
 - ✓ Southwest–Northeast: Aravalli Range
 - Feature: One of oldest fold mountains
 - Role: Barrier to Thar Desert expansion
 - ✓ Regional Names:
 - Patkai Bum: Arunachal Pradesh
 - Naga Hills: Nagaland
 - Manipur Hills: Manipur
 - Mizo (Lushai) Hills: Mizoram
 - Highest Peak: Mount Saramati
 - ◇ Height: 3826 m
 - ◇ Location: Nagaland
 - ✓ Orientation: North–South
 - Effect: Funnel for SW Monsoon (Bay of Bengal branch)
 - Result: Very heavy rainfall
 - Example: Mawsynram

- Fact: Wettest place in world
- ✓ Economic Linkage
 - Vegetation: Dense bamboo forests
 - Scheme: National Bamboo Mission

Q 29. Ans. B

Explanation:

- Anaimudi (2,695 metres) is the highest peak of the Peninsular Plateau and is located in the Anaimalai Hills, which are part of the southern Western Ghats spanning Kerala and Tamil Nadu.
- The Western Ghats are known by different local names and show an increase in elevation from north to south.
- Other prominent peaks include Doda Betta (2,637 metres) in the Nilgiri Hills, while Mahendragiri (1,501 metres) is the highest peak in the Eastern Ghats.

Q 30. Ans. B

Explanation:

- Anai Mudi — Southernmost
 - ✓ Location: Anaimalai Hills, Kerala
 - ✓ Elevation: 2,695 m
 - ✓ Significance:
 - ✓ Highest peak of the Western Ghats
 - ✓ Highest peak of the entire Peninsular India
 - ✓ Range: Western Ghats
 - ✓ Position: Located in southern Kerala, making it the southernmost among the given options.
- Shevroy Hills
 - ✓ Location: Salem district, Tamil Nadu
 - ✓ Major feature: Yercaud hill station
 - ✓ Mineral resource: Bauxite deposits
 - ✓ Position: Lies north of the Anaimalai Hills in Tamil Nadu.
 - ✓ Important directional correction:
 - ✓ Moving northward, Shevroy Hills occur before Javadi Hills.
- Javadi Hills
 - ✓ Location: Tiruvannamalai and Vellore districts, Tamil Nadu
 - ✓ Rock composition: Bluish-grey granite
 - ✓ Major feature: Vainu Bappu Observatory (astronomical observatory)
 - ✓ Position: Located north of the Shevroy Hills.

- ✓ Important note:
- ✓ Older references sometimes incorrectly reverse the order of Shevroy and Javadi Hills.
- Mahendragiri — Northernmost
 - ✓ Location: Gajapati district, Odisha
 - ✓ Elevation: 1,501 m
 - ✓ Range: Eastern Ghats
 - ✓ Position: Lies far north of Tamil Nadu hill ranges, making it the northernmost among the given options.
- Additional Facts:
 - ✓ Eastern Ghats — South to North Sequence:
 - Sirumalai — Tamil Nadu
 - Pachaimalai — Tamil Nadu
 - Shevroy — Tamil Nadu
 - Javadi — Tamil Nadu
 - Nagari Hills — Andhra Pradesh
 - Palkonda Range — Andhra Pradesh
 - Velikonda Range — Andhra Pradesh
 - Nallamala Hills — Andhra Pradesh (south of Krishna River)
 - Northern Circars / Mahendragiri — Odisha
 - ✓ Highest Peaks — Major Indian Ranges:
 - Himalayas (India): Kangchenjunga — 8,586 m — Sikkim
 - Western Ghats / Peninsular India:
 - ◇ Anai Mudi — 2,695 m — Kerala
 - Eastern Ghats:
 - ◇ Jindhagada Peak / Arma Konda — ~1,690 m — Andhra Pradesh
 - Aravalli Range:
 - ◇ Guru Shikhar — 1,722 m — Rajasthan
 - Satpura Range:
 - ◇ Dhupgarh — 1,350 m — Madhya Pradesh
 - Nilgiri Hills:
 - ◇ Doda Betta — 2,637 m — Tamil Nadu
- The Mikir Hills are not part of the Himalayas; they are a detached extension of the Peninsular Plateau, specifically related to the Chotanagpur Plateau.
- Separation feature:
 - They were separated from the main Peninsular block by the Malda Fault (Garo–Rajmahal Gap) during the Himalayan uplift.
 - This gap was later filled with Ganga–Brahmaputra alluvial sediments, forming parts of West Bengal and Bangladesh.
- ✓ Important geographical link:
 - Kaziranga National Park lies along the northern edge of the Mikir (Karbi Anglong) Hills.
 - During floods, animals migrate to the higher slopes of these hills for safety.
- Meghalaya:
 - ✓ Major hill ranges: Garo, Khasi, and Jaintia Hills
 - ✓ Geological nature: Detached extensions of the Peninsular Plateau
 - ✓ Important fact: Mawsynram, located in the Khasi Hills, is among the wettest places on Earth.
- Nagaland:
 - ✓ Major hill range: Naga Hills
 - ✓ Geographical nature: Part of the Purvanchal (Eastern Himalaya extensions)
 - ✓ Highest peak: Mount Saramati (3,826 m)
 - ✓ Boundary significance: Forms part of the India–Myanmar border.
- Arunachal Pradesh:
 - ✓ Major hill ranges (West → East):
 - ✓ Dafla, Miri, Abor, Mishmi Hills
 - ✓ Boundary range: Patkai Bum — forms an international boundary with Myanmar.
 - ✓ Naming basis: Hills are named after local tribal communities.
- Additional Facts:
 - ✓ Peninsular Plateau Extensions in the Northeast:
 - Detached plateaus:
 - ◇ Meghalaya Plateau
 - ◇ Karbi Anglong Plateau (Mikir Hills)
 - Separation feature:
 - ◇ Malda Fault (Garo–Rajmahal Gap)
 - Cause: Himalayan uplift and tectonic movement.

Q 31. Ans. B

Explanation:

- Mikir Hills:
 - ✓ The Mikir Hills are located in Assam, lying south of the Brahmaputra River.
 - ✓ They form part of the Karbi Anglong Plateau, situated mainly in the Karbi Anglong district of Assam.
 - ✓ Geological significance:

- Purvanchal Hills — Core Features
 - ◇ Region: Eastern extension of the Himalayas
 - ◇ Major hill groups:
 - ▶ Patkai Bum
 - ▶ Naga Hills
 - ▶ Mizo (Lushai) Hills
 - ▶ Manipur Hills
- Nature: Fold mountains
- General trend: North–South alignment.
 - ✓ Other Northeast Physiographic Facts:
 - ✓ Manipur Hills:
 - River origin: Barak River
 - Major lake: Loktak Lake — the largest freshwater lake in Northeast India
 - National Park: Keibul Lamjao — the only floating national park in the world.
 - ✓ Mizo Hills (Lushai Hills):
 - State: Mizoram
 - Highest peak: Phawngpui (Blue Mountain)

Q 32. Ans. A

Explanation:

- Kankar: Calcareous nodules (CaCO_3)
 - ✓ Soil: Bhangar (older alluvium)
 - ✓ Region: Indo-Gangetic Plains
 - ✓ Formation: Segregation of calcium carbonate over time
- Regur: Black cotton soil
 - ✓ Region: Deccan Trap (Maharashtra, Gujarat, MP)
 - ✓ Recent: PM-MITRA Parks
 - ✓ Focus: Textile hubs near cotton regions
 - ✓ Benefit: Reduced logistics cost
- Laterite: Leached soil
 - ✓ Condition: High temperature, high rainfall
 - ✓ Recent (2026): Red Mud policy
 - ✓ Ministry: Mines
 - ✓ Mandate: 100% utilization
 - ✓ Use: Cement, road construction
 - ✓ Regions: Odisha, Chhattisgarh
- Additional Information:
 - ✓ Soil Maturity:
 - Bhangar: Mature soil
 - Khadar: Immature (new alluvium)
 - Kankar: Indicator of soil age
 - ✓ Economic Linkage:
 - Bhangar: High calcium content

- Issue: Alkalinity
- Scheme: PM-PRANAM (2023–24)
- Focus: Reduce urea use
- Promote: Alternative fertilisers

Q 33. Ans. D

Explanation:

Andaman and Nicobar Islands:

- The highest number of offshore islands in India.
- Total islands, islets, and rocky outcrops: 836.
- Location: Bay of Bengal.
- Island type: Oceanic islands of tectonic origin.

Gujarat:

- Offshore islands: More than 144.
- Rank: Highest among mainland states.
- Key coastal feature: Highly indented coast — Gulf of Kachchh, Gulf of Khambhat.
- Important islands:
 - ✓ Aliabet — Narmada estuary
 - ✓ Piram Island
 - ✓ Beyt Dwarka

Maharashtra:

- Offshore islands: About 126.
- Rank: Second among mainland states.
- Important islands:
 - ✓ Salsette Island — Mumbai city located here
 - ✓ Elephanta Island — famous for rock-cut caves

Lakshadweep:

- Total islands: 32.
- Inhabited islands: 10.
- Location: Arabian Sea.
- Island type: Coral origin (atolls).
- Much fewer islands than in the Andaman & Nicobar Islands.
- Strategic Offshore Islands — East Coast
 - ✓ Abdul Kalam Island (Odisha):
 - Function: Missile testing range
 - ✓ Sriharikota (Andhra Pradesh):
 - Location: Mouth of Pulicat Lake
 - Function: ISRO launch centre
 - ✓ Pamban Island (Tamil Nadu):
 - Location: Gulf of Mannar
 - Major town: Rameswaram
 - ✓ Sagar Island (West Bengal):
 - Location: Mouth of the Hooghly River
 - Event: Ganga Sagar Mela

- Strategic Offshore Islands — West Coast
 - ✓ Willingdon Island (Kerala):
 - Type: Largest artificial island in India
 - Location: Kochi
 - ✓ Khadir Bet (Gujarat):
 - Location: Rann of Kutch
 - Archaeological importance: Dholavira (Indus Valley Civilisation site)

Q 34. Ans. B

Explanation:

Gulf of Kutch — Coral reefs present:

- Coral type: Fringing reefs (patchy)
- Location: Gujarat coast
- Significance: Northernmost limit of coral reefs in India
- Environmental stress factors:
 - ✓ High salinity
 - ✓ Large tidal range
 - ✓ Temperature fluctuations
- Despite harsh conditions, corals naturally occur here.

Gulf of Khambhat — Coral reefs absent:

- Major rivers entering: Narmada and Tapi
- Water condition: Highly turbid (muddy)
- Sediment load: Very high
- Key limiting factor:
 - ✓ Heavy silt deposition blocks sunlight
 - ✓ Freshwater inflow reduces salinity
- Result: Conditions unsuitable for coral growth

Gulf of Mannar — Coral reefs present:

- Coral type: Fringing reefs
- Location: Tamil Nadu coast between India and Sri Lanka
- Status: India's first Marine Biosphere Reserve
- Environmental condition:
 - ✓ Clear, shallow, warm marine waters
- Significance: Major marine biodiversity hotspot.

Sunderbans — Coral reefs absent:

- River system: Ganga–Brahmaputra delta
- Sediment load: Extremely high
- Water condition: Highly muddy and low salinity
- Major ecosystem: Mangrove forests, not coral reefs
- Result: Unsuitable for coral formation

Additional Facts:

- Essential Conditions for Coral Growth (Very High Yield)
- Corals require:
 - ✓ Clear water: Sunlight penetration is essential for photosynthesis
 - ✓ Temperature: Warm tropical waters (20°C–27°C)
 - ✓ Salinity: Marine salinity (27–40 ppt)
 - ✓ Depth: Shallow waters (generally <50 m)
- Major limiting factor in river mouths:
 - ✓ High sediment load prevents coral survival.

Major Coral Reef Regions of India (The “Big Four”)

- Gulf of Mannar (Tamil Nadu):
 - ✓ Reef type: Fringing reefs
 - ✓ Status: Marine Biosphere Reserve
- Andaman & Nicobar Islands:
 - ✓ Reef type: Mainly fringing, some barrier reefs
 - ✓ Feature: Largest and most diverse coral reefs in India
- Lakshadweep Islands:
 - ✓ Reef type: Atolls
 - ✓ Key distinction: Only atoll-type coral reefs in India
- Gulf of Kutch (Gujarat):
 - ✓ Reef type: Patchy fringing reefs
 - ✓ Feature: Northernmost coral reef region

Minor Coral Location:

- Malvan Coast — Maharashtra (Sindhudurg region)
- Netrani Island — Karnataka
- Gokarna coast — Karnataka

Mangrove vs Coral Region Trap:

- Mangrove-dominated region: Sunderbans
- Coral-dominated region: Gulf of Mannar
- River-dominated muddy region: Gulf of Khambhat
- Clear marine island region: Lakshadweep

Q 35. Ans. D

Explanation:

Statement 1 is correct:

- Shola forests are temperate forests found at higher elevations.

- Major locations: Nilgiri Hills, Anaimalai Hills, Palani Hills, Western Ghats.
- Forest type: Southern Montane Wet Temperate Forests.
- Typical altitude: Above 1,500 m.
- Cause: High altitude lowers temperature, creating temperate conditions within tropical latitudes.

Landscape structure:

- Occur as patches of dense evergreen trees in sheltered valleys.
- Surroundings: Montane grasslands on exposed hilltops.
- Ecosystem type: Shola–grassland mosaic.

Statement 2 is correct:

- Shola forests support temperate-type vegetation due to cool high-altitude conditions.
- Rhododendron species occur in Shola forests, and such vegetation shows similar ecological characteristics to Himalayan temperate flora.
- Shola forests occur in high rainfall mountain regions, not deserts.
- Require:
 - ✓ Cool climate
 - ✓ High rainfall
 - ✓ High altitude
- Found mainly in humid mountain zones of Southern India.

Additional Facts:

Shola Ecosystem — Key Features:

- Forest type: Southern Montane Wet Temperate Forest.
- Vegetation pattern:
 - ✓ Evergreen forest patches in valleys
 - ✓ Grasslands on hilltops
- Climate: Cool, moist conditions.

Major threats:

- Invasive species:
 - ✓ Eucalyptus
 - ✓ Wattle (Acacia)
 - ✓ Pine

Important Flora & Fauna Links

- Neelakurinji Flower:
 - ✓ Bloom cycle: Once every 12 years.
 - ✓ Habitat: Shola grasslands.
 - ✓ Region: Nilgiri Hills.

- Nilgiri Tahr:
 - ✓ Habitat: Shola–grassland ecosystem.
 - ✓ Conservation site: Eravikulam National Park.
 - ✓ Status: Endangered species.

Southern vs Northern Montane Forests

- Southern Montane Forests (Peninsular India)
 - ✓ Locations: Western Ghats, Nilgiris, Vindhyas.
 - ✓ Vegetation zones:
 - Subtropical (lower slopes)
 - Temperate (upper slopes — Sholas)
- Key feature: No alpine vegetation.

Northern Montane Forests — Himalayan Sequence

- 1,000–2,000 m:
 - ✓ Wet temperate forests — Oak, Chestnut
- 1,500–3,000 m:
 - ✓ Coniferous forests — Pine, Deodar, Silver Fir, Spruce
- 3,000–3,600 m:
 - ✓ Alpine transition — Juniper, Birch
- Above 3,600 m:
 - ✓ Alpine pastures — Mosses, Lichens
- Pastoral practice:
 - ✓ Transhumance by Gujjars, Bakarwals, Bhotiyas.

Q 36. Ans. A

Explanation:

Neem:

- Neem is a characteristic species of Tropical Thorn Forests.
- Adaptation type: Xerophytic plant suited to dry and semi-arid climates.
- Common regions: Rajasthan, Gujarat, Haryana, parts of Madhya Pradesh and Uttar Pradesh.
- Adaptation features:
 - ✓ Deep root system to access groundwater.
 - ✓ Thick leaves reduce water loss.

Palas:

- Palas (*Butea monosperma*) is a common species found in dry deciduous and thorn forest margins.
- Popular name: Flame of the Forest (bright orange-red flowers).

- Adaptation: Can survive in low rainfall and dry conditions.
- Distribution: Semi-arid and dry plateau regions.

Hurra:

- Hurra (*Terminalia chebula*) belongs to Tropical Deciduous Forests, not Thorn Forests.
- Forest type: Moist and Dry Deciduous Forests.
- Economic importance: Medicinal plant; commonly known as Harad (Ayurveda use).
- Rainfall requirement: Moderate rainfall, higher than Thorn Forest regions.

Teak:

- Teak (*Tectona grandis*) is a dominant species of Moist Deciduous Forests, not Thorn Forests.
- Rainfall requirement: 100–200 cm annually.
- Economic importance: Major commercial timber species.
- Distribution: Central India, Western Ghats, Eastern Ghats.

Additional Facts:

- Tropical Thorn Forests — Core Features (Very High Yield)
- Rainfall limit: Less than 50 cm annually
- Climate: Semi-arid to arid
- Major regions:
 - ✓ South-west Punjab
 - ✓ Haryana
 - ✓ Rajasthan
 - ✓ Gujarat
 - ✓ Western Madhya Pradesh
 - ✓ Western Uttar Pradesh
- Peninsular distribution:
 - ✓ Rain-shadow areas of the Western Ghats
 - ✓ Parts of Maharashtra
 - ✓ Karnataka
 - ✓ Andhra Pradesh
- Xerophytic Adaptations (Highly Tested)
- Typical plant adaptations:
 - ✓ Deep roots: Reach underground water sources.
 - ✓ Succulent stems: Store water.
 - ✓ Modified leaves: Leaves reduced to spines or thick waxy surfaces to reduce transpiration.

Characteristic Species of Tropical Thorn Forests

- Species:
 - ✓ Khejri (*Prosopis cineraria*) — State tree of

Rajasthan

- ✓ Babool (*Acacia*)
- ✓ Khair — Source of katha
- ✓ Neem
- ✓ Ber
- ✓ Wild Date Palm
- Bihar's Vegetation:
 - ✓ Dominant forest type: Tropical Deciduous Forests
- Moist Deciduous regions:
 - ✓ West Champaran (Someswar Hills)
 - ✓ Kishanganj
 - ✓ Purnia
- Major species:
 - ✓ Sal
 - ✓ Shisham
 - ✓ Semal
- Dry Deciduous regions:
 - ✓ Kaimur
 - ✓ Rohtas
 - ✓ Gaya
 - ✓ Jamui
- Major species:
 - ✓ Amaltas
 - ✓ Mahua
 - ✓ Palas
 - ✓ Khair

Q 37. Ans. A

Explanation:

- **Statement 1 is correct:** The original Indo-Brahma river system was a single large river that got dismembered into three systems:
 - ✓ Indus, Ganga, and Brahmaputra are due to tectonic and geological changes.
- **Statement 2 is correct:** The Indus River system developed in the western part of the subcontinent. It originates near Lake Mansarovar in Tibet and flows through Ladakh into Pakistan, finally draining into the Arabian Sea.
 - ✓ Its major tributaries, like the Jhelum, Chenab, Ravi, Beas, and Sutlej, form an extensive drainage network, indicating its well-developed western system after the dismemberment.
- **Statement 3 is incorrect:** The Brahmaputra River system developed in the eastern part, not the central part of India. It originates in Tibet as the Tsangpo, flows eastward, then takes a sharp bend (gorge) to enter India through Arunachal Pradesh.

- ✓ It flows through Assam, forming a wide braided channel and joins the Ganga in Bangladesh, highlighting its distinct eastern drainage evolution.

Q 38. Ans. B

Explanation:

Carbon dioxide, Methane, Chlorofluorocarbons, Nitrous oxide:

These gases are collectively known as Greenhouse Gases (GHGs) because they absorb outgoing longwave (infrared) radiation and trap heat in the atmosphere.

Major Greenhouse Gases:

- Carbon dioxide (CO₂):
 - ✓ Primary greenhouse gas driving global warming.
 - ✓ Baseline gas for measuring Global Warming Potential (GWP = 1).
 - ✓ Major sources: Fossil fuel burning, deforestation.
- Methane (CH₄):
 - ✓ Heat-trapping ability: 25–28 times stronger than CO₂ (100-year scale).
 - ✓ Major Indian sources:
 - Livestock digestion (enteric fermentation)
 - Paddy cultivation fields.
- Nitrous oxide (N₂O):
 - ✓ Heat-trapping ability: About 273 times stronger than CO₂.
 - ✓ Major source: Nitrogen-based fertilizers in agriculture.
- Chlorofluorocarbons (CFCs):
 - ✓ Fully synthetic greenhouse gases.
 - ✓ Also act as ozone-depleting substances.
 - ✓ Found in refrigerants and aerosols.

Additional Facts:

- Greenhouse Effect — Core Mechanism:
 - ✓ Incoming shortwave solar radiation passes through atmosphere.
 - ✓ Earth emits longwave infrared radiation.
 - ✓ Greenhouse gases absorb and re-radiate heat, warming lower atmosphere.
- Major Greenhouse Gases — Priority Order
 - ✓ Carbon dioxide (CO₂) — Most important anthropogenic gas

- ✓ Methane (CH₄) — Strong heat-trapping gas
- ✓ Nitrous oxide (N₂O) — Very high warming potential
- ✓ Fluorinated gases:
 - CFCs
 - HFCs
 - SF₆

Natural greenhouse gas:

- Water vapour — Most abundant greenhouse gas.
- Environmental Protocols — Gas Target Mapping:

Kyoto Protocol (1997):

- Targets Greenhouse Gases
- Covered gases:
 - ✓ CO₂
 - ✓ CH₄
 - ✓ N₂O
 - ✓ HFCs
 - ✓ PFCs
 - ✓ SF₆

Montreal Protocol (1987):

- Targets Ozone-Depleting Substances (ODS)
- Major gas: CFCs

Kigali Amendment (2016):

- Targets HFCs
- Aim: Reduce high-GWP refrigerant gases.

India — Climate Commitments:

- National Action Plan on Climate Change (NAPCC) — 2008
- Total missions: 8
- Key missions:
 - ✓ National Solar Mission
 - ✓ National Water Mission
 - ✓ National Mission for Sustainable Agriculture
 - ✓ National Mission for Green India
 - ✓ National Mission for Sustaining Himalayan Ecosystem
- Net-Zero Target:
 - ✓ India target year: 2070
 - ✓ Announced at: COP26 (Glasgow, 2021)

Q 39. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ Winter monsoon winds (Northeast Monsoon) are dry and anticyclonic in nature.

- ✓ Origin: Asian landmass (Siberia/Central Asia).
- ✓ Direction: Land to sea (northeast to southwest).
- ✓ Rainfall nature: Generally dry winds, suppress rainfall over most parts of India.
- ✓ Exception:
 - These winds cause rainfall on the Coromandel Coast (Tamil Nadu and south Andhra Pradesh).
 - Reason: Winds cross the Bay of Bengal, pick up moisture, then strike the eastern coast.
- Statement 2 is correct:
 - ✓ Winter rainfall in northwestern India is caused by temperate cyclones, called Western Disturbances.
 - ✓ Origin region: Mediterranean Sea.
 - ✓ Movement mechanism: Carried eastward by the Sub-Tropical Westerly Jet Stream.
 - ✓ Affected regions:
 - Punjab
 - Haryana
 - Rajasthan
 - Western Uttar Pradesh
 - Jammu & Kashmir
 - Himachal Pradesh

Additional Facts:

- Western Disturbances — Key Features:
 - ✓ Type: Extra-tropical cyclones.
 - ✓ Moisture sources:
 - Mediterranean Sea
 - Black Sea
 - Caspian Sea
 - ✓ Season: Winter months (December–March).
- Agricultural importance:
 - ✓ Essential for Rabi crops:
 - Wheat
 - Mustard
 - Gram
- Local name:
 - ✓ Winter rainfall in Rajasthan and Haryana: Mawath.
- Northeast Monsoon — Key Features
 - ✓ Season: October to December
 - ✓ Wind direction: Northeast → Southwest
 - ✓ Nature: Dry continental winds
- Major rainfall region:
 - ✓ Tamil Nadu coast
 - ✓ Parts of Andhra Pradesh

- Rainfall share:
 - ✓ Tamil Nadu receives 50–60% of its annual rainfall from Northeast Monsoon.

Jet Stream Role in Winter Rainfall:

Jet stream involved: Sub-Tropical Westerly Jet Stream (STWJ)

- Winter position:
 - ✓ Lies south of the Himalayas.
 - ✓ Guides Western Disturbances into northern India.
- Summer shift:
 - ✓ Moves north of the Himalayas.
 - ✓ Western Disturbances weaken over India.

Q 40. Ans. B

Explanation:

Kalbaisakhi: (Useful for tea, jute and rice cultivation)

- Nature: Violent pre-monsoon thunderstorms.
- Season: April–May.
- Regions affected: West Bengal and Assam.
- Agricultural importance:
 - ✓ Provides rainfall beneficial for Tea cultivation.
 - ✓ Supports Jute cultivation.
 - ✓ Helps Boro rice (pre-kharif rice).
- Local name in Assam: Bordoichila.
- Origin region: Chotanagpur Plateau; moves eastward.

Mango Shower: Pre-monsoon showers in Kerala and coastal Karnataka)

- Nature: Pre-monsoon rainfall.
- Regions: Kerala and Coastal Karnataka.
- Agricultural role:
 - ✓ Prevents premature falling of mangoes.
 - ✓ Helps mango ripening.
- Season: April–May.

Loo: (Dry, hot northern plains wind)

- Nature: Hot, dry wind.
- Season: May–June (summer).
- Regions affected:
 - ✓ Punjab
 - ✓ Haryana
 - ✓ Rajasthan
 - ✓ Uttar Pradesh
 - ✓ Bihar
- Origin area: Thar Desert region.

- Impact: Causes heat waves and heatstroke.

Blossom Shower: (Coffee flowers prosper in Kerala)

- Nature: Light pre-monsoon rainfall.
- Regions: Kerala and Karnataka.
- Agricultural importance:
 - ✓ Essential for coffee flowering.
- Alternate name: Cherry Blossom showers.

Additional Facts:

Local Pre-Monsoon Storms of India (Very High Yield)

- Eastern India:
 - ✓ Kalbaisakhi — West Bengal
 - ✓ Bordoichila — Assam (same storm, different local name)
- Northern Plains:
 - ✓ Loo — Hot dry summer wind
 - ✓ Andhi — Dust storms in Punjab, Haryana, Rajasthan
- Southern India:
 - ✓ Mango Showers — Aid mango ripening
 - ✓ Blossom Showers — Aid coffee flowering
- Seasonal Timing — Pre-Monsoon Events
 - ✓ Typical months: April–May
 - ✓ Cause:
- Intense surface heating
- Development of local low-pressure systems
- Outcome:
 - ✓ Thunderstorms
 - ✓ Dust storms
 - ✓ Localized rainfall
- Bihar-Specific fact:
 - ✓ Loo winds common during May–June.
 - ✓ Most affected districts:
 - Gaya
 - Aurangabad
 - Rohtas
 - Patna
- Major risk: Heatstroke during peak summer.

Q 41. Ans. C

Explanation:

A catchment is an area of land where water collects when it rains, often bounded by hills. As the water flows over the landscape, it finds its way into streams and down into the soil, eventually feeding the river.

Basin Name	Area (sq. km)
Indus (Up to border) Basin	453931.87
Ganga Basin	808334.44
Brahmaputra Basin	186421.60
Barak and others Basin	45622.41
Godavari Basin	302063.93
Krishna Basin	254743.31
Cauvery Basin	85624.44
Subarnarekha Basin	25792.16
Brahmani and Baitarni Basin	51893.68
Mahanadi Basin	139659.15
Pennar Basin	54243.43
Mahi Basin	38336.80
Sabarmati Basin	30678.59
Narmada Basin	92670.51
Tapi Basin	63922.91
West flowing rivers South of Tapi Basin	111643.87

Q 42. Ans. B

Explanation:

Malabar Coast — Southernmost

- Location: Kerala and southern Karnataka.
- Coast type: Narrow coastal plain.
- Special feature: Presence of backwaters (Kayals).
- Major water body: Vembanad Lake — the longest lake in India.
- Other feature: Ashtamudi Lake — Ramsar wetland.

Konkan Coast

- Location: Goa and Maharashtra.
- Extent: Between Mumbai and Goa.
- Coast nature: Narrow and rocky coastline.
- Major ports:
 - ✓ Mumbai
 - ✓ Marmagao

Kathiawar Coast

- Location: Saurashtra Peninsula region of Gujarat.
- Coast type: Peninsula coast projecting into the Arabian Sea.
- Associated feature: Lies between Gulf of Kachchh and the Gulf of Khambhat.

Kachchh Coast — Northernmost

- Location: Northern Gujarat.

- Key feature: Associated with the Rann of Kachchh region.
- Position: Northernmost coastal stretch of western India.

Additional Facts:

Western Coastal Plains — South to North Sequence (Very High Yield)

- Malabar Coast — Kerala
- Konkan Coast — Goa & Maharashtra
- Kathiawar Coast — Saurashtra (Gujarat)
- Kachchh Coast — Northern Gujarat

Western vs Eastern Coastal Plains (Very High Yield)

Western Coastal Plains — Key Features

- Coast type: Submergent coast.
- River mouth type: Estuaries dominate.
- Major west-flowing rivers:
 - ✓ Narmada
 - ✓ Tapi
 - ✓ Mandovi
 - ✓ Periyar
- Ports: Many natural harbours due to deep coastline.

Eastern Coastal Plains — Key Features

- Coast type: Emergent coast.
- River mouth type: Deltas dominate.
- Major east-flowing rivers:
 - ✓ Godavari
 - ✓ Krishna
 - ✓ Mahanadi
 - ✓ Cauvery
- Ports: Mostly artificial ports due to the shallow coast.

Eastern Coastal Plains — South to North Sequence

- Coromandel Coast — Tamil Nadu to southern Andhra Pradesh
- Andhra Coast — Central and northern Andhra Pradesh
- Utkal Coast — Odisha

Alternate historical term: Northern Circars (Andhra–Odisha coast).

Q 43. Ans. A

Explanation:

Statement 1 is correct:

- Chilika Lake is the largest coastal lagoon in India.

- Location: Odisha coast, along the Bay of Bengal.
- Water type: Brackish water lagoon.
- Global importance: One of the largest wintering grounds for migratory birds in Asia.

Key ecological facts:

- Ramsar Site: Included in Ramsar List of Wetlands.
- Montreux Record:
 - ✓ Added: 1993 (due to siltation).
 - ✓ Removed: 2002 — first Asian wetland restored and removed.
- Important species: Irrawaddy Dolphin — major habitat.
- Internal feature: Nalbana Bird Sanctuary located inside Chilika Lake.

Statement 2 is incorrect:

- India's only active volcano is located on Barren Island.
- Location: Andaman Sea, in Andaman and Nicobar Islands, not Lakshadweep.
- Volcanic nature: Stratovolcano.
- Tectonic setting: Located along subduction zone of the Indian Plate beneath the Burmese Plate.

Additional Facts:

Coastal Lagoons of India (Very High Yield)

- Chilika Lake:
 - ✓ Location: Odisha
 - ✓ Rank: Largest coastal lagoon in India
 - ✓ Water type: Brackish
- Pulicat Lake:
 - ✓ Rank: Second-largest brackish water lagoon
 - ✓ Location: Andhra Pradesh–Tamil Nadu border
 - ✓ Barrier island: Sriharikota Island
- Vembanad Lake:
 - ✓ Location: Kerala
 - ✓ Rank: Longest lake in India
 - ✓ Local name: Kayal
- Kolleru Lake (Trap Lake):
 - ✓ Location: Andhra Pradesh
 - ✓ Type: Freshwater lake, not lagoon
 - ✓ Position: Between Krishna and Godavari deltas.

Volcanic Islands of India:

- Barren Island:

- ✓ Location: Andaman Sea
- ✓ Status: Only active volcano in India
- ✓ Region: Andaman and Nicobar Islands
- Narcondam Island:
 - ✓ Location: Andaman Sea
 - ✓ Status: Dormant volcano
 - ✓ Ecological importance: Habitat of Narcondam Hornbill.

Island Type Distinction:

- Lakshadweep Islands:
 - ✓ Origin: Coral atolls
 - ✓ Location: Arabian Sea
 - ✓ Volcanic activity: Absent
- Andaman & Nicobar Islands:
 - ✓ Origin: Tectonic and volcanic
 - ✓ Location: Bay of Bengal / Andaman Sea
 - ✓ Volcanic activity: Present (Barren Island)

Q 44. Ans. D

Explanation:

Yamuna — Left-bank tributary:

- Yamuna is a right-bank tributary of the Ganga.
- Origin: Yamunotri Glacier, Uttarakhand (Western Himalayas).
- Flow direction: Generally south-east, then eastward.
- Confluence with Ganga: Prayagraj (Allahabad) — forms Triveni Sangam.
- Bank position logic:
 - ✓ Ganga flows west → east.
 - ✓ Rivers joining from south are right-bank tributaries.
- Significance: Longest and largest right-bank tributary of the Ganga.

Ghaghara — Left-bank tributary

- Bank type: Left-bank tributary of the Ganga.
- Origin: Tibetan Plateau (as Karnali River).
- Major tributaries: Karnali, Sharda rivers.
- Confluence: Joins Ganga near Chhapra (Bihar).
- Key fact: Largest tributary of Ganga by water volume.

Gandak — Left-bank tributary

- Bank type: Left-bank tributary.
- Origin: Nepal Himalayas.
- Formation: Confluence of Kaligandaki and Trishuli rivers.

- Entry into plains: Near Triveni (Bihar).
- Confluence with Ganga: Near Hajipur–Patna region.

Kosi — Left-bank tributary:

- Bank type: Left-bank tributary.
- Origin: Near Mount Everest region (Tibet–Nepal Himalayas).
- River nature: Braided river with frequent course changes.
- Confluence with Ganga: Near Katihar district (Bihar).
- Nickname: “Sorrow of Bihar” due to floods and channel shifting.

Additional Facts:

Left-Bank Tributaries of Ganga — West to East Sequence (Very High Yield)

- Ramganga
- Gomti
- Ghaghara
- Gandak
- Kosi
- Mahananda

Important exception:

- Gomti originates in plains — Fulhar Lake (Pilibhit, UP).

Right-Bank Tributaries of Ganga

- Yamuna — joins at Prayagraj
- Son — originates at Amarkantak Plateau
- Punpun — Bihar river
- Falgu — flows through Gaya (Bihar)

Yamuna Tributary:

These rivers do not join Ganga directly:

- Chambal
- Sind
- Betwa
- Ken

Correct classification:

- These are right-bank tributaries of the Yamuna, not direct tributaries of the Ganga.

Q 45. Ans. B

Explanation:

Longitudinal extent of about 30° is correct

- India extends widely in longitude, which causes time difference between eastern and western parts.
- Westernmost longitude: $68^{\circ}7'$ E at Ghuar Moti in Kachchh, Gujarat.
- Easternmost longitude: $97^{\circ}25'$ E at Kibithu in Anjaw district, Arunachal Pradesh.
- Total longitudinal extent: About 30 degrees.

Time difference logic:

- Earth rotates 1 degree of longitude in 4 minutes.
- A difference of 30 degrees produces about 120 minutes (2 hours) difference in local time.
- Therefore, the sun rises much earlier in eastern India than in western India.

Additional Facts:

- Indian Standard Time (IST)
- Standard meridian of India: $82^{\circ}30'$ E longitude
- Reason for selection:
 - ✓ Lies near the central part of India, reducing time differences across regions.
- IST difference from Greenwich Mean Time (GMT):
 - ✓ IST is 5 hours 30 minutes ahead of GMT.
- Reference location:
 - ✓ The $82^{\circ}30'$ E meridian passes near the Shankargarh area in Mirzapur district, Uttar Pradesh.
- States Through Which $82^{\circ}30'$ E Meridian Passes (North to South)
 - ✓ Uttar Pradesh
 - ✓ Madhya Pradesh
 - ✓ Chhattisgarh
 - ✓ Odisha
 - ✓ Andhra Pradesh

India's Longitudinal Extremes:

Westernmost point of India:

- ✓ Ghuar Moti, Kachchh district, Gujarat
- ✓ Longitude: $68^{\circ}7'$ E
- Easternmost point of India:
 - ✓ Kibithu, Anjaw district, Arunachal Pradesh
 - ✓ Longitude: $97^{\circ}25'$ E

Local Time Practice in Northeast India

- Local system used: Chaibagaan Time (Tea Garden Time)
- Region: Assam tea gardens
- Time difference:

✓ About 1 hour ahead of IST

● Reason:

- ✓ Sunrise occurs very early in northeastern India compared to western India.

Q 46. Ans. B

Explanation:

- Ghaghara:
- Tila, Seti, and Beri are tributaries of the Ghaghara River.
- Origin of Ghaghara: Mapchachungo Glacier, Tibet.
- Upper course name: Known as Karnali River in Nepal.
- Major Himalayan tributaries:
 - ✓ Tila
 - ✓ Seti
 - ✓ Beri
- Important feature: Cuts a deep gorge at Shishapani before entering plains.
- Additional tributary: Joined by Sarda (Kali) River.
- Final confluence: Meets the Ganga at Chhapra (Saran district), Bihar.

Key identification fact:

- Largest tributary of the Ganga by water volume.

Gandak:

- Gandak is formed by the union of:
 - ✓ Kali Gandaki
 - ✓ Trishuli River
- Origin region: Nepal Himalayas.
- Entry into plains: Near Triveni (Bihar).
- Confluence with Ganga: Sonpur/Hajipur (Bihar).

Kosi:

- Kosi is formed by seven Himalayan streams known as Saptakoshi.
- Major tributaries:
 - ✓ Sun Kosi
 - ✓ Arun
 - ✓ Tamur
- River nature: Braided river with frequent course shifting.
- Confluence with Ganga: Kursela (Katihar district), Bihar.
- Nickname: Sorrow of Bihar.

Son:

- Son River origin: Amarkantak Plateau, Madhya Pradesh.
- River type: Peninsular river.
- Major tributaries:
 - ✓ Rihand River
 - ✓ North Koel River
- Confluence with Ganga: Danapur (Patna district), Bihar.

Additional Facts:

- Ghaghara River:
 - ✓ Upper name: Karnali River (Nepal)
 - ✓ Lower course name in Uttar Pradesh: Sarayu River
 - ✓ Important city on banks: Ayodhya
- Boundary function:
 - ✓ Forms boundary between Uttar Pradesh and Bihar near entry into Bihar.
- Final confluence location:
 - ✓ Chhapra, Saran district, Bihar.
- Major Ganga Tributary Confluences in Bihar
 - ✓ Ghaghara meets Ganga at Chhapra (Saran district)
 - ✓ Gandak meets Ganga at Sonpur/Hajipur
 - ✓ Son meets Ganga at Danapur (Patna district)
- Punpun meets Ganga at Fatuha (Patna district)
- Kosi meets Ganga at Kursela (Katihar district)

Q 47. Ans. B

Explanation:

- Statement 1 is correct:
 - ✓ The Indian Desert (Thar Desert) is a northwestern extension of the Peninsular Plateau.
 - ✓ Geological nature: Ancient peninsular rock base buried under aeolian (wind-blown) sand deposits.
 - ✓ Rock structure below the desert: Similar to Aravalli and Peninsular Plateau rocks.

Supporting geological evidence:

- Akal Wood Fossil Park (Jaisalmer): Contains fossilised tree trunks about 180 million years old.
- Marine deposits near Brahmsar: Indicate that the region was once submerged under sea during the Mesozoic Era.

Statement 2 is correct:

- Luni River is the only major river system of the Indian Desert.
- River type: Ephemeral (seasonal) river dependent on monsoon rainfall.
- Origin: Pushkar Valley, Aravalli Range (Rajasthan).
- Formation: Initially formed by streams such as the Saraswati and Sabarmati.

Course characteristics:

- Water remains fresh up to Balotra (Rajasthan).
- Beyond Balotra: Water becomes saline due to desert salts.
- Drainage type: Inland drainage (endorheic).
- End point: Drains into Rann of Kutch (Gujarat).

Additional Facts:

- Indian Desert (Thar Desert) Features:
- Location: Western Rajasthan and adjoining Gujarat region.
- Climate type: Arid to semi-arid.
- Major wind action: Formation of aeolian sand dunes.

Dominant dune type:

- Barchans — crescent-shaped sand dunes common near Jaisalmer.

Local geographic terms:

- Marusthali: Core desert region with shifting sand dunes.
- Bagar (Bangar): Semi-arid transitional zone between Aravalli Range and desert.

Desert Drainage and Lakes

- Drainage type: Inland drainage dominant.
- Seasonal streams: Flow only during monsoon months.
- Important saline lakes (Playas):
 - ✓ Sambhar Lake: Largest inland salt lake in India.
 - ✓ Didwana Lake
 - ✓ Pachpadra Lake
- Definition:
 - ✓ Playa: Temporary saline lake formed in desert depressions.

Luni River:

- Origin: Aravalli Range near Pushkar
- Nature: Seasonal desert river
- Water quality change point: Balotra

- Final drainage: Rann of Kutch
- Meaning of name: "Luni" means salt river

Statement 3 is incorrect:

- The Indian Desert (Thar Desert) receives very low rainfall, generally below 150 mm annually in its core region.
- This extremely low rainfall creates highly arid conditions with sparse vegetation.
- The driest central part of the desert is called Marusthali, meaning the most arid and barren zone of the Thar Desert.

Statement 4 is correct:

- The surface of the Indian Desert shows undulating topography mainly shaped by wind (aeolian) action.
- The region is dotted with characteristic sand dune forms such as:
 - ✓ Barchans — crescent-shaped dunes formed due to unidirectional winds.
 - ✓ Longitudinal dunes (Seifs) — long sand ridges aligned parallel to prevailing wind direction.
- These dune types are widespread across western Rajasthan, especially around Jaisalmer and Barmer regions.

Additional Facts:

- Desert Zones of Rajasthan
- Moving westward from the Aravalli Range, the desert shows clear transitional zones:
- Aravalli Range:
 - ✓ Acts as a climatic barrier, limiting the reach of monsoon rainfall toward western Rajasthan.
- Rajasthan Bagar:
 - ✓ Semi-arid transitional zone west of the Aravallis.
 - ✓ Contains sparse scrub vegetation and seasonal streams.
- Marusthali:
 - ✓ Core arid region of the Thar Desert.
 - ✓ Dominated by sand dunes and barren terrain.

Q 48. Ans. D

Explanation:

Statement 1 is correct:

- The Deccan Plateau lies south of the Narmada River.

- Shape: Triangular (inverted triangle) landmass.
- Northern boundary: Narmada River valley separates it from the Central Highlands.
- Major rivers marking the northern margin: Narmada and Tapi flow westward in rift valleys.

Statement 2 is correct:

- The Deccan Plateau is separated from the Chotanagpur Plateau by major fault zones.
- Geological feature: Presence of rift valleys and structural faults.
- Extension link:
 - ✓ Chotanagpur Plateau is part of the broader Peninsular Plateau system.
 - ✓ Further northeast extension: Meghalaya Plateau.
 - ✓ Separation structure: Malda Fault (Garo–Rajmahal Gap) separates northeastern extensions from the main plateau.
- Statement 3 is correct:
 - ✓ The Deccan Plateau is higher in the west and slopes gently eastwards.
 - ✓ Western boundary: Western Ghats, average height 900–1600 m.
 - ✓ Eastern boundary: Eastern Ghats, average height around 600 m.
 - ✓ Slope direction: West to East.
- Drainage implication:
 - ✓ Most major rivers flow eastward into the Bay of Bengal.
 - ✓ Major east-flowing rivers:
 - Godavari
 - Krishna
 - Kaveri
 - Mahanadi

Additional Facts:

- Deccan Plateau — Geological Origin
- Northwestern region name: Deccan Trap
- Formation period: End of the Cretaceous period (about 66 million years ago)
- Formation process: Fissure volcanic eruptions
- Dominant rock type: Basalt
- Soil formed from basalt:
 - ✓ Black Soil (Regur)
- Suitable crops:
 - ✓ Cotton
 - ✓ Sugarcane
- Other rock regions:
 - ✓ Southern and eastern plateaus are dominated by Archaean granite and gneiss.

- ✓ Associated soils: Red and Yellow soils.

Boundaries of the Deccan Plateau:

- North: Satpura Range, with Mahadeo, Maikal, and Kaimur hills
- West: Western Ghats (Sahyadri) — continuous mountain wall
- East: Eastern Ghats — discontinuous and eroded
- South: Nilgiri Hills — meeting point of Eastern and Western Ghats.

Northeastern Extension of Peninsular Plateau:

- Plateau extensions:
 - ✓ Meghalaya Plateau (Garo, Khasi, Jaintia Hills)
 - ✓ Karbi-Anglong Plateau
- Separation structure:
 - ✓ Malda Fault (Garo–Rajmahal Gap)
- Formation:
 - ✓ Gap filled by Ganga–Brahmaputra sediments.
- Key fact:
 - ✓ These plateaus are not part of the Himalayas.

Q 49. Ans. B

Explanation:

- The Central Highlands are the northern part of the Peninsular Plateau lying north of the Narmada River. This region mainly includes the Malwa Plateau, Bundelkhand, and Baghelkhand, and it slopes from southwest to northeast.
- Key rivers draining the Central Highlands:
 - ✓ Chambal
 - ✓ Sind
 - ✓ Betwa
 - ✓ Ken
- All these rivers:
 - ✓ Originate in or near the Vindhyan and Malwa regions
 - ✓ Flow northeastward
 - ✓ Ultimately join the Yamuna River

Q 50. Ans. D

Explanation:

Devprayag is correct:

- The Bhagirathi River officially takes the name Ganga at Devprayag.

- Location: Tehri Garhwal district, Uttarakhand.
- Confluence at Devprayag:
 - ✓ Bhagirathi River
 - ✓ Alaknanda River
- After this confluence, the combined river is known as the Ganga River.

Key identification fact:

- Devprayag is the last and most important confluence of the Panch Prayag system.

Haridwar:

- At Haridwar, the Ganga enters the Northern Plains.
- Geographic importance: Marks the end of mountainous course and start of plains flow.
- Haridwar significance: Important religious and physiographic milestone, but not naming point.

Vishnuprayag:

- Confluence at Vishnuprayag:
 - ✓ Alaknanda + Dhauliganga
- It is the first confluence in the Panch Prayag sequence.
- Located in Chamoli district, Uttarakhand.

Gangotri:

- Gangotri is a source region, not a confluence point.
- Bhagirathi originates from Gangotri Glacier at Gaumukh.
- The river retains the name Bhagirathi until Devprayag.

Additional Facts:

Panch Prayag Sequence (North to South)

- Vishnuprayag: Alaknanda + Dhauliganga
- Nandaprayag: Alaknanda + Nandakini
- Karnaprayag: Alaknanda + Pindar
- Rudraprayag: Alaknanda + Mandakini
- Devprayag: Alaknanda + Bhagirathi → Ganga formed

Source Glaciers of Major Ganga Tributaries

- Bhagirathi:
 - ✓ Source: Gangotri Glacier
 - ✓ Exact origin point: Gaumukh
- Alaknanda:
 - ✓ Source: Satopanth Glacier
 - ✓ Located near Badrinath

- Mandakini:
 - ✓ Source: Chorabari Glacier
 - ✓ Associated shrine: Kedarnath

Q 51. Ans. B

Explanation:

- **Statement 1 is correct:** Jhelum originates from Verinag spring at Pir Panjal foothills (Kashmir valley) and traverses Srinagar-Wular Lake before Pakistan, forming part of the five Punjab rivers (Panjnad) that ultimately feed the Indus.
- **Statement 2 is correct:** Beas rises at Beas Kund near Rohtang Pass (4,000 m) in Himachal, carves Dhauladhar gorges, and meets Satluj at Harike, contributing to the perennial Himalayan drainage that irrigates Punjab plains via Bhakra-Nangal canals.
- **Statement 3 is incorrect:** Chenab is formed by Chandra-Bhaga at Tandi near Keylong (Himachal) but is actually the largest tributary of the Indus (1,180 km long, Chandrabhaga name and 504 km in India), not the smallest among Panchnad rivers (Jhelum, Ravi, Beas, Satluj).

Q 52. Ans. B

Explanation:

Doab: A tract of land lying between two rivers.

- The term Doab comes from Persian words:
 - ✓ Do means two
 - ✓ Ab means water
- Meaning: Land between two rivers.
- Geographic type: Interfluvial region between two river channels.
- Soil fertility: Very fertile due to regular deposition of alluvium during floods.
- Settlement pattern: Historically densely populated agricultural regions.

Additional Facts:

Major Doabs of India

- Ganga–Yamuna Doab
 - ✓ Location: Between Ganga and Yamuna rivers.
 - ✓ State: Uttar Pradesh.
 - ✓ Extent: From Shivalik foothills to Prayagraj.
- Sub-divisions:
 - ✓ Upper Doab: Meerut, Saharanpur, Aligarh

- ✓ Middle Doab: Agra, Mathura
- ✓ Lower Doab: Fatehpur, Kaushambi, Prayagraj
- Agricultural importance: Major region of wheat and sugarcane cultivation.

Punjab Doabs (River-Based Names)

- Bist Doab: Between Beas and Sutlej
- Bari Doab: Between Beas and Ravi
- Rechna Doab: Between Ravi and Chenab
- Chaj Doab: Between Chenab and Jhelum
- Sind Sagar Doab: Between Jhelum and Indus
 - ✓ Largest of all Punjab doab
 - ✓ Less fertile due to Thal Desert.

Raichur Doab (South India)

- Location: Between Krishna and Tungabhadra rivers.
- States: Karnataka and Telangana.
- Shape: Triangular region.

Historical importance:

- Frequent conflicts between Vijayanagara Empire and Bahmani Sultanate.

Economic importance:

- Rich in agriculture and mineral resources.

Q 53. Ans. C

Explanation:

- The **Eastern Ghats** are a discontinuous range of hills along the eastern coast of India. Unlike the Western Ghats, they are not very high or continuous.
- Jindhagada (1,690 metres) is the highest peak in the Eastern Ghats, located in Andhra Pradesh.
- **Anai Mudi**
 - ✓ Highest peak in South India (≈ 2695 m).
 - ✓ Located in Eravikulam National Park in Kerala.
- Gali Konda:
 - ✓ Peak in Andhra Pradesh (≈ 1643 m).
 - ✓ Located in the Eastern Ghats near Araku Valley.
- Mahendragiri:
 - ✓ Highest peak in Odisha (≈ 1501 m).
 - ✓ Located in the Eastern Ghats in Gajapati district.

Q 54. Ans. A

Explanation:

- Statement 1 is correct:
 - ✓ The distance between two longitudes decreases as we move towards the poles.
 - ✓ Reason: Longitudes are semicircles that converge at the North and South Poles.
 - ✓ Maximum distance between longitudes occurs at the Equator.
 - ✓ Distance between 1° longitude at Equator: About 111.3 km.
 - ✓ Distance between longitudes gradually decreases towards higher latitudes.
 - ✓ At the Poles, the distance between longitudes becomes zero because all longitudes meet.
- **Statement 2 is incorrect:**
 - ✓ Latitudes are parallel circles that never meet.
 - ✓ Distance between two latitudes is generally constant.
 - ✓ Distance between 1° latitude: About 111 km everywhere on Earth (standard textbook value).
- Important note:
 - ✓ Though Earth is slightly flattened at poles (oblate spheroid), this variation is very small and usually ignored in standard geography questions.
- Additional Facts:
 - ✓ Latitudes — Core Properties
 - Latitudes are parallel circles running east–west.
 - They never intersect each other.
 - Distance between successive latitudes: About 111 km.
 - Total number of latitudes: 180 (90° North + 90° South).
- Special latitude:
 - ✓ Equator (0° latitude) is the largest latitude.
 - ✓ It divides Earth into Northern and Southern Hemispheres.
- Longitudes — Core Properties
 - ✓ Longitudes are semicircles running north–south.
 - ✓ They converge at the poles.
 - ✓ Total number of longitudes: 360.
 - ✓ Distance pattern:
 - Maximum at Equator.
 - Decreases steadily toward poles.
 - Zero distance at poles.

● Great Circles — Key Facts

- ✓ Equator is the only latitude that forms a Great Circle.
- ✓ Any pair of opposite longitudes forms a Great Circle.
- ✓ Great Circles represent the shortest distance between two points on Earth.

Q 55. Ans. B

Explanation:

- Subansiri:
 - ✓ Subansiri is a right-bank tributary of the Brahmaputra River.
 - ✓ Origin: Tibet.
 - ✓ River type: Antecedent river cutting deep gorges through the Himalayas.
 - ✓ Status: Largest tributary of the Brahmaputra by volume.
 - ✓ Joining side: Joins Brahmaputra from the north (right bank).
- **Burhi Dihing:**
 - ✓ Burhi Dihing is a left-bank tributary of the Brahmaputra.
 - ✓ Origin: Patkai Bum Hills.
 - ✓ Flow region: Upper Assam plains.
 - ✓ River feature: Forms meanders and oxbow lakes.
 - ✓ Joining side: Joins from the south (left bank).
- Manas:
 - ✓ Manas River is a right-bank tributary of the Brahmaputra.
 - ✓ Origin: Bhutan Himalayas.
 - ✓ Ecological importance: Flows through Manas National Park.
 - ✓ National Park status:
 - UNESCO World Heritage Site
 - Tiger Reserve
- Joining side: Joins Brahmaputra from the north (right bank).
- **Dhansari:**
 - Dhansari River is a left-bank tributary of the Brahmaputra.
 - Origin: Laga Hills of Nagaland.
 - Flow region: Through Assam plains.
 - Joining side: Joins from the south (left bank).
- **Additional Facts:**
 - Right-Bank Tributaries of Brahmaputra (East to West)

- ✓ Subansiri
- ✓ Kameng
- ✓ Manas
- ✓ Sankosh
- ✓ Teesta
- Key boundary fact:
 - ✓ Sankosh River forms boundary between Assam and West Bengal.
- Historical fact:
 - ✓ Teesta River changed course after 1787 flood, shifting from Ganga basin to Brahmaputra basin.
- Left-Bank Tributaries of Brahmaputra (East to West)
 - ✓ Dibang
 - ✓ Lohit
 - ✓ Burhi Dihing
 - ✓ Dhansari
 - ✓ Kopili
- Infrastructure fact:
 - ✓ Bhupen Hazarika Setu (Dhola-Sadiya Bridge) is built across Lohit River.
- Majuli Island — Formation Facts
 - ✓ Majuli is the largest riverine island in the world.
 - ✓ Location: Assam.
- Formed between:
 - ✓ Main Brahmaputra River (south side)
 - ✓ Kherkutia Xuti channel joined by Subansiri River (north side).

Q 56. Ans. D

Explanation:

- Statement 1 is incorrect:
 - ✓ The Nine Degree Channel is located in the Arabian Sea, not the Bay of Bengal.
 - ✓ It separates Minicoy Island from the rest of the Lakshadweep Islands.
 - ✓ Correct separator for Andaman–Nicobar:
 - The Ten Degree Channel separates the Andaman Islands from the Nicobar Islands.
 - Exact location: Between Little Andaman (north) and Car Nicobar (south).
- Statement 2 is incorrect:
 - ✓ The Eight Degree Channel separates Minicoy Island (India) from the Maldives.
 - ✓ Correct separator within Lakshadweep:
 - The Eleven Degree Channel divides

Lakshadweep into:

- ◇ Amindivi Islands (north)
- ◇ Cannanore (Laccadive) Islands (south)

Additional Facts:

- Important Maritime Channels of India
 - ✓ Lakshadweep Region (Arabian Sea)
 - ✓ 8 Degree Channel:
 - Separates Maldives from Minicoy Island
 - ✓ 9 Degree Channel:
 - Separates Minicoy Island from the rest of Lakshadweep Islands
 - ✓ 11 Degree Channel:
 - Divides Amindivi Islands (north) and Cannanore Islands (south)
- Andaman and Nicobar Region (Bay of Bengal)
 - ✓ 10 Degree Channel:
 - Separates Andaman Islands and Nicobar Islands
 - ✓ Duncan Passage:
 - Separates South Andaman and Little Andaman
 - ✓ Coco Channel:
 - Separates North Andaman and Coco Islands (Myanmar)
 - ✓ Great Channel (Six Degree Channel):
 - Separates Great Nicobar and Sumatra (Indonesia)

India–Sri Lanka Separation:

- Palk Strait:
 - ✓ Lies between Tamil Nadu and Sri Lanka
- Gulf of Mannar:
 - ✓ Located south of Adam's Bridge (Ram Setu)
 - ✓ Lies between India and Sri Lanka

Q 57. Ans. C

Explanation:

Pir Panjal:

- Pir Panjal is the longest range of the Lesser Himalayas (Himachal Himalaya).
- The Lesser Himalayas consist of several important ranges, among which Pir Panjal is the longest and most significant.
- Geographically, it extends from the Jhelum River in the northwest to the upper Beas River region in Himachal Pradesh.
- It also forms a major natural barrier separating the Kashmir Valley from the plains to the south.

Dhaura Dhar:

- Dhaura Dhar is a prominent range of the Lesser Himalayas, mainly located in Himachal Pradesh.
- However, it is shorter than the Pir Panjal range.

Shiwalik:

- The Shiwalik Range belongs to the Outer Himalayas, not the Lesser Himalayas.
- It is the southernmost Himalayan range, composed mainly of loose sediments.
- Therefore, it cannot be the longest range of the Lesser Himalayas.

Zaskar:

- The Zaskar Range lies north of the Great Himalayas.
- It belongs to the Trans-Himalayan region, not the Lesser Himalayas.
- Hence, it is geographically unrelated to the Lesser Himalayan division.

Additional High-Value Facts:

- Lesser Himalayas — Major Ranges
 - ✓ The Lesser Himalayas include several important ranges:
 - Pir Panjal Range — Longest range
 - Dhaura Dhar Range — Prominent in Himachal Pradesh
 - Mahabharat Range — Extends through Nepal
- These ranges lie south of the Great Himalayas and north of the Shiwaliks.

Q 58. Ans. C

Explanation:

Statement 1 is correct:

- The Northeastern Plateau is an extension of the Peninsular Plateau.
- It was originally connected to the Chotanagpur Plateau.
- Separation occurred due to a major geological fault known as the Malda Fault (Garo–Rajmahal Gap).
- Location of fault: Between the Rajmahal Hills (Jharkhand) and the Meghalaya Plateau.
- The gap was later filled with alluvial deposits of the Ganga and Brahmaputra rivers, forming the plains of West Bengal and Bangladesh.

Statement 2 is correct:

- The Meghalaya Plateau is divided into three hill ranges arranged west to east:
 - ✓ Garo Hills (west)
 - ✓ Khasi Hills (central)
 - ✓ Jaintia Hills (east)
- The plateau is rich in mineral resources due to its ancient Gondwana rock structure.
- Major minerals found:
 - ✓ Coal
 - ✓ Limestone
 - ✓ Uranium

Additional Facts:

- Northeastern Plateau — Core Features
- Geological nature:
 - ✓ Ancient crystalline plateau similar to Peninsular Plateau rocks.
- Major plateaus included:
 - ✓ Meghalaya Plateau
 - ✓ Karbi Anglong Plateau
 - ✓ North Cachar Hills
- Important classification fact:
 - ✓ These plateaus are not part of the Himalayas.

Malda Fault (Garo–Rajmahal Gap)

- Location: Between Rajmahal Hills and Garo Hills.
- Formation cause:
- Occurred during Himalayan uplift.
- Present condition:
 - ✓ Filled with sediments of Ganga–Brahmaputra river system.
 - ✓ Khasi Hills — Rainfall Mechanism
- Major rainfall locations:
 - ✓ Mawsynram
 - ✓ Cherrapunji
- Reason for heavy rainfall:
 - ✓ Orographic rainfall caused by southwest monsoon winds striking southern slopes of Khasi Hills.
- Geographical shape:
 - ✓ Hills form a funnel-like structure facing the Bay of Bengal.
- Mineral Resources of Meghalaya Plateau
 - ✓ Major coal mining region:
 - ✓ Jaintia Hills
 - ✓ Major uranium deposit:
 - Domiasiat region (West Khasi Hills)
- Mining issue:

- ✓ Practice of rat-hole mining historically used due to thin coal seams.

Q 59. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ The Himalayas are young fold mountains formed due to tectonic collision between the Indian Plate and the Eurasian Plate.
 - ✓ They are deeply dissected by fast-flowing antecedent rivers such as:
 - Indus
 - Satluj
 - Brahmaputra
- These rivers existed before mountain uplift and cut deep gorges while the Himalayas were rising.
- Statement 2 is incorrect:
 - ✓ Western Himalayas (Kashmir region): About 400 km wide
 - ✓ Eastern Himalayas (Arunachal Pradesh): About 150 km wide
 - ✓ Thus, the Himalayas are widest in the west and narrowest in the east, not the opposite.
- Statement 3 is correct:
 - ✓ Altitudinal variation is greater in the Eastern Himalayas than in the Western Himalayas.
 - ✓ In the east, the mountains rise abruptly from the plains of Assam and Bengal, producing sharp elevation changes.
 - ✓ In the west, the rise is more gradual through parallel ranges such as:
 - Shiwalik Range
 - Lesser Himalayas
 - Greater Himalayas
- Additional Facts:
 - ✓ Width Variation of the Himalayas
- General pattern:
 - ✓ Western Himalayas: Wider and more spread out
 - ✓ Eastern Himalayas: Narrower but steeper
- Reason:
 - ✓ Stronger tectonic compression occurred in the eastern sector during Himalayan formation.
- Antecedent Rivers of the Himalayas:
 - ✓ Meaning of antecedent river:
 - ✓ A river that existed before mountain uplift and maintained its course by erosion.

- Major antecedent rivers:
 - ✓ Indus
 - ✓ Satluj
 - ✓ Brahmaputra
 - Key feature:
 - ✓ Formation of deep gorges, especially where rivers cut across major ranges.
 - Syntaxial Bends of the Himalayas:
 - ✓ Western Syntaxial Bend:
 - ✓ Located near Nanga Parbat
 - ✓ Associated with the Indus River
 - ✓ Eastern Syntaxial Bend:
 - ✓ Located near Namcha Barwa
 - ✓ Associated with the Brahmaputra River
- These bends mark the terminal limits of the Himalayan arc.

Q 60. Ans. A

Explanation:

- Champion Series → (c)
 - ✓ Located in Kolar and Raichur districts of Karnataka.
 - ✓ Famous for gold mines, especially the historic Kolar Gold Fields (KGF).
- Champaner Series → (d)
 - ✓ Located near Vadodara in Gujarat.
 - ✓ Known for green marble deposits.
 - ✓ Represents a green marble-bearing outlier of the Aravalli Range.
- Chilpi Ghat Series → (b)
 - ✓ Located near Balaghat district in Madhya Pradesh.
 - ✓ Associated with copper ore deposits, supplying the Malanjkhand Copper Plant.
- Rihalo Series → (a)
 - ✓ Extends from Delhi to Alwar (Rajasthan).
 - ✓ Known for rich marble deposits, including Makrana marble.

Additional Facts:

Dharwar Rock System — Key Features
Geological age:

- Formed about 2.5 to 1.8 billion years ago.

Rock nature:

- Composed of metamorphosed sedimentary rocks.
- Derived from erosion of Archaean rocks.

Economic importance:

- Rich in metallic minerals, especially:

- ✓ Iron ore
- ✓ Gold
- ✓ Manganese
- ✓ Copper
- ✓ Tungsten

Important Dharwar Mineral Belts:

Champion Series (Karnataka):

- Gold deposits
- Includes Kolar Gold Fields and Raichur belt

Chilpi Ghat Series (Madhya Pradesh):

- Copper deposits
- Linked to Malanjkhand Copper Project

Champaner Series (Gujarat):

- Famous for green marble

Rihalo Series (Delhi–Alwar belt):

- Known for Makrana marble, widely used in historical monuments

Chronology of Major Indian Rock Systems:

(Oldest to Youngest)

- Archaean — Oldest crystalline rocks
- Dharwar — Early sedimentary rocks, metallic minerals
- Cuddapah — Limestone and related deposits
- Vindhyan — Sandstone and building materials
- Gondwana — Coal deposits
- Tertiary — Associated with Himalayan formation

Q 61. Ans. B

Explanation:

- Karewas:
 - ✓ Lacustrine deposits: Kashmir Valley
 - ✓ Location: Slopes of Pir Panjal Range
 - ✓ Composition
 - Glacial clay: Sand, gravel, silt
 - ✓ Formation
 - Ancient lake: Vast Lake Shumar
 - Event: Entire Kashmir Valley submerged
 - Tectonic uplift: Pir Panjal uplifted
 - Drainage: Jhelum River drained lake
 - Result: Flat-topped, terrace-like mounds
 - ✓ Geological Age
 - Epoch: Pleistocene
 - Time span: 2.6 million to 11,700 years ago
 - ✓ Economic Importance
 - Major crop: Zafran (Saffron)

- Varieties: Lachha, Guchhi
- Other crops: Almonds, Walnuts, Apples
- ✓ Jhelum Factor
 - Role: Drain of Kashmir Valley
 - Exit point: Baramulla gorge

Additional Information

- ✓ Zaskar Range: Northeast of Kashmir Valley: No Karewas
- ✓ Karakoram Range: Trans-Himalayan, north: Glacial, no lacustrine deposits
- ✓ Kumaon Range: Uttarakhand:
- ✓ Recent Relevance
 - GI Tag: Kashmir saffron protection against Iranian adulteration
 - Infrastructure: Udhampur-Srinagar-Baramulla Rail Link
 - Impact: Karewa erosion, landslide risk

Q 62. Ans. A

Explanation:

- Bhore Ghat
 - ✓ Location: Western Ghats Maharashtra
 - ✓ Connects: Mumbai – Pune
 - ✓ Route: Mumbai–Chennai rail and road
- Goran Ghat
 - ✓ Location: Aravalli Range Rajasthan
 - ✓ Connects: Udaipur – Sirohi – Jalore
- Palghat Gap
 - ✓ Other name: Palakkad Gap
 - ✓ Location: Western Ghats
 - ✓ Between: Nilgiri Hills (north) – Anaimalai Hills (south)
 - ✓ Connects: Coimbatore – Kochi/Kozhikode
- Thal Ghat
 - ✓ Other name: Kasara Ghat
 - ✓ Location: Western Ghats, Maharashtra
 - ✓ Connects: Mumbai – Nashik
 - ✓ Role: Gateway to North India
- Additional Information
 - ✓ Western Ghats: Continuous range: passes are major natural breaks
 - ✓ Major passes: Thal, Bhore, Palghat
 - ✓ Climatology
 - Palghat Gap: Allows Southwest Monsoon penetration into western Tamil Nadu
 - Impact: Influences rainfall in Coimbatore region
- Other Important Pass
 - ✓ Senkottai Gap: Connects Quilon – Madurai

- Recent Relevance
 - ✓ NH-48: Expansion near Bhor Ghat
 - ✓ Project: “Missing Link” tunnels
 - ✓ Feature: Very wide tunnels to bypass steep gradients

Q 63. Ans. B

Explanation:

- Bara Lacha
 - ✓ Location: Zaskar Range
 - ✓ Route: Manali–Leh Highway
 - ✓ Altitude: ~5045 m
 - ✓ Connects: Lahaul – Leh (Ladakh)
 - ✓ Condition: Snow-covered Nov–mid May
- Bomdi La
 - Location: Arunachal Pradesh
 - ✓ Position: East of Bhutan
 - ✓ Connects: India – Lhasa (Tibet)
 - ✓ Condition: Closed in winter due to snowfall
- Burzail Pass
 - ✓ Location: Jammu & Kashmir
 - ✓ Connects: Srinagar – Kishanganga Valley
 - ✓ Significance: Historic route to Central Asia
 - ✓ Condition: Snow-covered in winter
- Lanak La
 - ✓ Location: Aksai Chin (Ladakh)
 - ✓ Altitude: ~5000 m
 - ✓ Connects: Ladakh – Lhasa
 - ✓ Status: Under Chinese control
 - ✓ Link: Road connects Xinjiang – Tibet
- Additional Information
 - ✓ “La”: Means pass in Tibetan language
 - ✓ Strategic Passes (Arunachal)
 - ✓ Bomdi La: Gateway to Tibet
 - ✓ Other passes: Diphu Pass, Yonggyap Pass
 - ✓ Geological Link
- Recent Relevance
 - ✓ BRO: Shinku La Tunnel construction
 - ✓ Purpose: All-weather Ladakh connectivity
 - ✓ Impact: Bypass Bara Lacha
 - ✓ Aksai Chin
 - ✓ Lanak La: Strategic sensitivity
 - ✓ Recent issue: Chinese infrastructure activity

Q 64. Ans. B

Explanation:

- Kumaon Himalaya:
 - ✓ Region: Between Sutlej – Kali rivers

- ✓ Location: Uttarakhand
- ✓ Includes: Garhwal and Kumaon divisions
- ✓ **Himalayan Regional Division**
 - Punjab (Kashmir) Himalaya
 - ◇ Rivers: Indus – Sutlej
 - ◇ Region: Jammu & Kashmir, Himachal Pradesh
 - ◇ Features: Karakoram, Zaskar
 - Nepal Himalaya
 - ◇ Rivers: Kali – Tista
 - ◇ Features: Everest, Annapurna
- ✓ Major Peaks (Uttarakhand: Himadri)
 - Nanda Devi
 - ◇ Height: 7,816 m
 - ◇ Fact: Highest peak entirely in India
 - Kamet
 - ◇ Height: 7,756 m
 - ◇ Location: Near Tibet border
 - Trishul
 - ◇ Height: 7,120 m
 - ◇ Type: Three peaks (trident shape)
 - Kedarnath Peak
 - ◇ Height: 6,940 m
 - ◇ Feature: Source region of Mandakini River
 - Key Concept
 - ◇ Himadri: Greater Himalayas
 - ◇ Feature: Highest peaks, permanent snow

Q 65. Ans. C

Explanation:

- Duncan Passage:
 - ✓ Type: Strait
 - ✓ Width: ~48 km
 - ✓ Location: Bay of Bengal
 - ✓ **Position**
 - North: Rutland Island (South Andaman)
 - South: Little Andaman
 - ✓ **Connectivity**
 - West: Bay of Bengal
 - East: Andaman Sea
 - Role: Maritime route between Bay of Bengal and Andaman Sea
 - ✓ **Geographical Features**
 - Includes islets: North Brother, South Brother, The Sisters
 - Nature: Deep-sea depression in Andaman–Nicobar chain

- ✓ Significance
 - Strategic: Key entry to Andaman Sea
 - Navigation: Important shipping route
 - Defense: Surveillance zone

Q 66. Ans. A

Explanation:

Statement 1 is correct:

- The Great Himalayas (Himadri) form the northernmost range of the Himalayan system.
- It is also the most continuous range, extending almost uninterrupted across the entire Himalayan arc.
- Though largely continuous, it is cut across by deep gorges formed by antecedent rivers such as:
 - ✓ Indus
 - ✓ Satluj
 - ✓ Brahmaputra

Statement 2 is correct:

- The average height of the Himadri is about 6,000 metres, making it the highest Himalayan range.
- It contains many of the world's highest peaks, including several prominent Himalayan summits.

Important peaks (west to east sequence):

- Nanga Parbat — western anchor of the Himalayas
- Nanda Devi — highest peak located entirely within India
- Dhaulagiri
- Annapurna
- Mount Everest — highest peak in the world
- Makalu
- Kanchenjunga — highest peak of India (excluding peaks in Pakistan-occupied regions)
- Namcha Barwa — eastern anchor of the Himalayan arc

Statement 3 is incorrect:

- The core of the Great Himalayas is not composed of sedimentary rocks.
- Instead, it consists mainly of hard crystalline rocks, particularly:
 - ✓ Granite
 - ✓ Gneiss

Correct placement of sedimentary rocks:

- Sedimentary formations are mainly found in the Trans-Himalayan (Tethys) zone, located north of the Himadri.
- These rocks originated from deposits of the ancient Tethys Sea, and marine fossils are found in this region.

Additional Facts:

- Rock Composition of Himalayan Zones
- Great Himalayas (Himadri):
 - ✓ Rock type: Granite and crystalline rocks
 - ✓ Nature: Very hard and resistant
- Trans-Himalayas (Tethys zone):
 - ✓ Rock type: Sedimentary rocks
 - ✓ Special feature: Presence of marine fossils
- Shiwalik Range:
 - ✓ Rock type: Unconsolidated sediments such as gravel and alluvium
 - ✓ Origin: Deposited by Himalayan rivers

Slopes of the Great Himalayas:

- Southern slope (towards India): Very steep and abrupt
- Northern slope (towards Tibet): Comparatively gentle
- This asymmetry is a characteristic structural feature of the Great Himalayan range.

Q 67. Ans. B

Explanation:

- Malda Gap (Garo–Rajmahal Gap): Key Facts
 - ✓ Type: Tectonic depression
 - ✓ Nature: Low-lying structural gap
 - ✓ **Location**
 - Separates: Meghalaya Plateau – Chotanagpur Plateau
 - West: Rajmahal Hills (Jharkhand)
 - East: Garo Hills (Meghalaya)
 - ✓ **Geological Context**
 - Meghalaya Plateau: Extension of Peninsular Plateau
 - ✓ **Formation**
 - Cause: Faulting during Himalayan orogeny
 - Process: Land subsidence between Rajmahal–Garo hills
 - Later fill: Alluvial deposits of Ganga–Brahmaputra system

- ✓ **Result**
- ✓ Feature: Flat plains of West Bengal and Bangladesh
- ✓ Effect: Detachment of Meghalaya Plateau from Peninsular block
- ✓ **Structural Link**
 - Break between: Rajmahal Hills – Garo Hills
 - Role: Physiographic boundary and transition zone
- ✓ Geological Evidence
 - Rock type similarity: Archaean gneiss, schist
 - Inference: Meghalaya + Chotanagpur once continuous landmass
- ✓ Meghalaya Plateau Structure
 - Hills (W→E): Garo – Khasi – Jaintia

Q 68. Ans. A

Explanation:

- North Eastern Plateau: Key Facts
 - ✓ Other name: Chotanagpur Plateau region
 - ✓ Type: Part of Peninsular Plateau
 - ✓ Location: East-Central India
 - ✓ **Geographic Extent**
 - Covers: Jharkhand, western West Bengal, Odisha, Chhattisgarh
 - ✓ Major Features
 - Plateaus: Hazaribagh, Ranchi
 - Hills: Rajmahal Hills
 - ✓ Drainage
 - Rivers: Damodar, Subarnarekha, Brahmani
 - ✓ Economic Importance
 - Nickname: Ruhr of India
 - Minerals: Coal, Iron ore, Manganese
 - Industrial centres: Jamshedpur, Bokaro, Rourkela
 - **Tectonic Link**
 - ◇ Past: Continuous with Meghalaya Plateau
 - ◇ Separation: Malda Gap (Garo–Rajmahal Gap)
- Additional Information
 - ✓ Central Highlands
 - States: UP, MP, Rajasthan
 - Region: Vindhyan belt
 - ✓ Deccan Plateau
 - States: Maharashtra, Telangana, Andhra Pradesh, Karnataka

- Boundary: South of Satpura
- ✓ Western/Central mix
 - States: Gujarat, MP, Rajasthan, UP
 - Region: Malwa Plateau, Aravalli influence

Q 69. Ans. B

Explanation:

- State-wise Data (Ministry of Ports Shipping and Waterways)
 - ✓ Gujarat
 - Coastline: 2,340.62 km
 - Rank: 1
 - ✓ Tamil Nadu
 - Coastline: 1,068.69 km
 - Rank: 2 (mainland)
 - ✓ Andhra Pradesh
 - Coastline: 1,053.07 km
 - Rank: 3
 - ✓ Maharashtra
 - Coastline: 877.97 km
 - Rank: 4
 - ✓ Key Update
 - Change: Tamil Nadu surpassed Andhra Pradesh
 - Reason: GIS-based reassessment
 - ✓ Methodology
 - Technique: GIS, high-resolution mapping
 - Reference line: High-Water Line (HWL)
 - Policy Impact
 - **Coastal Regulation Zone**
 - ◇ No change: Boundaries unchanged
 - ◇ Basis: High Tide Line (HTL)
 - ◇ Agency: National Centre for Sustainable Coastal Management
 - Strategic Importance
 - ◇ Directive: National Security Council Secretariat
 - ◇ Use: EEZ, territorial water management

Q 70. Ans. B

Explanation:

- Deccan Traps:
 - ✓ Type: Volcanic plateau
 - ✓ Region: West-central India
 - ✓ Name Origin

- ✓ “Trap”: Swedish word “trappa”
- ✓ Meaning: Stairs/steps
- ✓ **Formation**
 - Period: End of Cretaceous (~66 million years ago)
 - Process: Fissure eruptions
- ✓ Nature: Lava from cracks, not central cone
- ✓ Rock Type
 - Basaltic lava flows
 - Landscape: Step-like terraces
- ✓ Soil Formation
 - Soil: Black soil (Regur)
 - Minerals: Iron, magnesium, aluminum
 - Property: High moisture retention
 - Thickness
 - ◇ Maximum: Western Ghats (~2000 m)
 - Trend: Thins eastward and northward
 - Economic Importance
 - ◇ Crop: Cotton
 - Regions: Maharashtra, Gujarat
- ✓ Geological Link
 - Cause: Reunion Hotspot
- ✓ Additional Information
- ✓ Coastal Plains
 - Origin: Alluvial deposition
 - Rivers: Godavari, Krishna, Mahanadi, and Cauvery
- ✓ Fold Mountains
 - Example: Himalayas
 - Process: Plate collision (orogenic)
- ✓ Sedimentary Basins
 - Examples: Vindhyan, Cuddapah
 - Rocks: Limestone, sandstone
- ✓ Key Concepts
 - Process type: Volcanic
 - Not: Sedimentary or fold mountain
- ✓ Recent Relevance
 - Critical Minerals
 - Area: Deccan margins
 - Focus: Lithium exploration
- ✓ Carbon Sequestration
 - Method: Enhanced Rock Weathering
 - Material: Basalt dust
 - Effect: CO₂ capture as carbonates
- ✓ Type: Submarine plateau
- ✓ Location: Continental shelf, south of Kanniyakumari
- ✓ Area: ~10,000 sq km
- ✓ Oceanography
 - Feature: Upwelling of nutrient-rich waters
 - Result: Highly fertile fishing ground
- ✓ Economic Importance
 - Sector: Fisheries
 - Hydrocarbon: Oil and gas exploration blocks
- ✓ Policy Link
 - Scheme: Hydrocarbon Exploration and Licensing Policy (HELP)
 - Ministry: Ministry of Petroleum and Natural Gas
- ✓ Coastal Link
 - State: Tamil Nadu
 - Coastline: 1,068.69 km (2025 revision)
- ✓ Ecological Importance
 - Adjacent: Gulf of Mannar
 - Biodiversity: 60+ fish species, coral ecosystems
- Additional Information
 - ✓ Gulf of Kachchh
 - Location: Gujarat
 - Feature: Longest coastline (2,340.62 km)
 - Focus: Petroleum refining
 - ✓ Gulf of Khambhat
 - Location: Gujarat
 - Focus: Tidal energy, Kalpasar Project
 - ✓ Chilika Coast
 - Location: Odisha
 - Feature: Chilika Lake (Ramsar site)
 - Focus: Olive Ridley turtles
 - ✓ Maritime Jurisdiction
 - Territorial Waters
 - Limit: 12 nautical miles
 - ✓ Contiguous Zone
 - Limit: 24 nautical miles
 - ✓ Exclusive Economic Zone (EEZ)
 - Limit: 200 nautical miles
 - Status: Wadge Bank lies within EEZ
 - ✓ Coastline Revision (2025)
 - Old: 7,516.6 km
 - New: 11,098.81 km
 - Agency: National Hydrographic Office
 - Notified by: Ministry of Home Affairs

Q 71. Ans. C

Explanation:

- Wadge Bank:

Q 72. Ans. C

Explanation:

- Location: Gulf of Mannar
 - ✓ Kariyachalli Island: Key Facts
 - ✓ Type: Small, uninhabited offshore island
 - ✓ Group: Tuticorin group
 - ✓ State: Tamil Nadu
 - ✓ Protected Area
 - Park: Gulf of Mannar Marine National Park
 - Biosphere: Part of Gulf of Mannar Biosphere Reserve
 - Total islands: 21
 - ✓ Ecosystem
 - Features: Coral reefs, seagrass meadows
 - Importance: High marine biodiversity
 - ✓ State Coastal Data (2025)
 - Tamil Nadu coastline: 1,068.69 km
 - Rank: 2nd longest mainland coastline
 - Authority: Ministry of Ports Shipping and Waterways
- Additional Information
- Gulf of Kachchh
 - ✓ Location: Gujarat
 - ✓ Key islands: Pirotan, Narara
 - ✓ Feature: Marine National Park (Jamnagar)
- Gulf of Khambhat
 - ✓ Location: Gujarat
 - ✓ Islands: Alia Bet, Piram
 - ✓ Feature: High tides, heavy siltation
- Palk Bay
 - ✓ Location: North of Gulf of Mannar
 - ✓ Famous island: Pamban (Rameswaram)
 - ✓ Feature: Shallow, sheltered waters
- Key Concepts
- Biosphere Reserve
 - ✓ Year: 1989
 - ✓ Status: India's first Marine Biosphere Reserve
- Species
 - ✓ Dugong: Only herbivorous marine mammal in India
 - ✓ IUCN status: Vulnerable
- Coastal Regulation Zone
 - ✓ No change in boundaries
 - ✓ Basis: High Tide Line (HTL)
 - ✓ Agency: National Centre for Sustainable Coastal Management
- Recent Relevance
 - ✓ Policy: Blue Economy 2.0
 - ✓ Focus: Coral restoration, climate adaptation

Q 73. Ans. C

Explanation:

- Mount Koyob:
 - ✓ Location: South Andaman, Andaman and Nicobar Islands
 - ✓ Height: ~460 m
 - ✓ Rank: 4th highest peak (archipelago)
 - ✓ Major Peaks (North → South Trend)
- Saddle Peak
 - ✓ Location: North Andaman
 - ✓ Height: 732 m
 - ✓ Rank: Highest in archipelago
- Mount Diavolo
 - ✓ Location: Middle Andaman
 - ✓ Height: 515 m
- Mount Harriet
 - ✓ Location: South Andaman
 - ✓ Height: 365 m
- Mount Thullier
 - ✓ Location: Great Nicobar
 - ✓ Height: 642 m
 - ✓ Rank: Highest in Nicobar group
- Key Pattern
 - ✓ Trend: Height decreases north → south (Andaman group)

Q 74. Ans. D

Explanation:

Statement 1 is incorrect:

- Lakshadweep Islands are of coral origin, not tectonic mountain origin.
- They consist mainly of coral atolls and reefs, formed by the accumulation of calcium carbonate from coral organisms over submerged ridges.
- Andaman and Nicobar Islands: Tectonic origin; elevated portions of submarine mountains, forming an extension of the Arakan Yoma range of Myanmar.

Statement 2 is incorrect:

- Ten Degree Channel: Separates Andaman Islands from Nicobar Islands.
- Lakshadweep Islands:
 - ✓ Divided by:
 - Nine Degree Channel: Separates Minicoy Island from the rest of Lakshadweep.
 - Eleven Degree Channel: Separates

Amindivi Islands (north) from Cannanore (Laccadive) Islands (south).

Additional Facts:

- Lakshadweep Islands — Key Features
- Origin:
 - ✓ Entirely coral-based islands, mainly atolls with surrounding lagoons.
- Formation:
 - ✓ Built by coral polyps depositing calcium carbonate skeletons over submerged ridges.
- Water availability:
 - ✓ No rivers or streams due to porous coral limestone.
 - ✓ Freshwater obtained from underground freshwater lenses.

Important Islands of Lakshadweep:

- Kavaratti:
 - ✓ Administrative capital of Lakshadweep.
- Agatti Island:
 - ✓ It contains the only airport in Lakshadweep.
- Andrott Island:
 - ✓ Largest island in Lakshadweep by area.
- Pitti Island:
 - ✓ The uninhabited island was declared a Bird Sanctuary.

Minicoy Island — Special Features:

- Location:
 - ✓ Southernmost island of Lakshadweep.
- Channel separation:
 - ✓ 9 Degree Channel: Separates Minicoy from main Lakshadweep group.
 - ✓ 8 Degree Channel: Separates Minicoy from Maldives.
- Language distinction:
 - ✓ Malayalam is spoken across Lakshadweep.
 - ✓ Mahl language is spoken in Minicoy, similar to Divehi of Maldives.

Island Origin Comparison

- Lakshadweep Islands:
 - ✓ Coral origin
 - ✓ Atoll formation
- Andaman and Nicobar Islands:
 - ✓ Tectonic origin
 - ✓ Extension of submarine mountain ranges
 - ✓ Includes volcanic islands such as Barren Island.

Q 75. Ans: B

Explanation:

- ✓ Patkai Bum:
- ✓ Type: Easternmost extension of the Himalayas
- ✓ Region: Arunachal Pradesh, Nagaland
- ✓ Boundary: India – Myanmar
- ✓ **Himalayan Structure**
 - Feature: Syntaxial bend (sharp southward turn)
 - Extension: Purvanchal Hills (Eastern Highlands)
 - Position: Northernmost range of Purvanchal
- ✓ Geology
 - Rock type: Sandstone
 - Nature: Folded hills
- Additional Information
- Pir Panjal Range
 - ✓ Type: Lesser Himalayas
 - ✓ Region: J&K, Himachal Pradesh
 - ✓ Feature: Longest range of the Lesser Himalayas
- Zaskar Range
 - ✓ Type: Trans-Himalaya
 - ✓ Location: North of the Great Himalayas
- Dhauladhar Range
 - ✓ Type: Lesser Himalayas
 - ✓ Region: Himachal Pradesh
- Key Concept
 - ✓ Eastern Himalayas: Turn south → from Purvanchal Hills
- Strategic Importance
 - ✓ Terrain: Rugged, forested
 - ✓ Use: Mountain passes for connectivity and defence

Q 76. Ans: C

Explanation:

- Statement 1 is correct:
 - ✓ The Gandak River is formed by the confluence of the Kali Gandaki and Trishuli (Trishulganga) rivers.
 - ✓ These rivers originate in the high Himalayas of Nepal.
 - ✓ In Nepal, the Gandak is also known as the Narayani River.
- Statement 2 is correct:

- ✓ The Gandak River enters the Ganga plains in the Champaran region of Bihar, specifically near Triveni in West Champaran district.
- ✓ After flowing through northern Bihar, it joins the Ganga River near Sonpur, opposite Patna.
- ✓ This confluence site is culturally and geographically significant.
- Additional Facts:
 - ✓ Gandak River:
 - Origin:
 - ◇ Formed by the Kali Gandaki and Trishuli rivers in the Nepal Himalayas
 - ✓ Alternate name in Nepal:
 - Narayani River
 - ✓ Entry into India:
 - Triveni, West Champaran district, Bihar
 - ✓ Confluence with Ganga:
 - Sonpur, near Patna (Bihar)

Cultural and Ecological Significance:

- Saligram Stones:
 - ✓ Found in the Kali Gandaki riverbed.
 - ✓ These are fossil ammonites, considered sacred in Hindu worship.
- Valmiki National Park:
 - ✓ Located near the Gandak entry point in West Champaran.
 - ✓ It is the only National Park in Bihar.
- Sonpur Mela:
 - ✓ Held near the Gandak–Ganga confluence.
 - ✓ Known as one of the largest traditional animal fairs in Asia.
- Position Among Ganga Tributaries
 - ✓ The Gandak is a left-bank tributary of the Ganga.
 - ✓ Sequence of major left-bank tributaries from west to east:
 - Ramganga
 - Gomti
 - Ghaghara
 - Gandak
 - Kosi
 - Mahananda

Q 77. Ans. B

Explanation:

- Amarkantak:
 - ✓ Location: Anuppur district, Madhya Pradesh

- ✓ Coordinates: 22°40' N, 81°45' E
- ✓ Elevation: ~1,048 m
- ✓ Physiographic Setting
 - Meeting point: Vindhya Range – Satpura Range
 - Fulcrum: Maikal Hills
 - Position: Maikal Range, Central India
- ✓ Drainage Pattern
 - Type: Radial drainage
- ✓ River Origins
 - Narmada River: Flows west to Arabian Sea
 - Son River: Flows north-east
 - Johila River: Tributary of Son
- ✓ Environment
 - Forest: Dense Sal forests
- ✓ Geographical Extent
 - Plateau extends into: Chhattisgarh
- ✓ Significance
 - Title: Teerthraj (King of Pilgrimages)
- ✓ Recent Developments
 - Mining: Restricted
 - Eco-tourism: Sustainable pilgrimage initiative
 - Focus: Source protection, plastic-free zones

Q 78. Ans. A

Explanation:

- India–Bangladesh Border
 - ✓ Length: 4,096.7 km
 - ✓ Rank: Longest border India shares with any country
 - ✓ Nature
 - Pattern: Zig-zag
 - Terrain: Plains, riverine belts, hills
 - ✓ Bordering Indian States
 - West Bengal
 - ◇ Length: ~2,217 km (longest share)
 - ◇ Features: Sundarbans, Gangetic delta
 - Tripura
 - ◇ Border: Surrounded on three sides (N, W, S)
 - Meghalaya
 - ◇ Border: Along Garo, Khasi, Jaintia Hills (southern edge)
 - Mizoram
 - ◇ Region: Southeastern end
 - ◇ Feature: Chittagong Hill Tracts

- Assam
 - ◇ Segments: Barak Valley, Dhubri region
- Additional Information
 - ✓ Bihar
 - No border with Bangladesh
 - ✓ Nagaland
 - Border: Myanmar
 - ✓ Manipur
 - Border: Myanmar
 - ✓ Key Concept
 - Siliguri Corridor: Connects Northeast India with mainland India
 - ✓ Recent Developments
 - Border Security
 - ◇ Agency: Border Security Force
 - ◇ Technology: Smart fencing, BOLD-QIT
 - ◇ Areas: Dhubri (Assam), West Bengal
 - Infrastructure
 - ◇ Project: Agartala–Akhaura rail link
 - ◇ Impact: Reduced travel time to Kolkata
 - ◇ Policy: Act East connectivity

- Himalaya extension)
 - Link: Barail Range connects Himalayas → Northeast hills
- ✓ State-wise Highest Peaks (NE India)
- ✓ Manipur
 - Peak: Mount Iso
 - Height: 2,994 m
- ✓ Nagaland
 - Peak: Mount Saramati
 - Height: 3,826 m
 - Location: India–Myanmar border
- ✓ Mizoram
 - Peak: Phawngpui (Blue Mountain)
 - Height: 2,157 m
 - Range: Lushai Hills
- ✓ Meghalaya
 - Peak: Shillong Peak
 - Height: 1,965 m
 - Region: Shillong Plateau
- Key Concepts
- Syntaxial Bend
 - ✓ Feature: Himalayas bend southward
 - ✓ Result: Formation of Purvanchal Hills
- Watershed Role
 - ✓ Divides: Brahmaputra basin – Chindwin (Irrawaddy) basin
- Mount Iso vs Mount Saramati
 - ✓ Fact: Saramati much higher (>3,800 m)

Q 79. Ans. A

Explanation:

- **Statement 1 is correct:** The Eastern Ghats are discontinuous and low hills. They have been highly eroded by rivers such as the Mahanadi, Godavari, Krishna and Kaveri, which cut across them and break their continuity.
- **Statement 2 is incorrect:** Kalsubai and Kudremukh are not part of the Eastern Ghats. They belong to the Western Ghats. The Eastern Ghats include ranges like Javadi Palconda Nallamala and Mahendragiri hills.

Q 80. Ans. B

Explanation:

- Mount Iso (Tenpu/Tempu):
 - ✓ Location: Senapati district, Manipur
 - ✓ Height: ~2,994 m
 - ✓ Rank: Highest peak of Manipur
 - ✓ Range: Barail Range
 - ✓ Position: Near Nagaland border
 - ✓ Physiographic Context
 - System: Purvanchal Hills (Eastern

Q 81. Ans. A

Explanation:

- Hispara Glacier
 - ✓ Location: Karakoram Range
 - ✓ Length: ~49 km
 - ✓ Feature: Meets Biafo Glacier at Hispar La
- Rimo Glacier
 - ✓ Location: Rimo Muztagh (Karakoram), Ladakh
 - ✓ Drainage: Feeds Shyok River
- Gangotri Glacier
 - ✓ Location: Uttarkashi, Uttarakhand
 - ✓ River source: Bhagirathi River
 - ✓ Significance: Main source of Ganga system
- Zemu Glacier
 - ✓ Location: Sikkim
 - ✓ Position: Base of Kanchenjunga
 - ✓ River: Feeds Teesta River
 - ✓ Length: ~26 km
 - ✓ Status: Largest in Eastern Himalaya

Q 82. Ans. D

Explanation:

- West → East Sequence
 - ✓ Mahadev Range → Maikal Range → Chotanagpur Plateau → Khasi Hills
- ✓ Longitude Order
 - Mahadev Range
 - Longitude: ~78° E
 - Region: Satpura Range, Madhya Pradesh
- ✓ Maikal Range
 - Longitude: ~81° E
 - Region: MP–Chhattisgarh border
 - Link: Eastern Satpura extension
 - Feature: Includes Amarkantak Plateau
 - Rivers: Narmada, Son origin
- ✓ Chotanagpur Plateau
 - Longitude: ~85° E
 - Region: Jharkhand, West Bengal
 - Nickname: Ruhr of India
 - Resources: Coal, Iron, Manganese
 - Focus: Lithium, Cobalt exploration
- ✓ Khasi Hills
 - Longitude: ~91° E
 - Region: Meghalaya (Shillong Plateau)
- ✓ Geological Link
- ✓ Chotanagpur Plateau + Khasi Hills
 - Origin: Same Peninsular block
 - Separated by: Malda Gap (Rajmahal–Garo Gap)
 - Cause: Faulting + subsidence
 - Filling: Ganga–Brahmaputra alluvium

Q 83. Ans. C

Explanation:

- Kanchenjunga
 - ✓ Location: Sikkim–Nepal border
 - ✓ Rank: 3rd highest peak in world
 - ✓ Status: Highest peak in India (territory-wise)
- Kamet
 - ✓ Location: Chamoli, Uttarakhand
 - ✓ Range: Garhwal Himalaya
 - ✓ Rank: 2nd highest in Garhwal (after Nanda Devi)
- Namcha Barwa
 - ✓ Location: Tibet (near Arunachal Pradesh)
 - ✓ Feature: Eastern syntaxial bend of the Himalaya
 - ✓ River: Brahmaputra (Dihang) U-turn

- Saser Kangri
 - ✓ Location: Ladakh
 - ✓ Range: Saser Muztagh (Karakoram)
 - ✓ Feature: Easternmost Karakoram subrange

Q 84. Ans. B

Explanation:

- Delhi Ridge:
 - ✓ Extension: Northernmost extension of Aravalli Range
 - ✓ Role: Barrier between Thar Desert and Indo-Gangetic Plain
 - ✓ Nickname: Green lungs of Delhi
 - ✓ Includes: Asola Bhatti Wildlife Sanctuary
- **Aravalli Range:**
 - ✓ Age: Proterozoic (very old fold mountains)
 - ✓ Extent: ~800 km
 - ✓ Start: Himmatnagar (Gujarat)
 - ✓ End: Delhi Ridge
 - ✓ Highest peak: Guru Shikhar (1,722 m), Mount Abu
 - ✓ Rock type: Quartzite, metamorphic
- Additional Information
 - ✓ Vindhya Range
 - ✓ Location: Central India (MP)
 - ✓ Role: North boundary of Narmada
- **Satpura Range**
 - ✓ Location: Between Narmada and Tapi
- Saurashtra Hills
 - ✓ Location: Gujarat (Kathiawar)
 - ✓ Origin: Volcanic

Q 85. Ans. C

Explanation:

Statement 1 is correct:

- Mount Saramati (3,826 m) is the highest peak of Nagaland.
- It lies in the Naga Hills along the India–Myanmar border.
- It is the highest point not only of the Naga Hills but of the entire state of Nagaland.

Statement 2 is correct:

- Phawngpui (2,157 m) is the highest peak of Mizoram.
- It is located in the Lushai Hills (Mizo Hills).
- The peak lies within Phawngpui National Park, a protected ecological area.

- It is widely known by its alternate name, Blue Mountain.

Additional Facts:

- Important Peaks of Northeastern India
- Mount Saramati:
- Location: Naga Hills, Nagaland–Myanmar border
- Height: 3,826 m
- Status: Highest peak of Nagaland

Phawngpui (Blue Mountain):

- Location: Lushai (Mizo) Hills, Mizoram
- Height: 2,157 m
- Protected area: Phawngpui National Park

Betlingchhip (Tripura):

- Location: Jampui Hills
- Status: Highest peak of Tripura

Purvanchal Hills — North to South Sequence

The hill ranges forming the eastern extension of the Himalayas follow this order:

- Patkai Bum — Forms a boundary with Myanmar (Arunachal Pradesh region)
- Naga Hills — Includes Mount Saramati
- Manipur Hills — Surround the central Manipur basin
- Mizo (Lushai) Hills — Includes Phawngpui peak

Additional Regional Features:

Mizoram — Molasses Basin:

- Region composed of soft sedimentary rocks such as sandstone and shale.
- Creates rolling, highly folded terrain.

Loktak Lake (Manipur):

- Largest freshwater lake in Northeast India.
- Contains floating vegetation islands called phumdis.
- Houses Keibul Lamjao National Park, the only floating national park in the world.

Q 86. Ans. C

Explanation:

- Highest Peak (A&N Archipelago)
 - ✓ Name: Saddle Peak
 - Location: North Andaman

- Height: 732 m
- Park: Saddle Peak National Park
- ✓ Nicobar Group Peak
 - Name: Mount Thullier
 - Location: Great Nicobar
- ✓ Trend: Height decreases north → south (Andaman group)
- ✓ Vegetation
 - Type: Moist tropical evergreen forests (Saddle Peak region)

Q 87. Ans. C

Explanation:

- Katchal Island:
 - ✓ Location: Central Nicobar group
 - ✓ Union Territory: Andaman and Nicobar Islands
 - ✓ Sea Region
 - Situated in: Andaman Sea
 - Position: East of Bay of Bengal boundary
 - ✓ Geographical Context
 - North: Camorta, Nancowry Islands
 - Group: Central Nicobar
 - ✓ Topography
 - Nature: Relatively flat
 - Impact: Highly vulnerable to 2004 tsunami
 - ✓ Key Channels
- Ten Degree Channel
 - ✓ Separates: Andaman Islands – Nicobar Islands
- Sombrero Channel
 - ✓ Separates: Central Nicobar – Little & Great Nicobar

Q 88. Ans. C

Explanation:

Statement 1 is correct:

- The Konkan Coast extends from south of Surat (near Daman) in Gujarat to the Goa region.
- It forms the northern section of the Western Coastal Plains.
- This coastal stretch mainly covers Maharashtra and Goa.

Statement 2 is correct:

- The Konkan Coast is separated from the upland Maharashtra plateau by the west-facing escarpment of the Western Ghats (Sahyadri Range).

- The Western Ghats rise steeply from the coastal plain, forming a sharp boundary between the narrow coast and the interior plateau.
- This steep rise is responsible for the narrow width of the Konkan coastal plain.

Additional Facts:

- Location of the Konkan Coast
- Overall stretch:
 - ✓ South of Surat (Gujarat) to Goa
- Regional coverage:
 - ✓ Coastal Maharashtra
 - ✓ Coastal Goa
- Adjacent boundaries:
 - ✓ West: Arabian Sea
 - ✓ East: Western Ghats (Sahyadris)
- Nature of the Konkan Coastline:
 - ✓ Type of coast:
 - Submergent coastline
 - ✓ Key characteristics:
- Deeply indented shoreline
 - ✓ Presence of estuaries and creeks (locally called Khadis)
 - ✓ Suitable for development of natural harbours
- Examples of major natural harbours:
 - ✓ Mumbai
 - ✓ Marmagao

Soil and Crops of the Konkan Region:

- Dominant soil type:
 - ✓ Laterite soil
- Reason for formation:
 - ✓ Heavy rainfall leading to intense leaching
- Important crops:
 - ✓ Coconut
 - ✓ Cashew
 - ✓ Mango (especially Alphonso mango)

Rivers of the Konkan Coast:

- Nature of rivers:
 - ✓ Short
 - ✓ Swift flowing
 - ✓ Mostly seasonal
 - ✓ Form estuaries instead of deltas
 - ✓ Examples:
 - Ulhas River
 - Mandovi River
 - Zuari River

Transport Feature:

- Konkan Railway:
 - ✓ Connects Roha (Maharashtra) to Thokur near Mangaluru (Karnataka)
 - ✓ Passes through numerous tunnels and bridges due to rugged terrain.

Q 89. Ans. A

Explanation:

- Khasi Hills
 - ✓ Location: Meghalaya
 - ✓ Part of: Shillong Plateau
 - ✓ Key places: Cherrapunji, Mawsynram
- Baba Budan Hills – Karnataka
 - ✓ Location: Chikkamagaluru district
 - ✓ Range: Western Ghats
 - ✓ Significance: First coffee cultivation in India
 - ✓ Highest peak: Mullayanagiri
- Girnar Hills
 - ✓ Location: Junagadh, Gujarat
 - ✓ Feature: Older than Himalayas
 - ✓ Importance: Hindu and Jain pilgrimage site
- Palani Hills
 - ✓ Location: Tamil Nadu
 - ✓ Type: Extension of Western Ghats
 - ✓ Famous place: Kodaikanal
- Key Supporting Facts
- Meghalaya Plateau
 - ✓ Components: Garo, Khasi, Jaintia Hills
- Rajasthan
 - ✓ Main range: Aravalli Range
- Andhra Pradesh
 - ✓ Major ranges: Nallamala, Palkonda (Eastern Ghats)

Q 90. Ans. C

Explanation:

- **Statement 1 is correct:** The Greater Himalayas and Lesser Himalayas are separated by the Main Central Thrust (MCT). This marks a major tectonic boundary formed during phases of Himalayan uplift.
- **Statement 2 is correct:** The Lesser Himalayas and Shiwaliks are separated by the Main Boundary Thrust (MBT). This boundary developed during later stages of mountain building.
 - ✓ The Himalayas evolved through multiple phases of uplift. Around 45 million years

ago, a major uplift created rugged ranges like Pir Panjal Dhauladhar and Mahabharat.

- ✓ Intermontane basins such as Kashmir valley and Kathmandu valleys were formed between these ranges.
- ✓ The Shiwaliks were formed by thick terrestrial sediments during the Miocene to Pleistocene period.
- ✓ These deposits reflect continuous uplift and erosion of the Himalayas.
- ✓ The Himalayan Front Fault (HFF) marks the boundary with the Northern Plains and remains tectonically active.

Q 91. Ans. A

Explanation:

- Assertion (A) is correct:
- The Indian Desert (Thar Desert) region was submerged under the sea during the Mesozoic era, particularly during the Jurassic period.
- This indicates that the present desert region once formed part of the ancient Tethys Sea basin.
- Geological studies confirm that large parts of western Rajasthan were covered by shallow marine waters in the past.
- Reason (R) is correct:
 - ✓ Marine deposits such as limestone and sandstone have been found around Brahmsar near Jaisalmer.
 - ✓ These deposits contain marine fossils such as:
 - Ammonites
 - Marine gastropods
 - Cephalopods
- These organisms lived only in marine environments, proving that the area was once under seawater.

Why Reason (R) correctly explains Assertion (A):

- The discovery of marine sedimentary rocks and fossils in the Brahmsar–Jaisalmer region provides direct geological evidence that the area was submerged under the sea.
- Therefore, the presence of marine deposits is the scientific basis for concluding that the Indian Desert region was once marine-covered.

Additional Facts:

- Marine Evidence in the Thar Desert
- Brahmsar and Kuldhara region (near Jaisalmer):
- Contains extensive marine sedimentary deposits.
- Fossils of ammonites and marine organisms confirm a marine origin.
- Geographical term:
- The submergence of land under seawater is known as marine transgression.
- Lathi Formation — Groundwater Significance:
- Location:
 - ✓ Found in the Jaisalmer basin.
- Rock type:
 - ✓ Jurassic-age porous sandstone.
- Importance:
 - ✓ Acts as a major underground aquifer, storing groundwater in the desert region.

Other Supporting Geological Evidence:

- Akal Wood Fossil Park (Jaisalmer):
- Contains petrified tree fossils about 180 million years old.
- Indicates that the region once supported lush vegetation before becoming arid.

Barmer Basin:

- Location:
 - ✓ Barmer–Sanchole basin, Rajasthan.
- Importance:
 - ✓ Major petroleum and natural gas reserves exist here.
- These formed from ancient marine organic deposits buried over millions of years.

Q 92. Ans. B

Explanation:

- Barchans:
 - ✓ Type: Crescent-shaped sand dunes
 - ✓ Region: Thar Desert near Jaisalmer
 - ✓ Characteristics
 - Horns: Point downwind
 - Wind: Unidirectional, constant
 - Surface: Hard, flat
 - Sand supply: Limited
 - ✓ Formation Mechanism
 - Wind action: Sand moves over the crest
 - Slip-face: Steep leeward side deposition
 - Movement: Horns move faster than the centre

- Result: Crescent shape
- Additional Information
 - ✓ Longitudinal dunes (Seif)
 - Orientation: Parallel to wind
 - Condition: Variable wind direction or more sand
 - Location: Southern Thar
 - ✓ Parabolic dunes (Stabilised mounds)
 - Shape: Crescent
 - Horns: Point upwind
 - Cause: Vegetation anchoring
 - Location: Semi-arid margins
 - ✓ Star dunes
 - Shape: Central peak with multiple arms
 - Wind: Multidirectional
 - Occurrence: Rare in Thar
- Key Concepts
 - ✓ Process: Aeolian deposition
 - ✓ Erosional features: Mushroom rocks (rock pedestals)
- Thar Desert Facts
 - ✓ Rank: 9th largest subtropical desert
 - ✓ Distribution: ~60% Rajasthan, rest Gujarat, Punjab, Haryana, Pakistan
- Recent Relevance
 - ✓ Desertification: United Nations Convention to Combat Desertification reports
 - ✓ Trend: Canal irrigation stabilising Barchans
 - ✓ Example: Indira Gandhi Canal
 - ✓ Climate Change
 - Effect: Changing wind patterns
 - Impact: Eastward migration of Barchans

the main Aravalli alignment by the Banas River valley.

- Additional Facts:
 - ✓ Extent of the Aravalli Range
 - Southern limit:
 - ◇ Begins near Palanpur in Gujarat
 - Northern limit:
 - ◇ Extends through Rajasthan and Haryana up to the Delhi Ridge
 - ✓ Important landmark:
 - ◇ Raisina Hill in Delhi, where the Rashtrapati Bhavan stands, lies on the northern extension of the Aravallis.

Drainage Role of the Aravallis:

- The Aravalli Range acts as a major water divide in western India.
 - ✓ West-flowing rivers:
 - Luni
 - Sabarmati
 - ✓ Flow towards the Arabian Sea or inland basins.
 - East-flowing rivers:
 - ◇ Banas
 - Join the Chambal–Yamuna–Ganga system and drain into the Bay of Bengal.

Mineral Wealth of the Aravallis:

- Major mineral regions:
 - ✓ Khetri (Jhunjhunu district):
 - Famous for copper deposits
 - ✓ Zawar Mines (Udaipur):
 - Known for zinc, lead, and silver
 - ✓ Makrana:
 - Source of high-quality white marble used in major monuments such as the Taj Mahal.

Important Passes in the Aravalli Range:

- Haldighati Pass:
 - ✓ Site of the historic battle between Maharana Pratap and the Mughal forces (1576).
- Pipli Ghat:
 - ✓ Known as one of the highest passes in the Aravalli Range.
- Desuri Nal and Someshwar Nal:
 - ✓ Important passes connecting the Mewar and Marwar regions.

Q 93. Ans. C

Explanation:

- Statement 1 is correct:
 - ✓ The Aravalli Range is one of the oldest fold mountain systems in the world.
 - ✓ Over millions of years, intense erosion and denudation have reduced it to a residual (relict) mountain range.
 - ✓ Today, it appears as worn-down hills rather than sharp, high peaks.
- Statement 2 is correct:
 - ✓ Guru Shikhar (1,722 m) is the highest peak of the Aravalli Range.
 - ✓ It is located in the Mount Abu hills in Rajasthan.
 - ✓ The Mount Abu massif is separated from

Q 94. Ans. A**Explanation:**

- Archaeian Group:
 - Age: ~4 billion years
 - ✓ Nature: Crystalline, highly metamorphosed
 - ✓ Region: Tamil Nadu, Karnataka, Chhotanagpur Plateau, Nilgiris, Andhra Pradesh, Maharashtra, West Bengal, Odisha, Jharkhand, Chhattisgarh, Madhya Pradesh, Meghalaya, Mikir Hills, Bundelkhand, Aravallis
 - ✓ Significance: Basement complex
 - ✓ Minerals: Metallic and non-metallic minerals
- Dharwar System :
 - ✓ Origin: Erosion of Archaeian rocks
 - ✓ Nature: Metamorphosed sedimentary rocks
 - ✓ Region: Karnataka (Dharwar-Shimoga belt), Chhotanagpur Plateau, Upper Godavari, Aravallis
 - ✓ Minerals: Iron ore, manganese, mica, copper, zinc, gold, limestone
 - ✓ Key site: Kolar gold mines
- Cuddapah System:
 - ✓ Name origin: Cuddapah district (Andhra Pradesh)
 - ✓ Region: Lower valleys of Penganga, Godavari, Upper Narmada, Son, west of Aravallis near Jodhpur
 - ✓ Rocks: Limestone, sandstone, shale
 - ✓ Use: Building materials
- Vindhyan System:
 - ✓ Nature: Unmetamorphosed sedimentary rocks
 - ✓ Extent: Chittorgarh to Sasaram, extends to Hoshangabad
 - ✓ Region: Central Indian Highlands
 - ✓ Minerals: Sandstone, limestone, diamonds
 - ✓ Key areas: Panna, Kurnool
- Chronological Order:
 - ✓ Archaeian → Dharwar → Cuddapah → Vindhyan → Gondwana → Deccan Trap → Tertiary → Quaternary
- Economic Importance:
 - ✓ Dharwar: Major mineral belt of India
 - ✓ Vindhyan: Building stones, diamonds
- Recent Relevance:
 - ✓ Critical minerals: Exploration in Archaeian and Dharwar shields

- ✓ Geopark proposals: Vindhyan and Cuddapah basins
- ✓ Fossils: Stromatolites in sedimentary basins

Q 95. Ans. A**Explanation:**

- Assertion (A) is true:
 - ✓ India has selected 82°30' E longitude as its Standard Meridian.
 - ✓ This longitude determines Indian Standard Time (IST) for the entire country.
 - ✓ The standard meridian passes near Mirzapur in Uttar Pradesh, which is centrally located in India.
- Reason (R) is true:
 - ✓ Standard meridians across the world are chosen at multiples of 7°30' (7.5°).
 - ✓ This rule is based on the Earth's rotation:
- Key relation:
 - ✓ Earth rotates 360° in 24 hours
 - ✓ Therefore, 15° longitude = 1 hour
 - ✓ 7°30' longitude = 30 minutes

Thus, choosing meridians at multiples of 7°30' allows time to be expressed neatly in whole hours or half-hours from UTC.

Why (R) correctly explains (A):

- India selected 82°30' E (82.5° E) because:
 - ✓ It is a multiple of 7°30', satisfying the global standard time system.
 - ✓ It gives a time difference of 5 hours 30 minutes ahead of Greenwich Mean Time (UTC +5:30).

Calculation:

- $5^\circ \div 15^\circ = 5.5$ hours
- Therefore, IST = UTC +5:30

Hence, the reason directly explains why 82°30' E was chosen.

Q 96. Ans. A**Explanation:**

- The drainage pattern resembling the branches of a tree is known as "dendritic" the examples
- of which are the rivers of northern plain.
- When the rivers originate from a hill and flow in all directions, the drainage pattern is known
- as 'radial'. The rivers originating from the

Amarkantak range present a good example of it.

- When the primary tributaries of rivers flow parallel to each other and secondary tributaries join them at right angles, the pattern is known as 'trellis'.
- When the rivers discharge their waters from all directions in a lake or depression, the pattern is known as 'centripetal'

- East: Himalayas (major rivers)
- West: Afghan-Pak highlands (Kabul system)
- ✓ Strategic Context
 - Indus Waters System
 - Framework: Indus Waters Treaty

Q 97. Ans. C

Explanation:

- Drainage: Indus River system
 - ✓ Source regions: Sulaiman Range, Safed Koh Range, Hindu Kush
 - ✓ Key Western Tributaries
 - Kabul River
 - Origin: Hindu Kush
 - Largest western tributary
 - Confluence: Attock (joins Indus)
 - ✓ Gommal River
 - Origin: Afghanistan / Sulaiman Range
 - Region: South Waziristan
 - Confluence: Near Dera Ismail Khan
 - ✓ Kurram River
 - Origin: Safed Koh / Sulaiman Range
 - Route: Kurram Agency
 - Confluence: Near Isa Khel
 - ✓ Tochi River (Gambila)
 - Origin: North Waziristan / Sulaiman Range
- Additional Information
 - ✓ Eastern Tributaries
 - Ravi
 - Beas
 - Satluj
 - ✓ Origin: Indian Himalayas (Himachal Pradesh/Tibet)
 - ✓ System: Punjab rivers
 - ✓ Major Himalayan Tributaries (Also not western)
 - Jhelum: Pir Panjal Range
 - Chenab: Greater Himalayas
 - ✓ Upper Indus Tributaries
 - Shyok, Nubra, Zaskar
 - Region: Ladakh (Trans-Himalayan zone)
 - System: Upper Indus basin
 - ✓ Key Facts
 - Indus drainage system collects water from:

Q 98. Ans. C

Explanation:

Statement 1 is correct:

- The Kaveri River originates at Talakaveri in the Brahmagiri Hills of the Western Ghats.
- Location details:
 - ✓ Elevation: About 1,341 metres above sea level
 - ✓ District: Kodagu (Coorg)
 - ✓ State: Karnataka

Statement 2 is correct:

- The Kaveri River flows throughout the year because it receives rainfall from two different monsoon systems.
- This dual rainfall pattern makes the river almost perennial, unlike most peninsular rivers that are seasonal.

Rainfall sources:

- Upper catchment (Karnataka region): Receives rainfall from the Southwest Monsoon (June–September).
- Lower basin (Tamil Nadu region): Receives rainfall from the Northeast Monsoon (October–December).

This dual monsoon recharge ensures relatively continuous flow.

Additional Facts:

- Kaveri River — Basic Profile
- Source:
 - ✓ Talakaveri, Brahmagiri Hills, Western Ghats
- States covered:
 - ✓ Karnataka
 - ✓ Tamil Nadu
- Other basin regions:
 - ✓ Kerala
 - ✓ Puducherry (Karaikal region)
- Drainage direction:
 - ✓ Flows eastward into the Bay of Bengal.

Major Tributaries of the Kaveri:

Left-bank tributaries (north side):

- ✓ Hemavati
- ✓ Shimsha
- ✓ Arkavathy
- Right-bank tributaries (south side):
 - Kabini
 - Bhavani
 - Amaravati

Important River Features:

- Sacred river islands:
 - ✓ Srirangapatna
 - ✓ Shivanasamudra
 - ✓ Srirangam
- Important waterfalls:
 - ✓ Shivanasamudra Falls — site of one of India's earliest hydroelectric projects
 - ✓ Hogenakkal Falls — located near the Karnataka–Tamil Nadu border

Q 99. Ans. C

Explanation:

- The southernmost point of the Indian Union is Indira Point, located on Great Nicobar Island in the Andaman and Nicobar Islands.
- Its geographical coordinate is $6^{\circ} 45' N$ latitude.
- This point was partially submerged during the 2004 Indian Ocean Tsunami, caused by a major tectonic earthquake near Sumatra.
- The land around Indira Point sank by several metres due to tectonic subsidence, altering the coastline permanently.

Additional Facts:

Southern Extremes of India
Southernmost point of Indian Union:

- Indira Point
- Location: Great Nicobar Island
- Coordinate: $6^{\circ} 45' N$

Southernmost point of Indian mainland:

- Kanyakumari
- Location: Tamil Nadu
- Coordinate: $8^{\circ} 04' N$

Other Geographical Extremes of India

Northernmost point:

- Indira Col — $37^{\circ} 06' N$

Easternmost point:

- Kibithu (Arunachal Pradesh) — $97^{\circ} 25' E$

Westernmost point:

- Ghuar Mota (Gujarat) — $68^{\circ} 07' E$

Q 100. Ans. A

Explanation:

- Statement 1 is incorrect:
- The Godavari River originates from Trimbakeshwar in the Nashik district of Maharashtra.
- The Amarkantak Plateau is actually the source of other rivers, such as:
 - ✓ Narmada — flows west into the Arabian Sea
 - ✓ Son — flows north to join the Ganga
- Statement 2 is correct:
 - ✓ The Godavari River is generally not navigable along most of its course because it flows over the rocky terrain of the Deccan Plateau.
 - ✓ It becomes navigable mainly in its deltaic region, where the land becomes flat and water depth increases.
 - ✓ The Godavari also forms a large and fertile delta before draining into the Bay of Bengal.

Additional Facts:

- Godavari River — Basic Profile
 - ✓ Nickname:
 - Known as the Dakshin Ganga (Ganga of the South)
 - ✓ Source:
 - Trimbakeshwar, Nashik district, Maharashtra
 - ✓ Drainage direction:
 - Flows eastward into the Bay of Bengal
 - ✓ Length:
 - About 1,465 km, making it the longest Peninsular river.

Major Tributaries of the Godavari:

- Left-bank tributaries (north side):
 - ✓ Pranahita (largest tributary)
 - ✓ Indravati
 - ✓ Sabari
- Right-bank tributaries (south side):
 - ✓ Manjira
 - ✓ Pravara
 - ✓ Maner

Godavari Basin States:

- Major basin states:

- ✓ Maharashtra
- ✓ Telangana
- ✓ Andhra Pradesh
- ✓ Chhattisgarh
- ✓ Odisha
- Minor basin areas:
 - ✓ Madhya Pradesh
 - ✓ Karnataka

Important Projects on the Godavari

- Polavaram Project:
 - ✓ Major multipurpose irrigation project in Andhra Pradesh.
- Kaleshwaram Lift Irrigation Project:
 - ✓ Located in Telangana.
 - ✓ Built near the confluence of the Godavari and Pranahita rivers.

Q 101. Ans. B

Explanation:

Assertion (A) is correct:

- India extends longitudinally from about 68°7' E (Gujarat) to 97°25' E (Arunachal Pradesh).
- This gives a longitudinal extent of roughly 30 degrees.
- Since the Earth rotates 15 degrees per hour, a 30-degree difference produces about a 2-hour difference in local sunrise between the easternmost and westernmost parts of India.
- Therefore, sunrise occurs nearly 2 hours earlier in Arunachal Pradesh than in Gujarat.

Reason (R) is correct:

- The Earth rotates from west to east, which causes the Sun to appear to rise first in eastern locations and later in western locations.
- This explains why eastern places see the sunrise earlier than western places.

Why (R) is not the correct explanation of (A):

- The Assertion talks about the magnitude of the time difference (about 2 hours).
- The Reason explains only the direction of Earth's rotation, not the amount of time difference.
- To fully explain the 2-hour difference, the reason would need to include the rate of Earth's rotation (15° per hour) and India's 30° longitudinal extent.
- Hence, both statements are true, but the reason does not fully explain the assertion.

Additional Facts:

Longitude–Time Relationship

Key rules:

- Earth rotates 360° in 24 hours
- Therefore:
 - ✓ 15° longitude = 1 hour
 - ✓ 1° longitude = 4 minutes

Application to India:

- India's longitude span $\approx 30^\circ$
- Time difference:
 - ✓ $30 \times 4 \text{ minutes} = 120 \text{ minutes (2 hours)}$

Standard Time vs Local Time

Local time:

- Based on the Sun's actual position at a location.
- Would create different times across India.

Standard time (IST):

- Based on 82°30' E longitude.
- Used to maintain uniform time across the country.

Q 102. Ans. D

Explanation:

- Purvanchal Hills:
 - ✓ System: Eastern Hills (southward extension of Himalayas)
 - ✓ Location: India–Myanmar border
 - ✓ Origin: After Dihang gorge (syntaxial bend near Namcha Barwa)
 - ✓ Sequence (North → South)
- Patkai Bum
 - ✓ Location: Arunachal Pradesh
 - ✓ Feature: India–Myanmar boundary
 - ✓ Rock: Sandstone
- Naga Hills
 - ✓ Location: Nagaland
 - ✓ Highest peak: Mount Saramati (3,826 m)
- Manipur Hills
 - ✓ Location: Manipur
 - ✓ Feature: Contains Loktak Lake
 - ✓ Peak: Mount Iso (Tenpu)
- Mizo Hills (Lushai Hills)
 - ✓ Location: Mizoram
 - ✓ Highest peak: Phawngpui (2,157 m)
 - ✓ Feature: Molassis basin
- Key Characteristics
 - ✓ Relief

- Type: Low to moderate height
- Structure: Parallel ranges and valleys
- ✓ Geology
 - Rock type: Sedimentary (sandstone)
 - Origin: Tethys Sea deposits
- ✓ Vegetation
 - Type: Dense forests
- ✓ Drainage Divide
 - West-flowing rivers
 - System: Brahmaputra / Barak
- ✓ East-flowing rivers
 - System: Chindwin → Irrawaddy (Myanmar)

- Narmada
- Tapi
- Reason: Fault-controlled valleys

Q 104. Ans. A

Explanation:

- Gorakhpur: 26.76°N, 83.37°E
- Varanasi: 25.32°N, 82.99°E
- Patna: 25.61°N, 85.14°E
- Ranchi: 23.36°N, 85.33°E

Q 105. Ans. A

Explanation:

Cold winter winds affecting Asia originate from a strong high-pressure system formed over Siberia and Central Asia.

- During winter, the Eurasian landmass cools intensely, creating a very high-pressure zone called the Siberian High.
- Winds naturally flow from high pressure to low pressure, so cold, dry winds move outward from this region.
- The Himalayan Mountains act as a massive physical barrier, preventing these extremely cold winds from entering the Indian subcontinent.
- Because of this obstruction, India experiences comparatively milder winters than regions of Central Asia at similar latitudes.
- Thus, the Siberian and Central Asian high-pressure region is the correct origin of these cold winds.

Additional Facts:

- Role of the Siberian High in Winter Climate:
 - ✓ The Siberian High forms over Siberia and Central Asia during winter due to intense land cooling.
 - ✓ It is one of the strongest seasonal high-pressure systems in the world.
 - ✓ It generates cold, dry continental winds that move outward toward surrounding regions.

Role of the Himalayas as a Climatic Barrier:

- The Himalayas perform two major climatic functions:
 - ✓ Blocking Cold Winds in Winter
 - Prevent freezing continental winds from Central Asia from reaching India.

Q 103. Ans. B

Explanation:

- The general eastward drainage of major Peninsular rivers is primarily due to the structural tilt of the Peninsular plateau.
 - ✓ The Peninsular plateau is tilted from west to east and southeast.
 - ✓ This natural slope directs most rivers such as:
 - Godavari
 - Krishna
 - Kaveri
 - Mahanadi
 - ✓ Western Margin Subsidence
 - Period: Early Tertiary
 - Process: Subsidence of western edge
 - Result: Partial submergence under sea
 - ✓ Himalayan Uplift Impact
 - Cause: Uplift of Himalayas
 - Effect: Northern Peninsular edge subsided
 - Structure: Trough faulting
 - ✓ Rivers in Fault Valleys
 - Narmada River
 - Tapi River
 - Feature: Flow through rift valleys
 - Process: Sediment deposition in fractures
 - ✓ Block Tilting
 - Direction: Northwest –Southeast
 - Result: General eastward slope
 - ✓ Drainage Pattern
 - East-flowing rivers
 - Direction: Toward Bay of Bengal
 - Reason: SE tilt of plateau
 - ✓ Exceptions (West-flowing)

- Without this barrier, North India would experience extremely harsh winters similar to Central Asia.
- ✓ Intercepting Monsoon Winds in Summer
 - Force moist southwest monsoon winds to rise and produce rainfall.
 - Without the Himalayas, monsoon rainfall in northern India would be drastically reduced.

Arctic Ocean marginal belt:

- Extremely cold region, but not the direct source of winter winds affecting India.

Tibetan Plateau and adjoining Trans-Himalayan region

- Influences regional pressure patterns, but the primary cold wind origin remains Siberia–Central Asia.

Iranian Plateau and Caspian lowlands:

- Associated with western disturbances, not the cold continental winds blocked by the Himalayas.

Q 106. Ans. D

Explanation:

- Shillong Plateau: Precambrian crustal block
 - ✓ Nature: detached part of Indian Peninsular Shield
 - ✓ Includes: Mikir Hills
 - ✓ Rock type: granites and gneisses
- Barail Range: Tertiary Fold Mountains
 - ✓ Rock type: younger sedimentary rocks
 - ✓ Origin: Himalayan orogeny
 - ✓ Cause: Indian–Eurasian plate collision
 - ✓ Structural Relations
 - Barail Range: structural bridge
 - Location: between Shillong Plateau west and Naga Hills east
 - Function: drainage divide
- Additional Information
 - ✓ Chandubi Lake
 - ✓ Type: tectonic lake
 - ✓ Location: foot of Garo Hills
 - ✓ Formation: Great Assam Earthquake 1897
 - ✓ Process: subsidence of northern plateau edge

- Relevance: wetland conservation, earthquake-induced landforms
- Kulsi River
 - ✓ Type: tributary of Brahmaputra
 - ✓ Origin: northern slopes of Shillong Plateau
 - ✓ Flow: northward
 - ✓ Ecological importance: habitat of endangered Ganges River Dolphin
 - ✓ Threat: illegal sand mining
 - ✓ Impact: damage to breeding grounds
- Nongkhylllem Plateau
 - ✓ Type: structural sub-unit of central Meghalaya Plateau
 - ✓ Associated area: Nongkhylllem Wildlife Sanctuary
 - ✓ Recent finding: Eudiscopus denticulus
 - ✓ Significance: mammalian biodiversity hotspot
 - ✓ Structural Features (Static GK)
 - ✓ Shillong Plateau: Horst structure
 - ✓ Boundaries:
 - North: Brahmaputra Fault
 - South: Dauki Fault
 - East: Kopili Fault
 - Separation: from Chotanagpur Plateau
 - Feature: Malda Gap
 - Type: Ganga–Brahmaputra graben
 - Economic and Geopolitical Importance
 - Barail Range: terrain barrier
 - Effect: extreme terrain complexity
 - Critical link: Lumding–Badarpur rail line
 - Role: lifeline of Barak Valley
- Recent Geodynamic Activity
 - ✓ Kopili Fault: highly active
 - ✓ Location: between Shillong Plateau and Mikir Hills/Barail Range
 - ✓ Seismicity: multiple low to moderate earthquakes
 - ✓ Cause: tectonic compression with Burmese Arc
 - ✓ Hydropower Issue
 - River: Umngot River
 - Origin: Shillong Plateau
 - Issue: environmental clearance for small hydropower projects
 - Concern: impact on structural stability of southern slopes
 - Response: local protests

Q 107. Ans. C**Explanation:**

Jet streams play a steering role in bringing winter cyclonic depressions (Western Disturbances) to India.

- Western Disturbances originate as temperate cyclones over the Mediterranean region.
- These disturbances are carried eastward by the Subtropical Westerly Jet Stream.
- During winter, this jet stream flows south of the Himalayas, acting like an atmospheric pathway.
- As a result, these cyclonic systems are steered toward northwestern India, especially Punjab, Haryana, Rajasthan, and western Uttar Pradesh.
- They bring winter rainfall and snowfall, which is crucial for agriculture and water resources.
- Thus, the correct statement is that jet streams steer and guide cyclonic depressions toward northwestern India.

Additional Facts:

- Nature of Western Disturbances
 - ✓ Western Disturbances are extratropical (temperate) cyclones, not tropical cyclones.
 - ✓ They develop due to the interaction of cold polar air and warm subtropical air.
 - ✓ Their origin region is mainly the Mediterranean Sea.
 - ✓ They move from west to east, guided by upper-air westerlies.

Role of the Subtropical Westerly Jet Stream:

- In winter, the Subtropical Westerly Jet Stream shifts southward over northern India.
- It carries Western Disturbances across West Asia, Pakistan, and into India.
- When these disturbances encounter the Himalayas, they produce:
 - ✓ Rainfall in the northwestern plains
 - ✓ Snowfall in the Himalayan regions

Agricultural Importance of Winter Rainfall:

- Winter rainfall supports Rabi crops, especially:
 - ✓ Wheat
 - ✓ Mustard
 - ✓ Gram
- In northwestern India, this rainfall is locally known as Mahawat.
- Snowfall in the Himalayas helps recharge glaciers and rivers during summer.

Q 108. Ans. B**Explanation:**

The Meghalaya Plateau receives the highest rainfall in northeastern India due to strong orographic (relief) rainfall.

- The Bay of Bengal branch of the Southwest Monsoon carries large amounts of moisture toward northeastern India.
- When these winds reach the Meghalaya Plateau, they encounter the steep slopes of the Garo, Khasi, and Jaintia Hills.
- The hills are arranged in a funnel-like formation facing the Bay of Bengal.
- This forces the moisture-laden air to rise rapidly, causing cooling, condensation, and extremely heavy rainfall on the windward slopes.
- As a result, places such as Mawsynram and Cherrapunji (Sohra) receive some of the highest rainfall totals in the world.

Role of Orographic Rainfall in Meghalaya:

- Moist winds from the Bay of Bengal strike the plateau directly.
- The steep southern escarpment of the Khasi Hills forces sudden uplift of air.
- This produces intense and prolonged rainfall during the monsoon season.

Q 109. Ans. A**Explanation:**

- A river basin and a watershed differ mainly in their scale and extent.
- A river basin is a large drainage area drained by a major river and its tributaries.
 - ✓ It represents a macro-level drainage unit.
 - ✓ Example: Ganga Basin, Godavari Basin, Brahmaputra Basin.
- A watershed is a smaller drainage unit, usually drained by small streams, rivulets, or minor tributaries.
 - ✓ It represents a micro-level drainage unit.
 - ✓ Many small watersheds combine to form a large river basin.

Additional Facts:

- Hierarchical Relationship Between Basin and Watershed
 - ✓ Watershed → Sub-basin → River Basin

- ✓ Hundreds or thousands of watersheds collectively form one major river basin.
- ✓ This hierarchical structure is widely used in:
 - Water resource planning
 - Soil conservation
 - Flood control programs

Concept of Water Divide:

- A water divide is an elevated landform, such as:
 - ✓ Mountain
 - ✓ Ridge
 - ✓ Plateau edge
- It separates two drainage basins.
- Rainwater falling on opposite sides flows into different river systems.
- Indian Examples:
 - Ambala Ridge
 - ✓ Separates the Indus Basin from the Ganga Basin.
 - Western Ghats
 - ✓ Separate the west-flowing rivers (Arabian Sea) from the east-flowing rivers (Bay of Bengal).

Related Terminologies:

- Catchment Area
 - ✓ Total area from which a river collects water.
 - ✓ Often used interchangeably with drainage area in practical geography.
- Inland Drainage (Endorheic Basin)
 - ✓ Rivers do not reach the sea.
 - ✓ Example: Luni River draining into the Rann of Kutch.

Q 110. Ans. A

Explanation:

- **Statement 1 is correct:** The Kosi River is an antecedent river, meaning it existed before the uplift of the Himalayas and maintained its course by cutting through the rising mountains.
 - ✓ It originates north of Mount Everest in Tibet, where its headstreams drain the high Himalayan region.
 - ✓ The river carries a very high sediment load, which is why it frequently changes its course and is often called the “Sorrow of Bihar.”
- **Statement 2 is incorrect:** The Sun (Son) Kosi joins from the west, while the Tamur Kosi joins from the east.

- ✓ Along with the Arun Kosi (from the north), these rivers form the Sapta Kosi system before entering the plains of Bihar.

Q 111. Ans. A

Explanation:

- Correct decreasing order of length:
 - ✓ Godavari — about 1,465 km (longest Peninsular river)
 - ✓ Krishna — about 1,400 km
 - ✓ Mahanadi — about 851 km
 - ✓ Kaveri — about 800 km
- Thus, the correct sequence is:
- Godavari → Krishna → Mahanadi → Kaveri

Additional Facts:

- Godavari
 - ✓ Longest river of Peninsular India.
 - ✓ Originates from Trimbakeshwar near Nashik in Maharashtra.
 - ✓ Known as Dakshin Ganga due to its large basin and length.
 - ✓ Forms an extensive delta along the eastern coast.
- Krishna
 - ✓ The second-longest peninsular river.
 - ✓ Originates near Mahabaleshwar in Maharashtra.
 - ✓ Major tributaries include the Bhima and the Tungabhadra.
 - ✓ Forms a large delta along the Andhra coast.
- Mahanadi
 - ✓ Originates near Sihawa in Chhattisgarh.
 - ✓ Known historically as the “Sorrow of Odisha” due to frequent flooding before dam construction.
 - ✓ Forms a broad delta along the Odisha coast.
- Kaveri
 - ✓ Shortest among the four listed rivers.
 - ✓ Originates at Talakaveri in the Brahmagiri hills of Karnataka.
 - ✓ Known for its relatively steady flow compared to other Peninsular rivers.

Important Comparative Points:

- Godavari and Krishna together form the extensive Krishna–Godavari delta region along the eastern coast.
- Mahanadi has one of the largest deltas on the eastern coast of India.

- Kaveri, though shorter, is one of the most intensively utilised rivers for irrigation and hydroelectric power.

Q 112. Ans. A

Explanation:

- Statement 1 is Correct:
 - ✓ The Tapi (Tapti) River originates near Multai in the Betul district of Madhya Pradesh.
 - ✓ Its source lies in the Satpura Range, making it one of the important west-flowing rivers of Peninsular India.
- Statement 2 is Incorrect:
 - ✓ The statement wrongly claims that nearly 80% of the Tapi basin lies in Madhya Pradesh and Gujarat.
 - ✓ In reality, about 79% of the Tapi basin lies in Maharashtra, while:
 - Madhya Pradesh accounts for roughly 15%
 - Gujarat accounts for roughly 6%
 - ✓ Therefore, the basin is dominated by Maharashtra, not MP and Gujarat.
- Statement 3 is Incorrect:
 - ✓ The Tapi River flows through three states only:
 - Madhya Pradesh
 - Maharashtra
 - Gujarat
 - ✓ It does not flow through Rajasthan, making this statement incorrect.
- Additional Facts:
 - ✓ Origin and Course of the Tapi River
 - Origin: Multai, Betul district, Madhya Pradesh
 - Mountain range: Satpura Range
 - Flow direction: East to West
 - Mouth: Empties into the Arabian Sea near Surat in Gujarat.
- Relationship with the Narmada River
 - ✓ The Tapi and Narmada share several geographic similarities:
 - Both flow westwards across Peninsular India.
 - Both drain into the Gulf of Khambhat.
 - Both flow through rift valleys.
 - Both form estuaries, not deltas.
- Important Physiographic Sequence (North to South)

- ✓ Vindhya Range
- ✓ Narmada River
- ✓ Satpura Range
- ✓ Tapi River
- ✓ Ajanta Range

- Major Dams on the Tapi River:
 - ✓ Ukai Dam (Gujarat)
 - Forms the Vallabh Sagar Reservoir.
 - ✓ Kakrapar Dam (Gujarat)
 - Important for irrigation and power generation.
- Key City on the Tapi River
 - ✓ Surat (Gujarat)
 - Located near the river mouth.
 - One of India's major diamond and textile centres.

Q 113. Ans. A

Explanation:

- Tribes such as Gujjars, Bakarwals, and Bhotiyas inhabit the Himalayan mountainous regions, where the dominant vegetation is montane forests.
- Montane forests are found in high-altitude areas of the Himalayas.
- These regions support coniferous and mixed forests, along with alpine vegetation at higher elevations.
- The tribes practice transhumance pastoralism, moving seasonally with their livestock between high-altitude summer pastures and lower valleys in winter.

Q 114. Ans. C

Explanation:

- The Gandhi Sagar Dam is constructed on the Chambal River.
 - ✓ The Chambal River originates near Janapav (near Mhow) in Madhya Pradesh.
 - ✓ The Gandhi Sagar Dam is the first and uppermost dam built on the Chambal River.
 - ✓ It is located in the Mandsaur district of Madhya Pradesh.
 - ✓ The dam forms part of the Chambal Valley Project, which is a joint venture between Madhya Pradesh and Rajasthan for irrigation and hydroelectric power generation.
- Additional Facts:

- ✓ Sequence of Major Structures on the Chambal River (Upstream to Downstream order):
 - Gandhi Sagar Dam — Located in Madhya Pradesh
 - Rana Pratap Sagar Dam — Located at Rawatbhata, Rajasthan
 - Jawahar Sagar Dam — Located near Kota, Rajasthan
 - Kota Barrage — Located at Kota, Rajasthan (mainly for irrigation)
- Course of the Chambal River:
 - ✓ Origin: Janapav hills near Mhow in Madhya Pradesh
 - ✓ Flow direction: Initially flows northward, then turns northeast
 - ✓ States covered:
 - Madhya Pradesh
 - Rajasthan
 - Uttar Pradesh
- Confluence: Joins the Yamuna River near Etawah (Uttar Pradesh).

Important Geographic Features of the Chambal Basin:

- Chambal Ravines
 - ✓ Found along the lower course of the river.
 - ✓ Formed due to gully erosion in soft alluvial soils.
 - ✓ Historically known for difficult terrain and deep gullies.
- National Chambal Sanctuary
 - ✓ Established along the Chambal River.
 - ✓ Protects endangered species such as:
 - Gharial
 - Gangetic Dolphin
 - Freshwater turtles

Q 115. Ans. B

Explanation:

- Statement 1 is incorrect:
 - ✓ The Damuda Series does not belong to the Middle Gondwana Period.
 - ✓ It belongs to the Lower Gondwana Sequence, mainly of Permian age.
 - ✓ The Lower Gondwana sequence contains the major coal-bearing formations of India.
 - ✓ Includes important coal-bearing formations such as:
 - Karharbari Formation
 - Barakar Formation
 - Barren Measures
 - Raniganj Formation
 - Statement 2 is correct:
 - ✓ The coal seams of the Damuda Series are thicker, more continuous, and more extensive in eastern coalfields than in western regions.
 - ✓ The eastern river valleys, especially the Damodar Valley, contain India's most important coalfields.
 - ✓ Major eastern coalfields:
 - Jharia
 - Raniganj
 - Bokaro
 - Giridih
- These coalfields contain thick and high-quality coal seams, especially coking coal used in the steel industry.
- Additional Facts:
 - ✓ Gondwana Rock System — General Features
 - ✓ Geological period:
 - Mainly formed during the Carboniferous to Cretaceous periods.
 - ✓ Major resource:
 - Contains about 98 percent of India's coal reserves.
 - ✓ Major river valleys containing Gondwana rocks:
 - Damodar Valley
 - Mahanadi Valley
 - Godavari Valley
- Lower Gondwana Sequence (Major Coal-Bearing Section):
- Important groups in order:
 - ✓ Talchir Group:
 - Known for glacial deposits (tillite).
 - Indicates ancient glacial conditions.
 - ✓ Damuda Group:
 - Most important coal-bearing group.
 - Includes Barakar and Raniganj formations, major sources of coal.
 - ✓ Panchet Group:
 - Represents the uppermost part of the Lower Gondwana.
- ### Upper Gondwana Sequence:
- Geological periods:
 - ✓ Triassic, Jurassic, and Cretaceous.

- Importance:
 - ✓ Contains plant and fossil remains.
 - ✓ Generally contains very little coal compared to Lower Gondwana.

Q 116. Ans. D

Explanation:

- Social forestry means the management and protection of forests and afforestation on barren lands with the purpose of helping in the environmental, social and rural development. The National Commission on Agriculture (1976) has classified social forestry into three categories.
 - ✓ Urban forestry: pertains to the raising and management of trees on public and privately owned lands in and around urban centres such as green belts, parks, roadside avenues, industrial and commercial green belts, etc.
 - ✓ Rural forestry : Emphasis on promotion of agro-forestry and community-forestry.
 - ✓ Farm forestry: . It combines forestry with agriculture, thus, altering the simultaneous production of food, fodder, fuel, timber and fruit.
 - ✓ Community Forestry: It involves raising trees on public or community land such as village pastures, temple land, roadsides, canal banks, railway strips, and school premises. It enables even landless people to participate in tree-raising and access benefits otherwise restricted to landowners. Since Community Forestry is a sub-component of Rural Forestry, it is by extension, a part of Social Forestry.

Q 117. Ans. A

Explanation:

The vegetation described in the passage corresponds to Tropical Deciduous Forests, also known as Monsoon Forests.

Key features mentioned in the passage clearly match Tropical Deciduous Forests:

- Rainfall Range: These forests occur in regions receiving 70–200 cm of annual rainfall, which is the defining rainfall range of Tropical Deciduous vegetation.
- Seasonal Leaf Shedding: Trees shed their leaves

during the dry summer season to reduce water loss through transpiration.

- Dominant Species: The presence of teak, sal, and bamboo is a classic indicator of Tropical Deciduous Forests.
- Distribution: They are the most widespread forests in India, covering large parts of the country.
- Therefore, the vegetation described in the passage is Tropical Deciduous Forests.

Additional Facts:

- Why These Forests Are Called Monsoon Forests
 - ✓ Their growth pattern depends heavily on seasonal monsoon rainfall.
 - ✓ During the wet monsoon season, trees grow dense foliage.
 - ✓ During the dry summer, many trees shed leaves to conserve moisture.
- Two Important Sub-types of Tropical Deciduous Forests:
 - ✓ Moist Deciduous Forests
 - ✓ Rainfall: 100–200 cm
 - ✓ Found in:
 - Western slopes of the Western Ghats
 - Jharkhand, Odisha, Chhattisgarh
 - Himalayan foothill regions
 - ✓ Common Trees:
 - Teak
 - Sal
 - Shisham
 - Bamboo
 - ✓ Dry Deciduous Forests
 - ✓ Rainfall: 70–100 cm
 - ✓ Found in:
 - Central India
 - Parts of Uttar Pradesh, Bihar, and Madhya Pradesh
 - ✓ Common Trees:
 - Teak
 - Neem
 - Khair
 - Peepal
- Tropical Evergreen Forests
 - ✓ Require very high rainfall (above 200 cm).
 - ✓ Trees do not shed leaves seasonally.
 - ✓ Found mainly in the Western Ghats, the Andaman & Nicobar, and the northeastern states.
- Tropical Thorn Forests and Scrubs

- ✓ Found in low rainfall regions (below 70 cm).
- ✓ Vegetation consists mainly of thorny shrubs and xerophytes.
- Montane Forests
 - ✓ Found in mountain regions such as the Himalayas and high-altitude areas of Peninsular India.
 - ✓ Controlled mainly by altitude, not monsoon rainfall alone.
- Gondwana System
 - ✓ Famous for coal deposits.
- Tertiary System
 - ✓ Associated with the formation of the Himalayas.
- Quaternary System
 - ✓ Includes alluvial deposits of the Northern Plains.

Economic Importance of Dharwar Rocks

- The Dharwar system is highly significant because it contains major metallic mineral deposits such as:
 - ✓ Iron ore
 - ✓ Manganese
 - ✓ Copper
 - ✓ Gold (notably in the Kolar Gold Fields, Karnataka)

Key Characteristics of the Cuddapah Basin

- Located mainly in Andhra Pradesh.
- Known for its crescent-shaped structure.
- Contains limestone, slate, and asbestos, supporting cement industries in the region.

Q 119. Ans. A

Explanation:

Statement 1 is correct:

- India has a very large Exclusive Economic Zone (EEZ) extending up to 200 nautical miles from the baseline.
- The total area of India's EEZ is about 2.02 million square kilometres, which is indeed over 2 million sq km.
- This zone provides India exclusive rights to marine resources, including fisheries, offshore oil, and minerals.

Statement 2 is correct:

- The Great Channel separates Great Nicobar Island (India) from Sumatra Island (Indonesia).
- Indira Point, the southernmost point of India, lies on Great Nicobar Island near this channel.
- This channel is strategically important because it lies close to major international sea routes near the Strait of Malacca.

Statement 3 is incorrect:

- India's territorial waters extend only up to 12 nautical miles, not 24 nautical miles.

Q 118. Ans. D

Explanation:

Statement 1 is correct:

- The Dharwar System represents the first metamorphosed sedimentary rock system in India.
- These rocks were formed after the erosion of older Archaean rocks, whose sediments were later subjected to heat and pressure, resulting in metamorphism.
- Hence, Dharwar rocks are considered the earliest metamorphosed sedimentary rocks of India.

Statement 2 is correct:

- The Cuddapah Basin belongs to the Proterozoic era and forms part of the Purana rock system.
- It is geographically known for its distinct crescent-shaped structure.
- The basin is located mainly in Andhra Pradesh and lies over the eastern and central parts of the Dharwar craton.

Additional Facts:

Chronological Sequence of Major Rock Systems in India (Oldest to Youngest):

- Archaean System
 - ✓ Oldest rocks in India.
 - ✓ Form the basement complex of the Peninsular Plateau.
- Dharwar System
 - ✓ First metamorphosed sedimentary rocks.
 - ✓ Rich in metallic minerals.
- Cuddapah System
 - ✓ Part of the Purana rock group.
 - ✓ Mainly composed of sedimentary rocks.
- Vindhyan System
 - ✓ Also part of the Purana group.
 - ✓ Known for sandstones and limestones.

- The 24 nautical mile limit refers to the Contiguous Zone, where India has limited enforcement rights but not full sovereignty.

Additional Facts:

- Maritime Zones Defined Under UNCLOS
- Territorial Sea — Up to 12 Nautical Miles
 - ✓ Full sovereignty of the coastal state.
 - ✓ Includes water, airspace, and seabed.
 - ✓ Contiguous Zone — 12 to 24 Nautical Miles
- Limited rights for the enforcement of:
 - ✓ Customs laws
 - ✓ Immigration laws
 - ✓ Taxation laws
 - ✓ Pollution control
- Exclusive Economic Zone (EEZ) — Up to 200 Nautical Miles
 - ✓ Exclusive rights over:
 - Fishing
 - Offshore oil and gas
 - Mineral extraction
- Other countries retain navigation freedom.

Strategic Importance of the Great Channel

- Located near the western approach to the Strait of Malacca, one of the busiest shipping lanes in the world.
- Provides India a strategic advantage in monitoring maritime traffic between the Indian Ocean and the Pacific Ocean.

Q 120. Ans. A

Explanation:

- **Sir Creek** is a 96-km tidal estuary separating Gujarat (India) from Sindh (Pakistan). The dispute centres on the boundary line; India asserts the mid-channel boundary based on the internationally recognised Thalweg principle, while Pakistan claims the entire creek.
- **Kalapani** is located in the Pithoragarh district of Uttarakhand. The dispute between India and Nepal originates from differing interpretations of the source of the Kali River, which was designated as the western boundary of Nepal by the British in the 1816 Treaty of Sagauli.
- **Doklam** is a high-altitude plateau at the tri-junction of India (Sikkim), Bhutan, and China (Chumbi Valley). Its strategic importance lies in its elevation, which provides a vantage

point over India's vulnerable Siliguri Corridor (Chicken's Neck).

- **Pangong Tso** is a 135-km-long endorheic lake in eastern Ladakh. The Line of Actual Control (LAC) passes through the lake. The northern bank is marked by eight mountainous spurs jutting into the lake, geographically referred to as "Fingers," which form the basis of the territorial claims between India and China.

Q 121. Ans. A

Explanation:

Statement 1 is correct:

- The Karakoram Range extends southeastward from the Pamir Knot.
- This range enters the region of Ladakh and forms part of the Trans-Himalayan system.
- It includes some of the highest peaks in the world, such as K2, and acts as an important high mountain barrier in northern India.

Statement 2 is incorrect:

- The Hindu Kush Range does not extend eastward from the Pamir Knot.
- Instead, it extends westward and southwestward, forming the mountainous backbone of Afghanistan and parts of Pakistan.
- The range extending eastward from the Pamir Knot toward Tibet is the Kunlun Range, not the Hindu Kush.

Additional Facts:

- The Pamir Knot as a Mountain Hub
 - ✓ The Pamir Knot is a major mountain junction in Central Asia.
 - ✓ Several important mountain ranges radiate outward from this region in different directions.

Major Mountain Ranges Radiating from the Pamir Knot

- Southeastward
 - ✓ Karakoram Range
 - Followed further south by the Himalayas
- West and Southwest
 - ✓ Hindu Kush Range
 - Extends across Afghanistan and Pakistan.
- East

- ✓ Kunlun Mountains
 - Forms the northern boundary of the Tibetan Plateau.
- Northeast
 - ✓ Tian Shan Mountains
 - It extends into Central Asia and western China.

Trans-Himalayan Ranges:

- The following ranges belong to the Trans-Himalayan system:
 - ✓ Karakoram Range
 - ✓ Ladakh Range
 - ✓ Zaskar Range
- These ranges lie north of the main Himalayan ranges and are geologically older than the Himalayas.

Key Geographic Importance:

- The Pamir Knot is often referred to as the “Roof of the World” due to its very high elevation.
- It plays a critical role in the formation and distribution of major Asian mountain systems.

Q 122. Ans. A

Explanation:

- **Statement 1 is correct:** The Brahmaputra, one of the largest rivers of the world, has its origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake. From here, it traverses eastward longitudinally for a distance of nearly 1,200 km in a dry and flat region of southern Tibet, where it is known as the Tsangpo, which means ‘the purifier.’
- **Statement 2 is incorrect:** The Brahmaputra enters into Bangladesh near Dhubri and flows southward. In Bangladesh, the Teesta joins it on its right bank from where the river is known as the Jamuna. It finally merges with the river Padma, which falls in the Bay of Bengal.

Q 123. Ans. B

Explanation:

The British selected Darjeeling as a hill station and tea-growing centre mainly due to the absence of the Shiwalik formations.

- In the Darjeeling–Sikkim Himalayas, the Shiwalik range is absent or merges into the Lesser Himalayas.

- Because of this, the mountains descend abruptly into the plains, forming Duars (Door) formations.
- These Duars consist of:
 - ✓ Deep, well-drained soils
 - ✓ High rainfall due to the orographic effect
 - ✓ Gentle slopes suitable for plantations
- These physiographic conditions proved ideal for tea cultivation, which requires:
 - ✓ Heavy rainfall
 - ✓ Well-drained slopes
 - ✓ Humid climate

Additional Facts:

- The Duars of the Darjeeling–Sikkim Himalayas
 - ✓ The Duars are foothill plains formed where mountains descend into the plains.
 - ✓ The term “Duars” means “door”, as these areas serve as gateways between the plains and the hills.
 - ✓ The Teesta River system dominates this region and contributes to deep valley formation.

Key Regional Himalayan Landforms:

- Darjeeling–Sikkim Himalayas
 - ✓ Feature: Absence of Shiwaliks
 - ✓ Landform: Duars
 - ✓ Economic activity: Tea plantations
- Kashmir Himalayas
 - ✓ Feature: Karewa formations
 - ✓ Importance: Famous for saffron cultivation
- Himachal–Uttarakhand Himalayas
 - ✓ Feature: Dun valleys
 - ✓ Examples:
 - Dehradun
 - Patli Dun
- Arunachal Himalayas
 - ✓ Feature: Deep river gorges
 - ✓ Major rivers:
 - Kameng
 - Subansiri
 - Dibang
 - Lohit

Q 124. Ans. B

Explanation:

Statement 1 is Correct:

- Minor west-flowing rivers from the Western

Ghats are steep and fast-flowing.

- They originate at high elevations in the Western Ghats and travel a short distance to the Arabian Sea.
- This creates a steep gradient, resulting in rapid flow and high erosive power.

Statement 2 is Incorrect:

- These rivers do not form extensive deltas.
- Because they are short and swift, they carry limited sediment and lack broad coastal plains needed for delta formation.
- Instead, they typically form estuaries along the western coast.

Statement 3 is Correct:

- These rivers flow westward directly into the Arabian Sea.
- The Western Ghats act as a major water divide, directing rainfall on the western slopes toward the Arabian Sea.

Additional Facts:

- Key Characteristics of Minor West-Flowing Rivers
 - ✓ Short length, generally less than 100 km.
 - ✓ Steep gradient due to elevated origin.
 - ✓ Fast-flowing nature with strong erosional activity.
 - ✓ Formation of estuaries, not deltas.
 - ✓ High potential for hydroelectric power generation.

Important Examples of West-Flowing Rivers:

- Goa
 - ✓ Mandovi River
 - ✓ Zuari River
- Karnataka
 - ✓ Sharavati River
 - Forms the famous Jog Falls.
 - ✓ Netravati River
- Kerala
 - ✓ Periyar River
 - Longest river in Kerala.
 - ✓ Pamba River
- Gujarat
 - ✓ Shetrunji
 - ✓ Dhadhar

Harbours

- Presence of estuaries rather than deltas.
- Deep coastal waters allow ships to approach close to land.
- This supports major ports such as:
 - ✓ Mumbai
 - ✓ Kochi
 - ✓ Mormugao
- Role of the Western Ghats as a Water Divide
- Rain falling on the western slopes flows into the Arabian Sea.
- Rain falling on the eastern slopes feeds long rivers like:
 - ✓ Godavari
 - ✓ Krishna
 - ✓ Kaveri

Q 125. Ans. A

Explanation:

The Kayals of Kerala are best described as lagoons.

- In Kerala, the term Kayal refers to backwaters, which are long, narrow water bodies running parallel to the coast.
- These water bodies are separated from the Arabian Sea by sand bars or barrier islands, which is the defining feature of lagoons.
- The water in Kayals is typically brackish, as it contains both seawater and freshwater from rivers.
- Their structure and coastal formation clearly match the geographical definition of lagoons.

Additional Facts:

- Major Kayals of Kerala
 - ✓ Vembanad Kayal
 - Longest lake in India and the largest in Kerala.
 - Designated as a Ramsar Wetland Site.
 - Famous for the Nehru Trophy Boat Race, held during the Onam festival.
 - ✓ Ashtamudi Kayal
 - Second-largest backwater system in Kerala.
 - Named after its eight branching channels.
 - Important for tourism and inland water transport.

Why the Western Coast Has Good Natural Unique Feature: Kuttanad Region

- Located around Vembanad Lake.
- Known for below sea-level farming, similar to polder systems used in the Netherlands.
- Often called the “Rice Bowl of Kerala.”

- Known for mineral extraction activities in surrounding regions, but not associated with extremely high NaCl purity like Pachpadra.

Q 126. Ans. C

Explanation:

- Pachpadra Lake is uniquely known for producing high-quality salt with nearly 98% Sodium Chloride (NaCl) content.
- Pachpadra Lake, located in the Barmer district of Rajasthan, produces salt of exceptionally high chemical purity.
- The salt obtained here contains about 98% Sodium Chloride, making it suitable for direct consumption with minimal refining.
- Because of this purity, Pachpadra Lake is widely recognized as the most important source of high-quality edible salt among the saline lakes of Rajasthan.

Additional Facts:

- Major Saline Lakes of Rajasthan and Their Characteristics
 - ✓ Pachpadra Lake
 - ✓ Location: Barmer district
 - Key feature: Highest NaCl purity (around 98%)
 - Economic role: Produces high-grade edible salt
 - Known for traditional salt collection methods.
 - ✓ Sambhar Lake
 - Location: Spread across Jaipur, Ajmer, and Nagaur districts
 - Distinction: Largest inland saline lake in India
 - Ecological status: Recognized as a Ramsar Wetland Site
 - Important winter habitat for flamingos.
 - ✓ Didwana Lake
 - Location: Nagaur district
 - Key feature: High Sodium Sulfate (Na_2SO_4) content
 - Use: Salt mainly used in industrial applications, not suitable for direct human consumption.
 - ✓ Degana Lake
 - Location: Nagaur district

Why These Lakes Are Saline:

These lakes occur in arid regions with inland drainage, where:

- Water collects in closed basins (playas).
- High evaporation leaves behind salt deposits.
- Over time, dissolved minerals concentrate, forming saline lakes.

Q 127. Ans. B

Explanation:

Statement 1 is incorrect:

- The Rohilkhand Plain forms part of the Upper Ganga Plain in northern Uttar Pradesh (including Bareilly and Moradabad).
- However, it does not border the Aravalli Range.
- Its northern boundary is formed by the Himalayan foothills (Shivaliks/Kumaon region), while the Aravalli Range lies far to the southwest.

Statement 2 is correct:

- The Awadh Plain lies in the Middle Ganga Plain, mainly in central Uttar Pradesh.
- It is drained primarily by the Ghaghara River, along with tributaries such as the Gomti.

Statement 3 is correct:

- The Rarh region is located west of the Bhagirathi–Hooghly River in West Bengal.
- It is characterized by lateritic soils, representing a transition zone between the Chotanagpur Plateau and the alluvial plains of Bengal.
- Therefore, this statement is correct.

Additional Facts:

- Sub-divisions of the Gangetic Plain
- The Gangetic Plain is divided into three major regional sections:
 - Upper Ganga Plain
 - Located mainly in western Uttar Pradesh.
 - Includes:
 - ✓ Ganga–Yamuna Doab
 - ✓ Rohilkhand Plain
 - Northern boundary: Shivalik foothills.

Middle Ganga Plain

- Covers central Uttar Pradesh and Bihar.
- Includes:
 - ✓ Awadh Plain
 - ✓ Magadh Plain
 - ✓ Mithila Plain
- Characterized by:
 - ✓ Very gentle slope
 - ✓ Frequent flooding and river shifting

Lower Ganga Plain

- Located mainly in West Bengal and Bangladesh.
- Includes:
 - ✓ Rarh Plain
 - ✓ Barind Tract
 - ✓ Sundarbans Delta

Special Feature of the Rarh Region

- Lies west of the Bhagirathi–Hooghly River.
- Contains lateritic (reddish) soils, unlike the fertile alluvial soils of eastern Bengal.
- Acts as a transition zone between plateau and delta regions.

Q 128. Ans. A

Explanation:

Statement 1 is Correct:

- The badland topography of the Malwa Plateau is mainly formed due to deep gully erosion.
- During heavy monsoon rains, surface runoff cuts into the soft soil, forming rills, which later deepen into gullies.
- Over time, these gullies expand into ravines, creating rugged and highly dissected terrain known as badlands or “bihads”.
- This type of erosion is especially prominent along the Chambal River basin.

Statement 2 is Incorrect:

- The Chambal River does not originate from the Amarkantak Plateau.
- It originates near Janapav (close to Mhow) in the Vindhya Range of Madhya Pradesh.
- The Amarkantak Plateau is actually the source of rivers such as:
 - ✓ Narmada (flows west)
 - ✓ Son (flows northeast)
 - ✓ Tributaries of the Mahanadi

Additional Facts:

Formation of Badland Topography

Badlands develop through a progressive erosion process:

- Sheet Erosion
 - ✓ Thin layer of topsoil removed uniformly.
- Rill Erosion
 - ✓ Small channels form due to flowing water.
- Gully Erosion
 - ✓ Channels deepen into permanent gullies.
 - ✓ Leads to formation of ravines, the main feature of badlands.
- Chambal Ravines Region:
 - ✓ Found mainly along the Chambal River basin.
 - ✓ Spread across:
 - Madhya Pradesh (Morena, Bhind)
 - Rajasthan (Dholpur)
 - Uttar Pradesh (Etawah)
- These ravines form one of India's most extensive badland landscapes.

The Amarkantak Plateau as a Radial Drainage Centre:

The Amarkantak Plateau is known for radial drainage, where rivers flow in different directions:

- West: Narmada River
- North-East: Son River
- South: Tributaries of the Mahanadi
- North: Johila River

Why Badlands Are Problematic:

- Soil becomes unsuitable for agriculture.
- Land is fragmented into deep trenches and gullies.
- Requires reclamation measures such as:
 - ✓ Afforestation
 - ✓ Check dams
 - ✓ Soil conservation methods

Q 129. Ans. C

Explanation:

- The Trans-Gangetic Plain is not rain-fed and does NOT have low groundwater use.
- The Trans-Gangetic Plain includes:
 - ✓ Punjab
 - ✓ Haryana
 - ✓ Delhi
 - ✓ Chandigarh

- ✓ Ganganagar district of Rajasthan
- This region is known as:
 - ✓ India's most intensively irrigated agricultural zone
 - ✓ The core of the Green Revolution region
- Agriculture here depends heavily on:
 - ✓ Extensive canal irrigation systems
 - ✓ Large-scale tube well irrigation
- Groundwater situation:
 - ✓ Extremely high groundwater extraction
 - ✓ Many districts categorized as over-exploited or "dark zones"
 - ✓ Water table is declining rapidly due to continuous cultivation of:
 - Paddy (rice)
 - Wheat

Western Himalayan Region supports horticulture:

- Soils include:
 - ✓ Skeletal soils
 - ✓ Mountain meadow soils
- Suitable for:
 - ✓ Apple cultivation
 - ✓ Walnut and cherry
 - ✓ Saffron cultivation in Karewa soils (notably in Kashmir)

Middle Gangetic Plain supports rice cultivation

- Covers:
 - ✓ Eastern Uttar Pradesh
 - ✓ Bihar
- Receives:
 - ✓ Higher rainfall (~120 to 150 cm)
- Agriculture pattern:
 - ✓ Dominated by rice-based farming

Planning Commission created 15 Agro-Climatic Zones

- India has:
 - ✓ 15 Agro-Climatic Zones (ACZ)
- Defined by:
 - ✓ Planning Commission (1988)
- Based on:
 - ✓ Climate
 - ✓ Topography
 - ✓ Soil characteristics

Q 130. Ans. B

Explanation:

Statement 1 is correct:

Heavy river discharge into the Bay of Bengal creates layered waters, limiting mixing with cooler deep water.

- The Bay of Bengal receives massive freshwater discharge from rivers such as:
 - ✓ Ganga
 - ✓ Brahmaputra
 - ✓ Mahanadi
 - ✓ Godavari
- This creates:
 - ✓ A low-salinity surface layer
 - ✓ Stratified (layered) water structure
- Effect:
 - ✓ Prevents vertical mixing with cooler deep water
 - ✓ Keeps sea surface temperature high (above 26.5°C)
 - ✓ Warm surface water supports cyclone formation and intensification

Statement 2 is incorrect:

Strong vertical wind differences over the Arabian Sea help the strengthening of tropical cyclones.

- Vertical wind shear refers to changes in wind speed and direction with height.
- Strong vertical wind shear:
 - ✓ Disrupts cyclone structure
 - ✓ Tilts the vertical column of rising air
 - ✓ Prevents cyclone intensification
- Arabian Sea condition:
 - ✓ Historically has higher vertical wind shear
 - ✓ This reduces cyclone formation, not increases it

Statement 3 is correct:

Weak low-pressure systems from the South China Sea often enter the Bay of Bengal, supporting cyclone formation.

- Many cyclones in the Bay of Bengal:
 - ✓ Begin as remnants of typhoons from the South China Sea
- Process:
 - ✓ These systems cross Southeast Asia
 - ✓ Enter the Bay of Bengal
 - ✓ Re-intensify over warm ocean waters
- This contributes significantly to:
 - ✓ Higher cyclone frequency in the Bay of Bengal

Q 131. Ans. C

Explanation:

Champakara and Udyogmandal canals

- National Waterway 3 (NW-3) is located entirely in Kerala.

- It is based on the West Coast Canal system.
- Two important feeder canals integrated into NW-3 are:
 - ✓ Champakara Canal
 - Length: about 14 km
 - Connects Kochi port with Ambalamugal industrial region
 - ✓ Udyogmandal Canal
 - Length: about 23 km
 - Connects Kochi with Eloor industrial belt
- These canals help:
 - ✓ Link industrial zones with inland waterways
 - ✓ Facilitate transport of fertilizers, chemicals, and petroleum products

Buckingham and Commamur canals:

- Associated with National Waterway 4 (NW-4)
- Located in:
 - ✓ Andhra Pradesh
 - ✓ Tamil Nadu
- Part of the Godavari–Krishna delta canal system

Kakinada and Eluru canals:

- Also part of National Waterway 4 (NW-4)
- Located in Andhra Pradesh
- Linked with Godavari and Krishna river systems

Hijli Tidal and Orissa canals

- Associated with National Waterway 5 (NW-5)
- Located in:
 - ✓ Odisha
 - ✓ West Bengal
- Connected with Brahmani and Mahanadi delta systems.

Q 132. Ans. A

Explanation:

- Cauvery River: Tamil Nadu, Karnataka, Kerala, and Puducherry
 - ✓ Major dispute mainly between Karnataka and Tamil Nadu
 - ✓ Kerala is included because important tributaries originate there.
 - ✓ Puducherry (Karaikal region) depends on Cauvery delta waters.
- Krishna River: Maharashtra, Karnataka, Andhra Pradesh, and Telangana
 - ✓ Initially a three-state dispute.

- ✓ Became a four-state dispute after the 2014 creation of Telangana.
- ✓ Water sharing governed through Krishna Water Disputes Tribunal (KWDT).
- Sutlej–Yamuna Link (SYL): Punjab, Haryana, and Rajasthan
 - ✓ Dispute over sharing waters of Ravi and Beas rivers.
 - ✓ Haryana demands canal completion to receive allocated water.
 - ✓ Punjab opposes construction citing water scarcity.
- Mahadayi (Mandovi) River: Goa, Karnataka, and Maharashtra
 - ✓ Dispute mainly due to Kalasa–Banduri Nala diversion project.
 - ✓ Karnataka seeks diversion to Malaprabha basin.
 - ✓ Goa fears reduction in downstream flow.

Additional Facts: Constitutional Framework of River Disputes

- Article 262 of the Constitution
 - ✓ Empowers Parliament to:
 - Provide mechanisms for resolving inter-state river disputes.
 - Exclude Supreme Court jurisdiction if necessary.
 - ✓ Purpose:
 - Prevent prolonged litigation.
 - Encourage specialized tribunals.
- Inter-State River Water Disputes Act, 1956
 - ✓ Enacted under Article 262.
 - ✓ Provides for:
 - Creation of Water Disputes Tribunals.
 - Settlement of disputes between states.
 - ✓ Important feature:
 - Tribunal decisions are final and binding.

Q 133. Ans. B

Explanation:

- Jog Falls, located on the Sharavati River in Karnataka, is a classic plunge-type waterfall.
- Key geomorphological feature:
 - ✓ In a plunge waterfall, water:
 - Falls vertically downward
 - Loses contact with the rock face during the fall
 - Drops freely due to formation of an overhang.

- Formation mechanism:
 - ✓ River flows over:
 - A hard, resistant rock layer
 - Overlying a softer rock layer
 - ✓ The softer layer gets eroded faster.
 - ✓ This creates an undercut cliff, leading to free fall of water.

Additional Facts: The Structure of Jog Falls

Jog Falls is not a single stream. It consists of four distinct segments, each with unique characteristics:

- Raja
 - ✓ Tallest segment
 - ✓ Falls in a single uninterrupted vertical column
- Roarer
 - ✓ Falls with tremendous force
 - ✓ Meets Raja midway with loud sound
- Rocket
 - ✓ Shoots downward in jet-like streams
- Rani
 - ✓ Graceful and winding
 - ✓ Appears like a flowing ribbon
- Total height:
 - ✓ About 253 meters
 - ✓ One of the highest untiered waterfalls in India
- Additional Facts: Height and Ranking of Indian Waterfalls
 - ✓ Jog Falls
 - Highest untiered (single-drop) waterfall in India.
- Kunchikal Falls (Karnataka)
 - ✓ Highest waterfall in India overall.
 - ✓ Located on Varahi River.
 - ✓ Classified as a tiered waterfall.

Additional Facts: Hydroelectric Importance

- Linganamakki Dam
 - ✓ Located upstream of Jog Falls.
 - ✓ Regulates water flow of the Sharavati River.
- Effects:
 - ✓ During dry season:
 - Waterfall flow reduces significantly.
- During monsoon (July–September):
 - ✓ Dam gates opened.
 - ✓ Falls regain full volume.
- Mahatma Gandhi Hydroelectric Project
 - ✓ Commissioned in 1948.
 - ✓ One of the earliest major hydroelectric projects in India.
 - ✓ Uses waters of the Sharavati River.

Q 134. Ans. A

Explanation:

Statement 1 is correct:

The region predominantly spans the rugged, densely forested terrains of Chhattisgarh, Odisha, and Andhra Pradesh.

- Dandakaranya is a large plateau and forested region located in:
 - ✓ Chhattisgarh (especially Bastar region)
 - ✓ Odisha (Koraput, Kalahandi districts)
 - ✓ Andhra Pradesh (northern parts)
 - ✓ Some extensions also reach Maharashtra and Telangana
- Key physical features:
 - ✓ Dense tropical forests
 - ✓ Dissected plateau terrain
 - ✓ Located between:
 - Chotanagpur Plateau (north)
 - Eastern Ghats (east)

Statement 2 is incorrect:

- Dandakaranya is not known for coal reserves.
- The region is famous for:
 - ✓ High-grade iron ore
 - ✓ Tin deposits
 - ✓ Other metallic minerals
- India's major Gondwana coal reserves are located in:
 - ✓ Damodar Valley
 - Jharia
 - Raniganj
 - Bokaro
 - ✓ Mahanadi Valley
 - ✓ Son Valley
- Important mineral fact:
 - ✓ Bailadila Hills (Chhattisgarh) contain:
 - Some of the highest-grade hematite iron ore in India.

Additional Facts: Mineral Wealth of Dandakaranya Bailadila Iron Ore Deposits

- Located in:
 - ✓ Bastar district, Chhattisgarh
- Known for:
 - ✓ High-grade hematite iron ore
 - ✓ Significant export-quality ore
- Transport link:
 - ✓ Kirandul–Kothavalasa railway line
 - ✓ Connects mines to:
 - Visakhapatnam Port

- Major export destinations:
 - ✓ Japan
 - ✓ South Korea

Tin Production in India:

- Chhattisgarh is the only tin-producing state in India.
- Tin ore:
 - ✓ Cassiterite
- Main location:
 - ✓ Bastar region

Additional Facts: Drainage System:

- Indravati River
 - ✓ Major river draining the Dandakaranya region.
 - ✓ Tributary of:
 - Godavari River
- Important feature:
 - ✓ Forms Chitrakote Falls in Bastar.
- Chitrakote Falls
 - ✓ Known as:
 - “Niagara Falls of India”
 - ✓ Located on:
 - Indravati River
 - ✓ Type:
 - Block-type waterfall

Additional Facts: Geomorphology of Dandakaranya:

Dissected Plateau Nature

- Not a flat plateau.
- Characterized by:
 - ✓ Deep river valleys
 - ✓ Steep ridges
 - ✓ Uneven terrain

Result:

- Difficult accessibility
- High forest cover
- Significant tribal population

Q 135. Ans. A

Explanation:

Statement 1 is correct:

- The Rayalaseema Plateau contains the Cuddapah Basin, a well-known Proterozoic sedimentary basin.
- The basin is crescent-shaped, which is a commonly tested geomorphological feature.

- It is rich in important sedimentary rocks such as:
 - ✓ Limestone — widely used in the cement industries
 - ✓ Quartzite — used in construction and industrial applications
- The abundance of limestone has led to the development of major cement industries in the Rayalaseema region.

Statement 2 is correct:

- The Telangana Plateau is composed mainly of ancient Archaean gneisses, typical of the Indian Shield region.
- Due to the hard, rocky terrain and undulating surface, groundwater storage is limited.
- As a result, the region traditionally relies on tank irrigation, where rainwater is stored in small reservoirs or tanks.
- This irrigation method has historical importance and has been revived through initiatives such as Mission Kakatiya.

Statement 3 is incorrect:

- The statement incorrectly claims that Rayalaseema receives significantly higher Southwest Monsoon rainfall than Telangana.
- In reality, Rayalaseema is one of the driest regions of India, located in the rain shadow of the Western Ghats.
- Telangana receives relatively more Southwest Monsoon rainfall than Rayalaseema.

Additional Facts:

- Rayalaseema Plateau — Key Features
 - ✓ Location:
 - Southern part of Andhra Pradesh
 - ✓ Important geological feature:
 - Cuddapah Basin — crescent-shaped sedimentary basin
 - ✓ Major minerals:
 - Limestone
 - Barytes
 - Quartzite
 - ✓ Drainage system:
 - This region is drained by the Krishna, Tungabhadra, and Pennar river systems.
 - ✓ Climate feature:
 - Lies in the rain shadow zone of the Western Ghats, resulting in low rainfall.

Telangana Plateau — Key Features:

- Rock structure:
 - ✓ Dominated by Archaean gneissic rocks
- Drainage system:
 - ✓ Drained mainly by the Godavari and Krishna river systems.
- Irrigation pattern:
 - ✓ Heavy reliance on tank irrigation due to limited groundwater storage.
- Major mineral region:
 - ✓ Singareni coalfields are an important source of coal.

Rainfall Contrast Between the Two Regions

- Telangana Plateau:
 - ✓ Receives moderate rainfall from the Southwest Monsoon.
- Rayalaseema Plateau:
 - ✓ Receives comparatively low Southwest Monsoon rainfall.
 - ✓ Sometimes receives additional rainfall from the Northeast Monsoon, but overall remains drought-prone.

Q 136. Ans. A

Explanation:

- Bhur formations are elevated, sandy tracts found mainly in the Upper Ganga Plain, especially in western Uttar Pradesh.
- They are formed by aeolian (wind-blown) deposition of sand, particularly during dry summer months when strong winds carry loose sand from riverbeds.
- These sandy deposits usually occur over the Bhangar (old alluvial) terraces.
- Because the soil is sandy and highly porous, it does not retain moisture well and is therefore less fertile for agriculture compared to the surrounding alluvial plains.

Additional Facts:

- Bhur — Key Characteristics
- Location:
 - ✓ Found mainly in the Upper Ganga Plain, especially in districts such as:
 - Bijnor
 - Moradabad
 - Rampur
- Formation process:

- ✓ Created by wind deposition of sand (aeolian action).
- Soil nature:
 - ✓ Sandy
 - ✓ Highly porous
 - ✓ Low water retention capacity
- Agricultural value:
 - ✓ Generally less fertile than the surrounding alluvial soils.

Q 137. Ans. B

Explanation:

Statement 1 is correct:

- The Chakara (mudbank) phenomenon involves the suspension of fine clay and mud particles in coastal waters.
- This suspended mud absorbs wave energy, reducing turbulence and creating unusually calm water patches even during rough sea conditions.
- This calm zone forms naturally during the Southwest Monsoon and supports intensive fishing activity.

Statement 2 is incorrect:

- The statement wrongly places the Chakara phenomenon on the Coromandel Coast during the Northeast Monsoon.
- In reality, the Chakara phenomenon occurs:
 - ✓ Along the Malabar Coast (Kerala)
 - ✓ During the Southwest Monsoon (June to September)
- It is particularly observed near Alappuzha and Kollam along Kerala's coast.

Additional Facts:

- Chakara (Mudbanks):
 - ✓ Location:
 - Found along the Malabar Coast of Kerala, especially near Alappuzha and Kollam.
 - ✓ Season of occurrence:
 - Occurs during the Southwest Monsoon.
 - ✓ Main feature:
 - Formation of calm water patches amidst rough monsoon seas.
 - ✓ Economic importance:
 - Supports intensive fishing, especially for prawns and fish, during monsoon months.

Possible Causes of Chakara Formation:

- Wave-mud interaction theory:
 - ✓ The Southwest Monsoon swells churn up and suspend bottom sediments, creating a viscous fluid mud layer that then dissipates wave energy.
- Subterranean water theory:
 - ✓ Underground water pressure forces fine mud upward through the seabed.
- Backwater discharge theory:
 - ✓ Sediments from Kerala's backwaters (Kayals) enter the sea during heavy monsoon discharge.

Malabar vs Coromandel Coast: Key Differences

- Malabar Coast:
 - ✓ Faces the Arabian Sea
 - ✓ Receives heavy Southwest Monsoon rainfall
 - ✓ Known for backwaters (Kayals)
 - ✓ Experiences the Chakara phenomenon
- Coromandel Coast:
 - ✓ Faces the Bay of Bengal
 - ✓ Receives major rainfall from the Northeast Monsoon
 - ✓ Does not experience Chakara mudbanks

Q 138. Ans. D

Explanation:

The Gulf of Mannar is a micro-tidal region, meaning it has a very low tidal range.

- Tidal power generation requires macro-tidal conditions, where there is a large difference between high tide and low tide.
- Therefore, the Gulf of Mannar is not suitable for tidal energy generation.

The Gulf of Kutch holds India's highest tidal energy potential due to its extreme tidal range.

- The Gulf of Kutch has one of the highest tidal range in India, reaching up to about 8 metres.
- Its funnel-shaped coastline enhances tidal movement, making it the most suitable location in India for tidal power projects.

Major west-flowing rivers like the Narmada and Tapi drain directly into the Gulf of Khambhat.

- The Gulf of Khambhat (Cambay) receives waters from major west-flowing rivers such as:
 - ✓ Narmada
 - ✓ Tapi
 - ✓ Mahi
 - ✓ Sabarmati

- These rivers form large estuaries rather than deltas due to steep gradients.

The Gulf of Mannar is geographically characterised by a rich marine biosphere reserve and coral reefs.

- The Gulf of Mannar is known for its rich biodiversity, including:
 - ✓ Coral reefs
 - ✓ Seagrass beds
- It was designated as India's first Marine Biosphere Reserve.

Additional Facts:

- Gulf of Kutch — Key Features
 - ✓ Location:
 - Northwestern coast of Gujarat
 - ✓ Main characteristic:
 - Highest tidal range in India
 - ✓ Economic importance:
 - Suitable for tidal energy development
 - Contains major ports such as Kandla and Mundra
 - ✓ Ecological feature:
 - Houses Marine National Park (Jamnagar).

Gulf of Khambhat — Key Features

- Location:
 - ✓ Between the Kathiawar Peninsula and mainland Gujarat
- Major rivers draining into it:
 - ✓ Narmada
 - ✓ Tapi
 - ✓ Mahi
 - ✓ Sabarmati
- Geographical feature:
 - ✓ Characterised by broad estuaries due to fast-flowing rivers.

Gulf of Mannar: Key Features

- Location:
 - ✓ Between the Tamil Nadu coast and Sri Lanka
- Important ecological feature:
 - ✓ India's first Marine Biosphere Reserve
- Major marine species:
 - ✓ Dugong (Sea Cow)
 - ✓ Coral reef communities
- Tidal condition:
 - ✓ Micro-tidal, unsuitable for tidal energy generation.

Q 139. Ans. A**Explanation:**

Incorrect Statement: Peninsular rivers maintain a perennial flow regime throughout the year due to groundwater-based flow.

- This statement is incorrect because peninsular rivers are generally seasonal (non-perennial).
- They lack glacial sources and depend almost entirely on monsoon rainfall.
- During the dry summer months, their discharge reduces sharply, and many smaller rivers may dry up, leaving isolated pools in rocky channels.
- Groundwater base flow alone is not sufficient to sustain perennial flow in most peninsular rivers.

Additional Facts:

- Himalayan Rivers: Flow Characteristics

Nature:

- Mostly perennial rivers

Water sources:

- Snowmelt
- Glacier melt
- Monsoon rainfall

Flow pattern:

- Continuous flow throughout the year
- Peak discharge during the summer and monsoon

Examples:

- Ganga
- Brahmaputra
- Indus

Peninsular Rivers: Flow Characteristics

- Nature:
 - ✓ Mostly seasonal rivers
- Water source:
 - ✓ Rainfall during the monsoon
- Flow pattern:
 - ✓ Very low discharge in summer
 - ✓ Sudden increase during the monsoon
 - ✓ Rapid decline after monsoon withdrawal
- Examples:
 - ✓ Godavari
 - ✓ Krishna
 - ✓ Narmada
 - ✓ Mahanadi

Q 140. Ans. C**Explanation:**

- Aravalli Range:
 - ✓ The Aravalli Range is one of the oldest fold mountains in the world, formed during the Precambrian era.
 - ✓ It originated due to compressional tectonic forces, where crustal plates converged and folded the rocks.
 - ✓ Over millions of years, it has undergone intense erosion, reducing its height significantly.
 - ✓ Because of this long erosion history, it is now classified as a relict (residual) fold mountain, not a block mountain.
- Satpura Range:
 - ✓ The Satpura Range is a classic block mountain (horst).
 - ✓ It was formed due to faulting, where land blocks were uplifted along fractures in the Earth's crust.
 - ✓ It lies south of the Narmada Rift Valley, which represents a graben (down-faulted block).
- Vindhyan Range:
 - ✓ The Vindhyan Range is also associated with fault-block formation.
 - ✓ It forms the northern boundary of the Narmada Rift Valley.
 - ✓ Like the Satpuras, it represents an uplifted block (horst) created by crustal faulting.
- Nilgiri Hills:
 - ✓ The Nilgiri Hills are considered fault-block mountains.
 - ✓ They were uplifted due to tectonic faulting along the margins of the Deccan Plateau.
 - ✓ They form the junction of the Western Ghats and Eastern Ghats.
- Key Concept Summary:
 - ✓ Fold Mountains formation mechanism:
 - ✓ Formed by the compression of tectonic plates.
- Examples:
 - ✓ Himalayas (young fold mountains)
 - ✓ Aravalli Range (old fold mountains)
- Important point:
 - ✓ Old fold mountains like the Aravallis are now heavily eroded and classified as relict mountains.

Block Mountains

- Formation mechanism:
 - ✓ Formed by faulting, where large crustal blocks are uplifted or dropped.
- Key associated terms:
 - ✓ Horst → Raised block (mountain)
 - ✓ Graben → Dropped block (rift valley)
- Examples in India:
 - ✓ Satpura Range
 - ✓ Vindhyan Range
 - ✓ Nilgiri Hills

Q 141. Ans. C

Explanation:

Statement 1 is incorrect:

- In a Positive Indian Ocean Dipole (IOD):
 - ✓ Western Indian Ocean (near East Africa) becomes warmer than normal.
 - ✓ Eastern Indian Ocean (near Indonesia/Australia) becomes cooler than normal.

Statement 2 is incorrect:

- During a Negative IOD:
 - ✓ Western Indian Ocean becomes cooler than normal.
 - ✓ The thermocline becomes shallower, not deeper, in the western region.
 - ✓ This reduces convection and weakens moisture supply, often leading to deficient monsoon rainfall in India.

Statement 3 is correct:

- During a Positive IOD:
 - ✓ Warm waters in the western Indian Ocean cause strong convection and rising air.
 - ✓ This enhances moisture transport toward India.
 - ✓ If a Positive IOD coincides with El Niño, it can partially offset El Niño's weakening effect on the monsoon, reducing drought risk.

Key Concept Summary:

- Positive Indian Ocean Dipole (Positive IOD)
 - ✓ Sea surface temperature pattern:
 - Western Indian Ocean → Warmer
 - Eastern Indian Ocean → Cooler
 - ✓ Atmospheric effect:
 - Strong convection over the western Indian Ocean

- Enhanced monsoon rainfall over India
- Can neutralise El Niño drought effects
- ✓ Thermocline pattern:
 - Deeper in the west
 - Shallower in the east
- Negative Indian Ocean Dipole (Negative IOD)
 - ✓ Sea surface temperature pattern:
 - Western Indian Ocean → Cooler
 - Eastern Indian Ocean → Warmer
 - ✓ Atmospheric effect:
 - Moisture shifts toward Indonesia and Australia
 - Weak Indian monsoon
 - Increased drought risk in India
 - ✓ Thermocline pattern:
 - Shallower in the west
 - Deeper in the east

Q 142. Ans. C

Explanation:

- **Pair 1 is correct:** Bhakra Nangal Project — Forms the Gobind Sagar reservoir on the Sutlej River.
 - ✓ The Bhakra Nangal Dam is constructed across the Sutlej River.
 - ✓ The reservoir created behind the dam is called Gobind Sagar.
 - ✓ It is one of India's major multipurpose river valley projects, providing irrigation, hydropower, and flood control.
- **Pair 2 is incorrect:** Tehri Dam Project — Located precisely at the confluence of the Bhagirathi and Alaknanda rivers.
 - ✓ This pair is incorrect.
 - ✓ The confluence of the Bhagirathi and Alaknanda rivers is called Devprayag, where the river officially takes the name Ganga.
 - ✓ The Tehri Dam is not located there.
 - ✓ It is built on the Bhagirathi River at its confluence with the Bhilangna River, upstream of Devprayag.
- **Pair 3 is correct:** Sardar Sarovar Project — Rajasthan receives a share of the irrigation water but no share of the generated hydroelectric power.
 - ✓ This pair is correct.
 - ✓ The Sardar Sarovar Project is built on the Narmada River.
 - ✓ Rajasthan receives irrigation water through the Narmada Canal.

- ✓ However, Rajasthan does not receive any share of the hydroelectric power generated by the project.
- **Pair 4 is correct:** Hirakud Project — Primarily constructed across the Mahanadi river to mitigate flooding in the coastal delta region of Odisha.
 - ✓ This pair is correct.
 - ✓ The Hirakud Dam is constructed across the Mahanadi River in Odisha.
 - ✓ One of its primary purposes was flood control, as the Mahanadi frequently caused destructive floods in the delta region before the dam was built.
- Karakoram Range
 - ✓ Northernmost
 - ✓ Includes Saltoro Ridge
 - ✓ Contains major glaciers such as the Siachen Glacier
- Ladakh Range
 - ✓ Lies between the Shyok River and the Indus River
- Zaskar Range
 - ✓ Located south of the Indus River
- Pir Panjal Range
 - ✓ Part of the Lesser Himalayas
 - ✓ Surrounds the Kashmir Valley

Additional High-Value Facts:

Strategic Importance of Saltoro Ridge

- Controls access to the Siachen Glacier region.
- Major passes located here include:
 - ✓ Sia La
 - ✓ Bilafond La
- The ridge plays a critical role in defining military positions in the region.

Q 144. Ans. B

Explanation:

Statement B is correct: The river rises in the Dagshai hills of the Shiwaliks and ends as an inland-draining river in the Thar Desert.

- The Ghaggar River originates in the Dagshai Hills of the Shiwalik Range in Himachal Pradesh.
- It flows through Punjab, Haryana, and Rajasthan.
- In Rajasthan, the river loses its flow and disappears into the sands of the Thar Desert, making it an inland-draining (endorheic) river system.
- Its downstream continuation in Pakistan is known as the Hakra River.

Statement A is incorrect: It is a year-round Himalayan river that acts as a major left-bank tributary of the Indus.

- The Ghaggar is not a perennial Himalayan river.
- It is a seasonal (ephemeral) river, flowing mainly during the monsoon.
- It does not join the Indus River; instead, it disappears in the desert.

Statement C is incorrect: It flows across the Malwa Plateau and is the main cause of gully erosion in Rajasthan.

Q 143. Ans. A

Explanation:

- Saltoro Ridge:
 - ✓ The Saltoro Ridge is a sub-range of the Karakoram Range.
 - ✓ It lies west of the Siachen Glacier and forms its southwestern boundary.
 - ✓ The ridge separates the Siachen Glacier basin from the valleys lying to the west.
 - ✓ Because it is structurally part of the Karakoram system, the correct answer is Karakoram Range.
- Ladakh Range:
 - ✓ The Ladakh Range lies south of the Karakoram Range.
 - ✓ It is separated from the Karakoram by the Shyok River valley.
 - ✓ Since Saltoro Ridge lies within the Karakoram system, it cannot belong to the Ladakh Range.
- Zaskar Range:
 - ✓ The Zaskar Range lies further south, beyond the Indus River.
 - ✓ Therefore, it is geographically distinct from the Saltoro Ridge.
- Pir Panjal Range:
 - ✓ The Pir Panjal Range is also part of the Lesser Himalayas.
 - ✓ It lies even further south and encloses the Kashmir Valley.
 - ✓ It is completely separate from the Karakoram mountain system.

Key Geographical Framework:

North-to-South Sequence of Major Ranges in This Region

- The river associated with gully erosion (ravines) in the Malwa Plateau is the Chambal River, not the Ghaggar.
- The Chambal basin is famous for its deep ravines, especially in Madhya Pradesh and Rajasthan.

Statement D is incorrect: The river system flows entirely within the boundaries of India.

- While the Ghaggar flows in India, its continuation, called the Hakra River, extends into Pakistan, where it dissipates in the Cholistan Desert.
- Therefore, the system crosses international boundaries.

Summary:

- Ghaggar–Hakra River System — Essential Facts
 - ✓ Origin:
 - Dagshai Hills, Shiwalik Range, Himachal Pradesh
 - ✓ Nature:
 - Seasonal (ephemeral) river
 - ✓ Drainage type:
 - Inland drainage (endorheic system)
 - Disappears in the Thar Desert
 - ✓ Course through India:
 - Himachal Pradesh
 - Punjab
 - Haryana
 - Rajasthan
 - ✓ Continuation outside India:
 - Known as the Hakra River in Pakistan
 - Disappears in the Cholistan Desert

Additional High-Value Facts:

- Historical Importance
 - ✓ Many Indus Valley Civilisation (Harappan) sites developed along the Ghaggar basin.
 - ✓ Important archaeological sites include:
 - Kalibangan (Rajasthan)
 - Banawali (Haryana)

Q 145. Ans. B

Explanation:

- **Pair A is correct:** Wheeler Island (Kalam Island) — Located off the coast of Odisha and used as India's main missile testing centre.
 - ✓ Wheeler Island, officially called Dr. Abdul Kalam Island, lies off the Odisha coast.

- ✓ It houses the Integrated Test Range (ITR), India's principal facility for testing ballistic missiles such as Agni and Prithvi.

- **Pair B is incorrect:** Sagar Island — It is used as an important monitoring point for cyclones coming from the Arabian Sea.

- ✓ Sagar Island is located in the Ganga delta in West Bengal, at the mouth of the Hooghly River.
- ✓ It lies in the Bay of Bengal, not near the Arabian Sea.
- ✓ Therefore, it cannot monitor cyclones from the Arabian Sea, which lies on the western side of India.

- **Pair C is correct:** Pamban Island — Located between India and Sri Lanka and marks the starting point of Rama Setu near Dhanushkodi.

- ✓ Pamban Island (also known as Rameswaram Island) lies between India and Sri Lanka.
- ✓ The town of Dhanushkodi, located at its eastern tip, marks the starting point of the chain of limestone shoals known as Rama Setu (Adam's Bridge).

- **Pair D is correct:** Sriharikota Island — A barrier island in Andhra Pradesh that separates Pulicat Lake from the Bay of Bengal.

- ✓ Sriharikota is a classic barrier island along the Andhra Pradesh coast.
- ✓ It separates Pulicat Lake from the Bay of Bengal.
- ✓ The island hosts the Satish Dhawan Space Centre, India's primary satellite launch site.

Q 146. Ans. A

Explanation:

- **Statement 1 is correct:** The Kosi River is characterised by an exceptionally high sediment load derived from the fragile and tectonically active Himalayas.
 - ✓ The Kosi River originates in the young Himalayan mountains, which are geologically unstable and prone to erosion.
 - ✓ Due to steep slopes and fragile rocks, the river carries huge quantities of sediment, including silt, sand, and pebbles.
 - ✓ This heavy sediment load is one of the main reasons for its unstable and shifting course.
- **Statement 2 is correct:** The sudden decrease in channel gradient as the river enters the plains of Bihar leads to rapid aggradation (deposition) of the riverbed.

- ✓ When the Kosi flows from the steep Himalayan region into the flat plains of North Bihar, its velocity reduces sharply.
- ✓ The river loses the energy required to transport sediment.
- ✓ As a result, rapid deposition (aggradation) occurs, raising the riverbed and increasing flood risk.
- **Statement 3 is incorrect:** The consistent westward migration of the Kosi is primarily attributed to the tectonic tilting of the Indo-Gangetic Foredeep towards the west.
 - ✓ This statement is incorrect because the westward shifting of the Kosi is mainly due to sediment-driven processes, not tectonic tilting.
 - ✓ The river frequently changes course because it deposits sediment and blocks its own path, forcing it to seek a new, lower route.
 - ✓ This repeated shifting occurs across its large alluvial fan, not because of consistent tectonic movement.
- **Statement 4 is correct:** “Avulsion” in the Kosi river occurs when the riverbed becomes higher than the surrounding floodplains due to continuous siltation.
 - ✓ This statement correctly defines avulsion.
 - ✓ Continuous deposition raises the riverbed above the surrounding land, making it unstable.
 - ✓ During heavy floods, the river breaks embankments and suddenly shifts to a new channel at a lower level.
 - ✓ The Kosi River is a classic example of an avulsing river.
- ✓ During hot seasons, capillary action pulls dissolved salts (such as sodium, magnesium, and calcium salts) upward from the groundwater to the soil surface.
- ✓ When the water evaporates, it leaves behind a white salt crust, making the soil toxic and unsuitable for crops.
- **Statement 2 is correct:** The application of gypsum is a common method to reclaim Usar lands, as it helps reduce soil alkalinity and restore soil fertility.
 - ✓ Gypsum (calcium sulfate) is the most widely used chemical amendment for reclaiming alkaline (sodic) Usar soils.
 - ✓ In these soils, excess sodium binds tightly to clay particles, making the soil compact and impermeable.
 - ✓ When gypsum is applied:
 - Calcium replaces sodium on clay particles.
 - The released sodium is then leached downward below the root zone using irrigation water.
- This process restores soil structure and improves agricultural productivity.

Additional facts: Salt-Affected Soils

- Local Names of Salt-Affected Soils
 - ✓ These soils are known by different regional terms across North India:
 - ✓ Reh
 - ✓ Kallar
 - ✓ Usar
 - ✓ Thur
 - ✓ Chopan
 - ✓ Karl

Key Physical Process:

Q 147. Ans. D

Explanation:

- **Statement 1 is correct:** Usar lands are formed due to the upward movement of saline and alkaline salts to the surface, caused by poor drainage and a high water table.
 - ✓ ‘Usar’ is the traditional term used for salt-affected soils (both saline and alkaline), especially in the Ganga Plains such as western Uttar Pradesh, Punjab, and Haryana.
 - ✓ These soils commonly develop in areas with extensive canal irrigation, where excess water leads to waterlogging and a rise in the groundwater table.
- The upward movement of salts occurs due to capillary action, not leaching.
- Capillary action allows water to move upward against gravity through fine soil pores.
- This process becomes intense when:
 - ✓ The water table is high
 - ✓ Evaporation rates are high
 - ✓ Drainage is poor
- Important distinction:
 - ✓ Capillary action → brings salts upward
 - ✓ Leaching → washes salts downward

Saline vs. Alkaline Soil Reclamation

- Saline soils
 - ✓ Contain neutral soluble salts
 - ✓ Reclaimed mainly by leaching with fresh water
- Alkaline (sodic) soils
 - ✓ Contain excess sodium
 - ✓ Require gypsum application first, followed by leaching

Biological Reclamation Method

- After gypsum treatment, farmers often grow salt-tolerant green manure crops.
- The most commonly used crop is:
 - ✓ Dhaincha (Sesbania)
 - ✓ A nitrogen-fixing legume
 - ✓ Improves soil organic matter
 - ✓ Helps lower soil pH
 - ✓ Enhances long-term soil fertility

Q 148. Ans. B

Explanation:

- **Landfall Island:** Sits at the very top of the island chain. It is strategically significant because it is the closest Indian territory to Myanmar, separated from Myanmar's Coco Islands by the Coco Channel.
- **North Andaman Island:** The main northern landmass, located just south of Landfall Island (separated by the Cleugh Passage). It houses Saddle Peak, the highest point in the archipelago.
- **Narcondam Island:** It is located in the Andaman Sea, far to the east of the main island chain, and is a dormant volcano.
- **Smith Island:** Located slightly further south, just off the east coast of North Andaman.

Q 149. Ans. A

Explanation:

- **Statement 1 is correct:** Wular Lake is classified as a tectonic lake formed due to subsidence of the earth's crust within a synclinal trough.
 - ✓ Wular Lake, located in Jammu and Kashmir, is a tectonic lake.
 - ✓ It was formed due to subsidence (downward movement) of the earth's crust in a synclinal trough within the Kashmir Valley.

- ✓ Although the Jhelum River feeds the lake, its basin originated primarily due to tectonic activity, not river erosion.

- **Statement 2 is incorrect:** The Tulbul Navigation Project is located at the point where the Jhelum River enters Wular Lake.
- Tulbul Navigation Project is located at the outlet (exit) of Wular Lake, not at the entry point.
- It is situated near Sopore, where the Jhelum River flows out of Wular Lake.
- The project functions as a navigation lock-cum-control structure, designed to regulate water release from the lake to maintain sufficient depth in the Jhelum River during winter months.

Additional Facts:

- Jhelum River — Key Geographical Points
 - ✓ Origin:
 - The Jhelum River originates from the Verinag Spring at the foothills of the Pir Panjal Range.
 - ✓ Unique feature:
 - Unlike most Himalayan rivers, the Jhelum forms large meanders within the Kashmir Valley because the valley floor is relatively flat.
 - ✓ Flow through Wular Lake:
 - The Jhelum enters Wular Lake from the southeast.
 - It exits from the western side, continuing toward Pakistan.

Wular Lake:

- Acts as a natural flood reservoir for the Jhelum River.
- Helps regulate seasonal flooding in the Kashmir Valley.
- It is one of the largest freshwater lakes in India.

Lake Types:

- Tectonic lakes
 - ✓ Formed due to crustal movements
 - ✓ Example: Dal Lake, Wular Lake
- Glacial lakes
 - ✓ Formed by glacier erosion
 - ✓ Examples: Roopkund
- Crater lakes
 - ✓ Formed by a meteorite impact
 - ✓ Example: Lonar Lake

- Lagoon lakes
 - ✓ Formed along coasts by sand barriers
 - ✓ Examples: Chilika Lake, Pulicat Lake

Q 150. Ans. B

Explanation:

Centripetal Drainage:

- The term centripetal means centre-seeking.
- In a centripetal drainage pattern, the surrounding land is higher, forming a basin or depression in the middle.
- Rivers and streams flow inward from all directions toward this central low point.
- Water usually collects in a lake, playa, or inland basin.
- Indian Examples:
 - ✓ Loktak Lake (Manipur) — A classic basin receiving drainage from surrounding hills.
 - ✓ Sambhar Lake (Rajasthan) — A closed inland basin where streams flow inward.

Radial drainage:

- Rivers flow outward from a central high point, such as:
 - ✓ Volcanic cones
 - ✓ Domes

- ✓ Plateau peaks

- It is the opposite of centripetal drainage.
- Indian Example:
 - ✓ Amarkantak Plateau, from where rivers flow in different outward directions.

Trellis drainage:

- Forms in regions with alternating hard and soft rock layers.
- Tributaries join the main river at right angles.
- Common in folded mountain regions.

Parallel drainage:

- Develops where there is a uniform steep slope.
- Streams flow almost parallel to each other.
- Indian Example:
 - ✓ Short west-flowing rivers from the Western Ghats into the Arabian Sea.

Dendritic Drainage:

- Key feature:
 - ✓ Resembles branches of a tree
- Formation condition:
 - ✓ Uniform rock structure
- Examples:
 - ✓ Ganga River system
 - ✓ Godavari River basin

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